

Advanced Motorsport Engineering

Laboratories Report

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1 LAB 0 – Basics

1.1 Introduction

Before building anything as complex as a full lap-time simulator, it is worth understanding the basic forces that govern vehicle performance and the simple analytical models that can extract useful information from them. That is the purpose of Lab 0. All five exercises use a **point mass model**: the car is reduced to a single mass with no geometry, no suspension, and no load distribution front-to-rear except where explicitly added. This is a drastic simplification, but it already captures the most important physics and produces surprisingly accurate estimates of real-world performance limits.

The five forces that appear throughout are aerodynamic drag and downforce (both scaling with the square of velocity), rolling resistance (proportional to vertical load), engine traction (derived from torque, gear ratio, and rolling radius), and longitudinal load transfer (the shift of weight between axles during acceleration or braking). In equation form:

$$F_{\text{Drag}} = \tfrac{1}{2}\rho\, c_x S\, v^2, \qquad F_{\text{Downforce}} = \tfrac{1}{2}\rho\, c_z S\, v^2, \qquad F_{\text{roll}} = f_{\text{roll}}\, F_z \tag{1.1}$$

$$F_{\text{engine}} = \frac{\tau_{\text{tot}}\, T(\omega)}{R_{\text{rol}}}, \qquad \Delta F_{z,\text{long}} = \frac{m\, a_x\, h_{\text{COG}}}{\text{Wheelbase}} \tag{1.2}$$

The aerodynamic balance and efficiency, two quantities that will come up repeatedly, are defined as:

$$\text{Bal} = \frac{F_{\text{Downforce,front}}}{F_{\text{Downforce}}} = \frac{c_{z,\text{front}}}{c_z}, \qquad \text{Eff} = \frac{F_{\text{Downforce}}}{F_{\text{Drag}}} = \frac{c_z}{c_x} \tag{1.3}$$

The five exercises build progressively on these ideas: constant-radius cornering (Ex. 1), minimum lap time on a simple oval (Ex. 2), coast-down parameter identification (Ex. 3), wheel load computation under braking (Ex. 4), and aerodynamic parameter estimation from a constant-speed test (Ex. 5).

1.2 Exercise 1 – Constant Radius Cornering

1.2.1 Problem Statement

A car with the parameters of Table 1.1 goes around a constant-radius corner. The maximum cornering speed is found for two radii ($R = 150\,\text{m}$ and $R = 30\,\text{m}$), comparing the case with and without aerodynamic downforce.

Table 1.1: Vehicle parameters – Exercise 1

Parameter	**Value**	**Unit**
Mass	500	kg
Friction coefficient μ	0.9	–
Downforce at 50 m/s	2700	N

1.2.2 Methodology

In steady-state cornering at constant radius, the centripetal acceleration must be sustained by tyre friction. For a point mass, the lateral equilibrium gives:

$$m\,\frac{v^2}{R} = \mu\,F_z = \mu\,(mg + F_{\text{Downforce}}) \tag{1.4}$$

Downforce scales as $F_{\text{Downforce}} = k\,v^2$ with $k = F_{\text{down,ref}}/v_{\text{ref}}^2 = 2700/50^2 = 1.08\,\text{N}\,\text{s}^2/\text{m}^2$. Substituting and solving for v yields a closed-form expression:

$$v = \sqrt{\frac{\mu\,m\,g}{\dfrac{m}{R} - \mu\,k}} \tag{1.5}$$

Without downforce the formula reduces to the familiar $v = \sqrt{\mu\,g\,R}$. Notice that as the downforce coefficient k increases or the radius grows large, the denominator shrinks towards zero and the limiting speed diverges – downforce effectively removes the speed limit at very large radii.

1.2.3 Results

Table 1.2: Maximum cornering speeds – Exercise 1

R [m]	$v_{\text{with DF}}$ [m/s]	$v_{\text{with DF}}$ [km/h]	$v_{\text{no DF}}$ [m/s]	$v_{\text{no DF}}$ [km/h]
150	43.24	155.7	36.41	131.1
30	16.77	60.4	16.28	58.6

The contrast between the two radii is striking. At 150 m the downforce gain is 24.6 km/h (+18.8%), a meaningful performance advantage that would be clearly felt by the driver. At 30 m the gain shrinks to just 1.8 km/h (+3.1%), barely worth mentioning. The reason is simple: a slow corner generates little downforce because $F_{\text{Downforce}} \propto v^2$, so at 60 km/h the aerodynamic grip contribution is almost negligible compared to the static weight.

Figure 1.1 shows the comparison directly, while Figure **??** traces the two curves over a continuous range of radii. The gap between them opens rapidly above $R \approx 50$ m and grows unboundedly as the limiting condition ($\mu\,k\,R/m \to 1$) is approached.

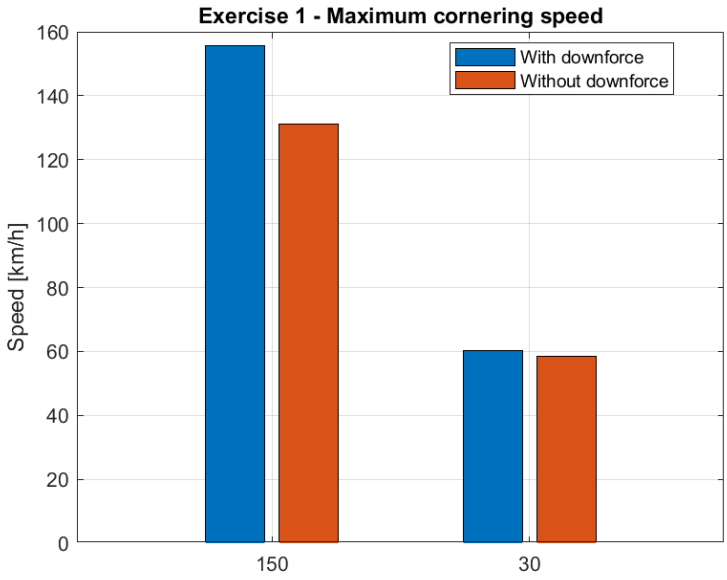


Figure 1.1: Maximum cornering speed with and without downforce for $R = 150\,\mathrm{m}$ and $R = 30\,\mathrm{m}$.

1.3 Exercise 2 – Minimum Lap Time on an Oval Trajectory

1.3.1 Problem Statement

A point mass car ($\mu = 1.1$) must complete an oval in minimum time. The track has two 300 m straights and two semicircular corners of radius $R = 79.58\,\mathrm{m}$.

1.3.2 Methodology

The approach is a simplified quasi-static simulation that separates the problem into four phases. The corner speed follows directly from lateral equilibrium: $v_{\mathrm{corner}} = \sqrt{\mu\, g\, R}$. On the straight the car accelerates at $a_x^+ = \mu\, g$ from the corner exit and then brakes at $a_x^- = \mu\, g$ to arrive at the next corner entry at exactly v_{corner}. The straight is split symmetrically between the two phases. Combined cornering and braking/acceleration conditions (the friction ellipse) are neglected, which slightly overestimates performance but keeps the model closed-form.

1.3.3 Results

Table 1.3: Minimum lap time results – Exercise 2

Quantity	Value	Unit
Corner speed	29.30	m/s
Corner speed	105.5	km/h
Maximum speed on straight	64.00	m/s
Maximum speed on straight	230.4	km/h
Time per straight	6.43	s
Time per corner (semicircle)	8.53	s
Total lap length	1100	m
Minimum lap time	**29.92**	s

The minimum lap time comes out at 29.92 s, with the car reaching a peak of 230 km/h at the mid-straight and slowing to 105 km/h through the corners. It is worth noting that despite the straights being only 300 m long, the car still manages to climb from corner speed to 230 km/h and back: the high value of $\mu = 1.1$ allows very aggressive acceleration and braking. The full speed profile is shown in Figures 1.2 and 1.3.

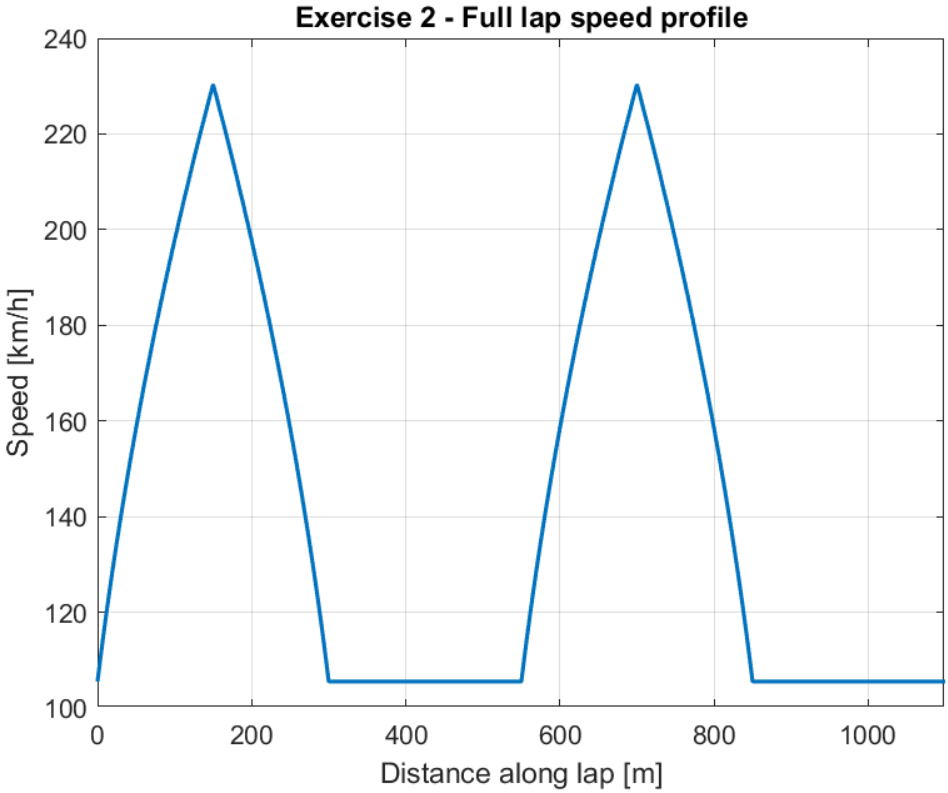


Figure 1.2: Full speed profile along the oval trajectory.

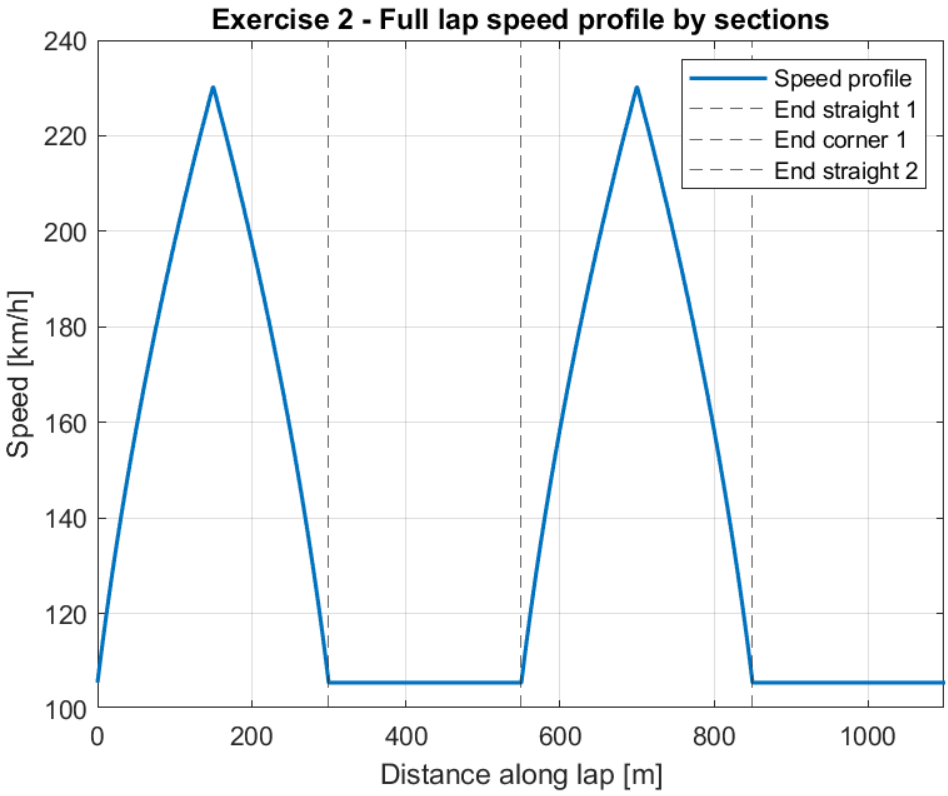


Figure 1.3: Speed profile with section annotations (acceleration, braking, corner).

1.4 Exercise 3 – Coast-Down Parameter Identification

1.4.1 Problem Statement

A car of mass $m = 1380\,\text{kg}$ performs a coast-down test: the clutch is disengaged at high speed and the decelerating speed profile is recorded. Given that $c_z S = 2\,\text{m}^2$, the aerodynamic drag coefficient $c_x S$ and rolling resistance coefficient f_{roll} are to be identified by fitting the equations of motion to the measured data.

1.4.2 Methodology

During a coast-down on a flat road there is no engine force and no road grade, so the deceleration is entirely due to drag and rolling resistance. The equation of motion is:

$$-m\,a_x = F_{\text{Drag}} + F_{\text{roll}} = \tfrac{1}{2}\rho\,c_x S\,v^2 + f_{\text{roll}}\big(mg + \tfrac{1}{2}\rho\,c_z S\,v^2\big) \tag{1.6}$$

Grouping terms by powers of v reveals that the right-hand side is a linear polynomial in v^2:

$$-m\,a_x = \underbrace{\tfrac{1}{2}\rho\,(c_x S + f_{\text{roll}}\,c_z S)}_{A}\,v^2 + \underbrace{f_{\text{roll}}\,mg}_{B} \tag{1.7}$$

This linearity is the key to the identification: if the speed profile is differentiated to obtain $a_x(t)$, a simple first-degree `polyfit` of $-m\,a_x$ against v^2 returns the coefficients A and B directly. The two physical parameters then follow as:

$$f_{\text{roll}} = \frac{B}{mg}, \qquad c_x S = \frac{2A}{\rho} - f_{\text{roll}}\,c_z S \tag{1.8}$$

One subtlety is worth highlighting: the subtraction of $f_{\text{roll}}\, c_z S$ in the second expression is essential. Without it, $c_x S$ would be overestimated because a portion of the v^2 deceleration actually comes from the increased rolling resistance caused by aerodynamic downforce pressing the tyre harder onto the road. Because $c_z S$ must be known in advance to separate the two, this prior knowledge is listed as a given in the problem.

In practice, the speed data is first smoothed with a Savitzky–Golay filter before differentiation to limit the amplification of measurement noise.

1.4.3 Results

Table 1.4: Identified aerodynamic and rolling parameters – Exercise 3

Parameter	**Value**	**Unit**
Fit coefficient A	0.8617	$\mathrm{N\,s^2/m^2}$
Fit coefficient B	137.7	N
$c_x S$	1.386	$\mathrm{m^2}$
f_{roll}	0.01017	–

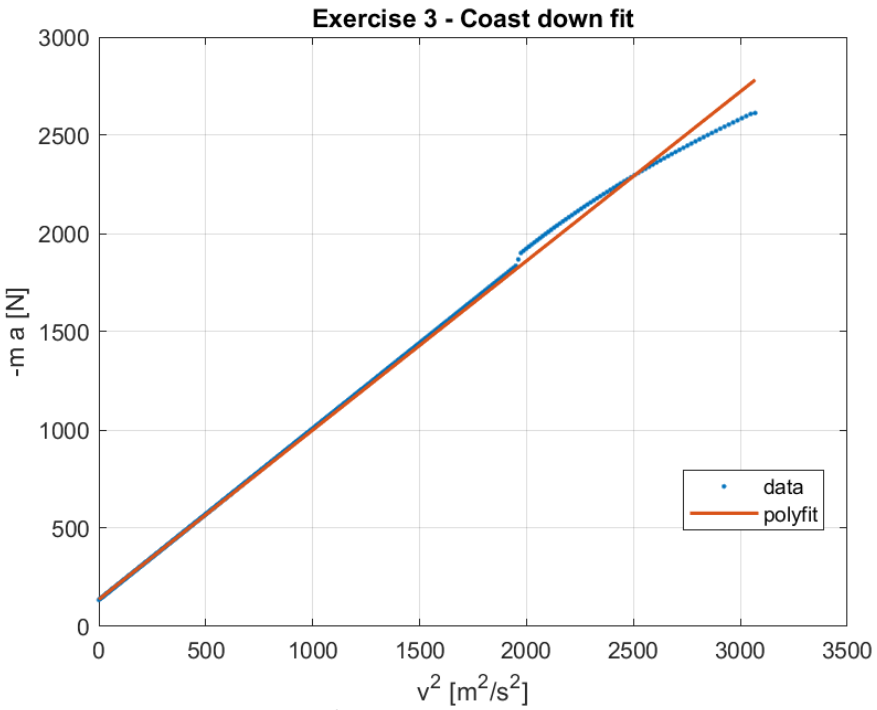


Figure 1.4: Linear fit of $-m\,a_x$ vs. v^2 via `polyfit`. The slope gives A and the intercept B.

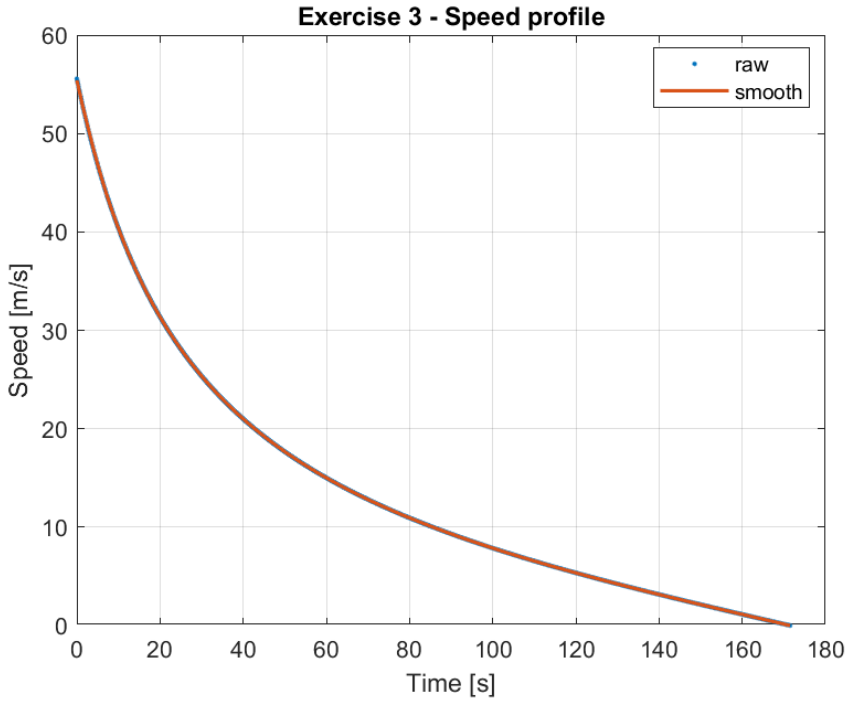


Figure 1.5: Measured coast-down speed profile, raw and after Savitzky–Golay smoothing.

1.5 Exercise 4 – Vertical Loads Under Braking

1.5.1 Problem Statement

A race car brakes from 200 km/h with a longitudinal deceleration of $-17\,\mathrm{m/s^2}$. The vertical force on each individual wheel is to be found, accounting for the static weight distribution, aerodynamic downforce, and the longitudinal load transfer induced by the deceleration.

Table 1.5: Vehicle parameters – Exercise 4

Parameter	**Value**	**Unit**
Mass	1250	kg
Weight distribution (front)	46	%
CoG height	0.35	m
Track width	1.6	m
$c_z S$	3.2	m^2
Aerodynamic balance (front)	51	%
Wheelbase	2.65	m

1.5.2 Methodology

Three effects superpose to determine the axle loads. The static weight sets the baseline; aerodynamic downforce at 200 km/h adds to it according to the aero balance; and the braking deceleration transfers load forward via the longitudinal load transfer $\Delta F_{z,\mathrm{long}}$:

$$F_{z,\mathrm{front}} = WD \cdot mg + Bal \cdot F_{\mathrm{Downforce}} + \Delta F_{z,\mathrm{long}} \tag{1.9}$$

$$F_{z,\mathrm{rear}} = (1 - WD) \cdot mg + (1 - Bal) \cdot F_{\mathrm{Downforce}} - \Delta F_{z,\mathrm{long}} \tag{1.10}$$

with:

$$\Delta F_{z,\mathrm{long}} = \frac{m\, a_x\, h_{\mathrm{COG}}}{\mathrm{Wheelbase}} \tag{1.11}$$

Left/right symmetry is assumed throughout, so each wheel on a given axle carries half the axle load.

1.5.3 Results

Table 1.6: Vertical loads per wheel under braking at 200 km/h, $a_x = -17\,\mathrm{m/s^2}$ – Exercise 4

Quantity	Value	Unit
Static front axle load	5640.8	N
Static rear axle load	6621.8	N
Total downforce	6049.4	N
Front downforce	3085.2	N
Rear downforce	2964.2	N
Longitudinal load transfer (to front)	2806.6	N
Front Left / Front Right wheel load	5766.3	N
Rear Left / Rear Right wheel load	3389.7	N

The deceleration transfers nearly 2807 N from the rear axle to the front, which is a large fraction of the rear static load. The result is that the rear wheels carry only 3390 N each – significantly less than the front wheels at 5766 N. This asymmetry is precisely why brake bias matters in race engineering: if the braking force were split equally front-to-rear, the rear tyres would lock long before the front ones.

Figure 1.6 shows the final wheel loads and Figure 1.7 illustrates how each of the three contributions adds up on each axle.

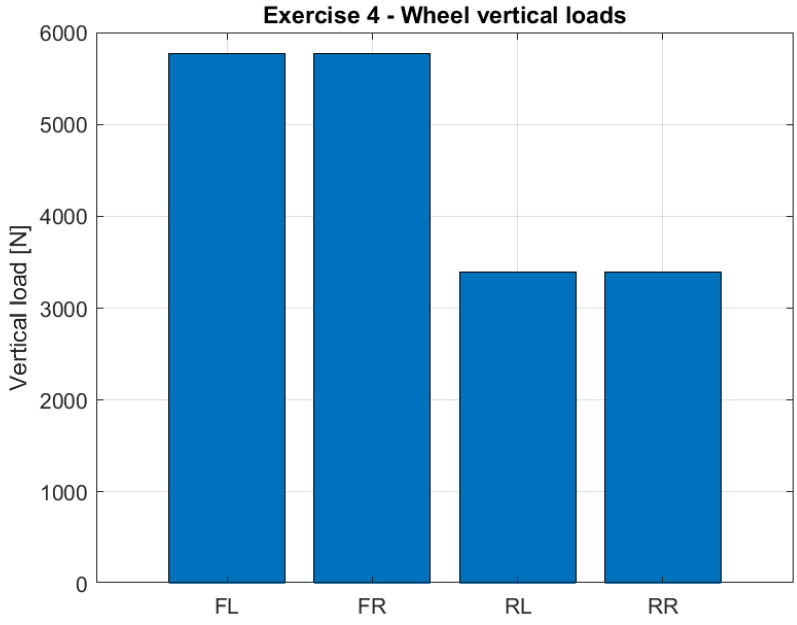


Figure 1.6: Vertical load on each wheel under braking.

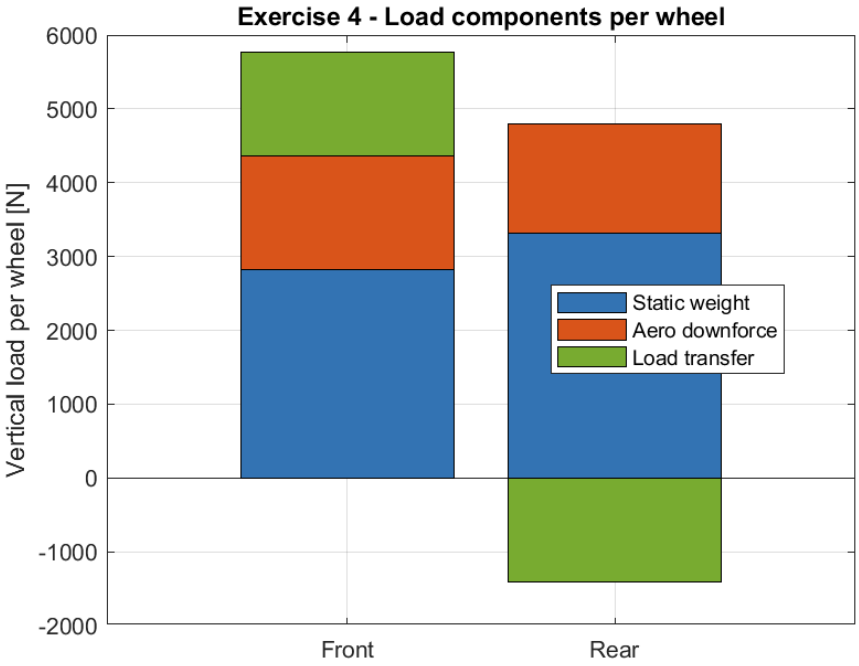


Figure 1.7: Decomposition of vertical load per wheel into static weight, aerodynamic downforce, and longitudinal load transfer. The load transfer increases the front load and reduces the rear.

1.6 Exercise 5 – Constant Speed Aerodynamic Estimation

1.6.1 Problem Statement

A car runs at a constant 250 km/h while suspension travel sensors record the deflection of the front and rear springs relative to the static ride height. From these deflections, together with the known engine output at that speed, the aerodynamic balance, $c_z S$, $c_x S$, and efficiency are estimated.

Table 1.7: Vehicle parameters – Exercise 5

Parameter	Value	Unit
Front spring rate	350	N/mm
Rear spring rate	200	N/mm
Front motion ratio	1.82	–
Rear motion ratio	1.30	–
Front suspension travel (from static)	6.4	mm
Rear suspension travel (from static)	11.5	mm
Tire rolling radius	345	mm
Total gear ratio	4.45	–
Engine torque	360	N m
Vehicle mass	1380	kg
Air density	1.225	kg/m^3
Rolling resistance coefficient	0.011	–
Vehicle speed	250	km/h

1.6.2 Methodology

The estimation proceeds in two independent steps: one reading the downforce from the suspension sensors, the other reading the drag from the engine power balance.

Step 1 – Downforce from suspension travel

The sensors are zeroed at low speed, so any compression observed at 250 km/h is purely aerodynamic. The motion ratio R_m relates wheel displacement to spring displacement through the suspension geometry: a wheel load F_w produces a spring force $R_m \cdot F_w$, so the wheel load is recovered as:

$$F_w = \frac{K_s \, \Delta x}{R_m} \tag{1.12}$$

Summing over both wheels per axle:

$$F_{\mathrm{DF,front}} = 2\,\frac{K_{s,f}\,\Delta x_f}{R_{m,f}} = 2\,\frac{350\,\mathrm{N/mm} \times 6.4\,\mathrm{mm}}{1.82} = 2461.5\,\mathrm{N} \tag{1.13}$$

$$F_{\mathrm{DF,rear}} = 2\,\frac{K_{s,r}\,\Delta x_r}{R_{m,r}} = 2\,\frac{200\,\mathrm{N/mm} \times 11.5\,\mathrm{mm}}{1.30} = 3538.5\,\mathrm{N} \tag{1.14}$$

The total downforce, aerodynamic balance, and $c_z S$ follow directly:

$$F_{\mathrm{DF,tot}} = F_{\mathrm{DF,front}} + F_{\mathrm{DF,rear}}, \qquad \mathrm{Bal} = \frac{F_{\mathrm{DF,front}}}{F_{\mathrm{DF,tot}}}, \qquad c_z S = \frac{2\,F_{\mathrm{DF,tot}}}{\rho\, v^2} \tag{1.15}$$

Step 2 – Drag from the longitudinal force balance

At constant speed the net longitudinal force is zero, so the engine traction force exactly balances drag and rolling resistance:

$$F_{\mathrm{Drag}} = F_{\mathrm{engine}} - F_{\mathrm{roll}} \tag{1.16}$$

The engine traction force is:

$$F_{\mathrm{engine}} = \frac{\tau_{\mathrm{tot}}\, T_{\mathrm{engine}}}{R_{\mathrm{rol}}} = \frac{4.45 \times 360}{0.345} = 4643.5\,\mathrm{N} \tag{1.17}$$

Rolling resistance acts on the full vertical load, including the aerodynamic downforce already found in Step 1 (exactly the same coupling seen in Exercise 3):

$$F_{\mathrm{roll}} = f_{\mathrm{roll}}(mg + F_{\mathrm{DF,tot}}) = 0.011 \times (1380 \times 9.81 + 6000) = 214.9\,\mathrm{N} \tag{1.18}$$

Finally, $c_x S$ and the efficiency:

$$c_x S = \frac{2\,F_{\mathrm{Drag}}}{\rho\, v^2}, \qquad \mathrm{Eff} = \frac{F_{\mathrm{DF,tot}}}{F_{\mathrm{Drag}}} = \frac{c_z}{c_x} \tag{1.19}$$

1.6.3 Results

Table 1.8: Estimated aerodynamic parameters – Exercise 5

Quantity	**Value**	**Unit**
Front downforce	2461.5	N
Rear downforce	3538.5	N
Total downforce	6000.0	N
Aerodynamic balance (front)	41.0	%
Engine traction force	4643.5	N
Rolling resistance force	214.9	N
Drag force	4428.6	N
$c_z S$	2.031	m^2
$c_x S$	1.499	m^2
Aerodynamic efficiency c_z/c_x	1.355	–

At 250 km/h the car is generating 6000 N of downforce – roughly half its own weight – with slightly more bias towards the rear (59% rear, 41% front). The aerodynamic efficiency of 1.36 means the car gains 1.36 N of downforce for every 1 N of drag penalty, which is the fundamental trade-off any aerodynamicist is trying to push.

What makes this exercise instructive beyond the numbers is the method itself: by combining two simple on-car measurements – suspension travel and engine speed – it is possible to characterise the aero package without access to a wind tunnel. Figure 1.8 summarises the force breakdown at the test condition.

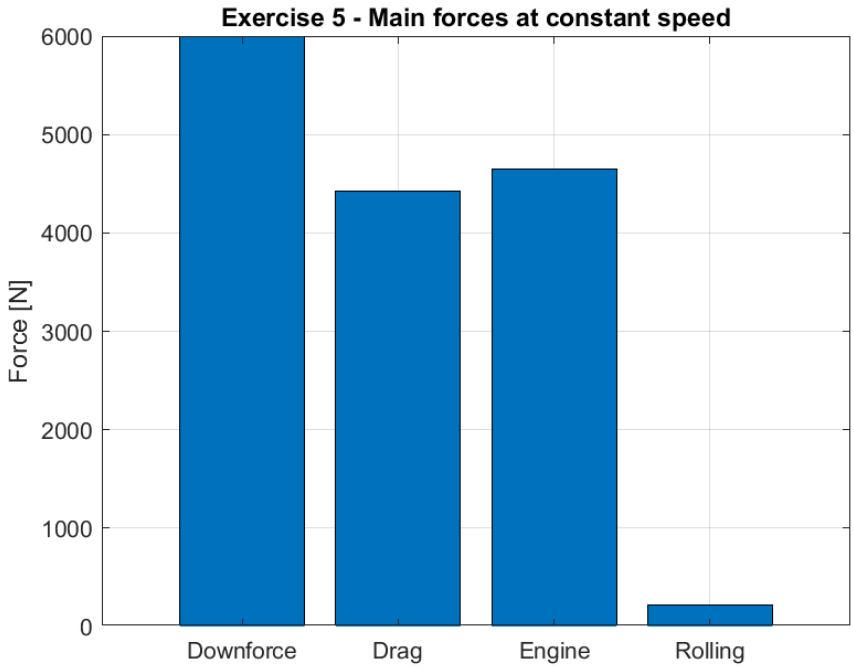


Figure 1.8: Force breakdown at 250 km/h: engine traction, drag, rolling resistance, and total downforce.

2 LAB 1 – Tyres

Introduction

Tyres are the only contact point between a racing car and the road, which makes understanding their behaviour essential before any lap-time simulation can be trusted. What distinguishes a tyre from a simple friction interface is that its grip is not fixed: it depends on temperature, sliding speed, and normal load, all of which vary continuously around a lap. A tyre that is too cold or one that has overheated both deliver less grip than a tyre running in its optimal temperature window – and the lap-time cost can be substantial.

This laboratory approaches tyre physics from two complementary angles. Exercise 1 couples a thermal model of the tyre to a vehicle dynamics simulator and asks how set-up choices such as toe angle, weight distribution, and aerodynamic balance affect the thermal equilibrium and, ultimately, the lap time. Exercise 2 takes measured contact-patch data from an instrumented car and works backwards to identify the friction law $\mu(v_{\text{slip}}, T_{\text{tread}})$ directly from the evidence.

The central quantity throughout is the effective friction coefficient:

$$\mu \;=\; \frac{\sqrt{F_x^2 + F_y^2}}{|F_z|} \;=\; f(v_{\text{slip}},\, T_{\text{tread}},\, F_z), \tag{2.1}$$

which compresses the combined in-plane tyre force into a single dimensionless number. The fact that μ depends on three separate variables – sliding speed, temperature, and load – is precisely what makes tyre optimisation non-trivial, and both exercises circle around different facets of that dependence.

2.1 Exercise 1 – Tyre Thermal Model and Set-up Sensitivity

2.1.1 Theoretical Background

Two-node tyre thermal model

The thermal model represents each tyre as two lumped masses: the *tread* (the thin outer rubber layer that contacts the road and generates frictional heat) and the *carcass* (the structural body that exchanges heat with the air inside). This two-node structure captures the essential physics – the tread heats up fast because it is thin and in direct contact with the heat source, while the carcass acts as a thermal reservoir that warms up slowly – without the complexity of a full finite-element thermal analysis.

The governing equations are:

$$\dot{T}_{\text{tread}} = \frac{Q_{\text{fric}} + Q_{\text{cond,TC}} - Q_{\text{conv,TA}} - Q_{\text{cond,TT}}}{m_{\text{tread}}\, c_{\text{tread}}}, \tag{2.2}$$

$$\dot{T}_{\text{carcass}} = \frac{Q_{\text{def}} - Q_{\text{cond,TC}}}{m_{\text{carcass}}\, c_{\text{carcass}}}, \tag{2.3}$$

where each thermal power term has a clear physical origin, summarised in Table 2.1.

Table 2.1: Thermal power contributions in the two-node tyre model.

Term	Expression	Physical mechanism
Q_{fric}	$p_1\, V(\lvert F_x\, \xi\rvert + \lvert F_y \tan\alpha\rvert)$	Sliding friction at contact patch
$Q_{\text{cond,TT}}$	$h_{tt}\, A_{cp}\, (T_{\text{tread}} - T_{\text{track}})$	Conduction tyre $\rightarrow$ track
$Q_{\text{conv,TA}}$	$h_{\text{forc}}\, A_{\text{conv}}\, (T_{\text{tread}} - T_{\text{air}})$	Forced convection tyre $\rightarrow$ air
$Q_{\text{cond,TC}}$	$p_3\, A_{\text{tot}}\, (T_{\text{carcass}} - T_{\text{tread}})$	Conduction carcass $\leftrightarrow$ tread
Q_{def}	$RRC \cdot F_z \cdot V$	Structural hysteresis (rolling resistance)

In the friction term, $\xi = F_x/(C_x\, C_{xFz}\, F_z)$ is the longitudinal slip estimate, α the effective slip angle, and $p_1 = 0.64$ a calibrated efficiency factor. Forced convection uses a turbulent flat-plate correlation:

$$h_{\text{forc}} = \frac{K_{\text{air}}}{L}\, 0.0239 \left(\frac{V\, L}{\nu_{\text{air}}}\right)^{0.805}, \tag{2.4}$$

where $L = (D^{-1}{+}W^{-1})^{-1}$ is the harmonic-mean tyre characteristic length and ν_{air} the kinematic viscosity of air.

Vehicle dynamics and speed model

Vehicle speed at each track position is limited by whichever is reached first: the traction limit (friction circle) or the engine power limit. The lateral-acceleration-limited speed is:

$$V_c = \sqrt{\frac{Mg}{M\gamma/\mu_m - C_{lA}}}, \tag{2.5}$$

where γ is the local track curvature, μ_m the mean friction coefficient of the two axles, and $C_{lA} = \frac{1}{2}\rho C_l A$ the aerodynamic lift coefficient (negative for downforce). The normal loads on each axle account for weight distribution and the downforce split:

$$F_z^F = \frac{W_F - C_{lA}V^2 \cdot AB}{2}, \qquad F_z^R = \frac{W_R - C_{lA}V^2 \cdot (1 - AB)}{2}. \tag{2.6}$$

Static toe angle

Static toe is a pre-alignment of the wheel relative to the car's centreline. A toe angle δ_{toe} generates a lateral force $F_y = C_\alpha\, \lvert\delta_{\text{toe}}\rvert$ per tyre, which the tyre then partially cancels with a compensating slip angle. The net result is a parasitic drag force per axle:

$$F_{\text{drag,toe}} = 2\, F_y\, \sin(\alpha + \lvert\delta_{\text{toe}}\rvert), \tag{2.7}$$

which costs top speed but continuously feeds additional heat into the tread through the Q_{fric} term. This double effect – drag penalty paired with enhanced warm-up – is what makes toe a thermally useful but aerodynamically expensive parameter.

2.1.2 Simulation Set-up

The key vehicle and tyre parameters are listed in Tables 2.2 and 2.3. The full system state $\mathbf{x} = [s,\ T_{\text{tr}}^F,\ T_{\text{ca}}^F,\ T_{\text{tr}}^R,\ T_{\text{ca}}^R,\ V]^\top$ is integrated with `ode15s` over 250 s, covering several laps on each track.

Table 2.2: Vehicle parameters used in the simulations.

Parameter	Symbol	Value	Unit
Total mass	M	600	kg
Drag coefficient	C_d	0.50	–
Lift coefficient	C_l	-1.20	–
Reference area	A	1.20	m^2
Engine power	P	200	kW
Rolling resistance	RRC	0.01	–
Air density	ρ	1.20	$\mathrm{kg/m}^3$

Table 2.3: Tyre parameters (front / rear where different).

Parameter	Symbol	Front	Rear	Unit
Diameter	D	0.66	0.66	m
Width	W	0.355	0.380	m
Tread mass	m_{tr}	2.9	2.9	kg
Carcass mass	m_{ca}	15.5	17.5	kg
Tread heat capacity	c_{tr}	3600	3600	J/kg/K
Carcass heat cap.	c_{ca}	1600	1600	J/kg/K
Contact patch width	w_{cp}	0.20	0.22	m
Track-tyre HTC	h_{tt}	10 000		$\mathrm{W/m}^2\mathrm{/K}$
Reference slip angle	α_{ref}	7		deg

2.1.3 Effect of Static Toe Angle on Lap Time

Two meteorological scenarios are studied on both tracks:

- **Cold Night:** $T_{\mathrm{air}} = 15\,^{\circ}\mathrm{C}$, $T_{\mathrm{track}} = 20\,^{\circ}\mathrm{C}$, $T_{\mathrm{tyre},0} = 25\,^{\circ}\mathrm{C}$.
- **Hot Day:** $T_{\mathrm{air}} = 30\,^{\circ}\mathrm{C}$, $T_{\mathrm{track}} = 50\,^{\circ}\mathrm{C}$, $T_{\mathrm{tyre},0} = 70\,^{\circ}\mathrm{C}$.

The static toe angle is swept symmetrically over $\delta_{\mathrm{toe}} \in [-2, +2]$ in 0.5° steps. By symmetry of the heating model (which depends only on $|\delta_{\mathrm{toe}}|$), only the non-negative half is simulated and the results are mirrored.

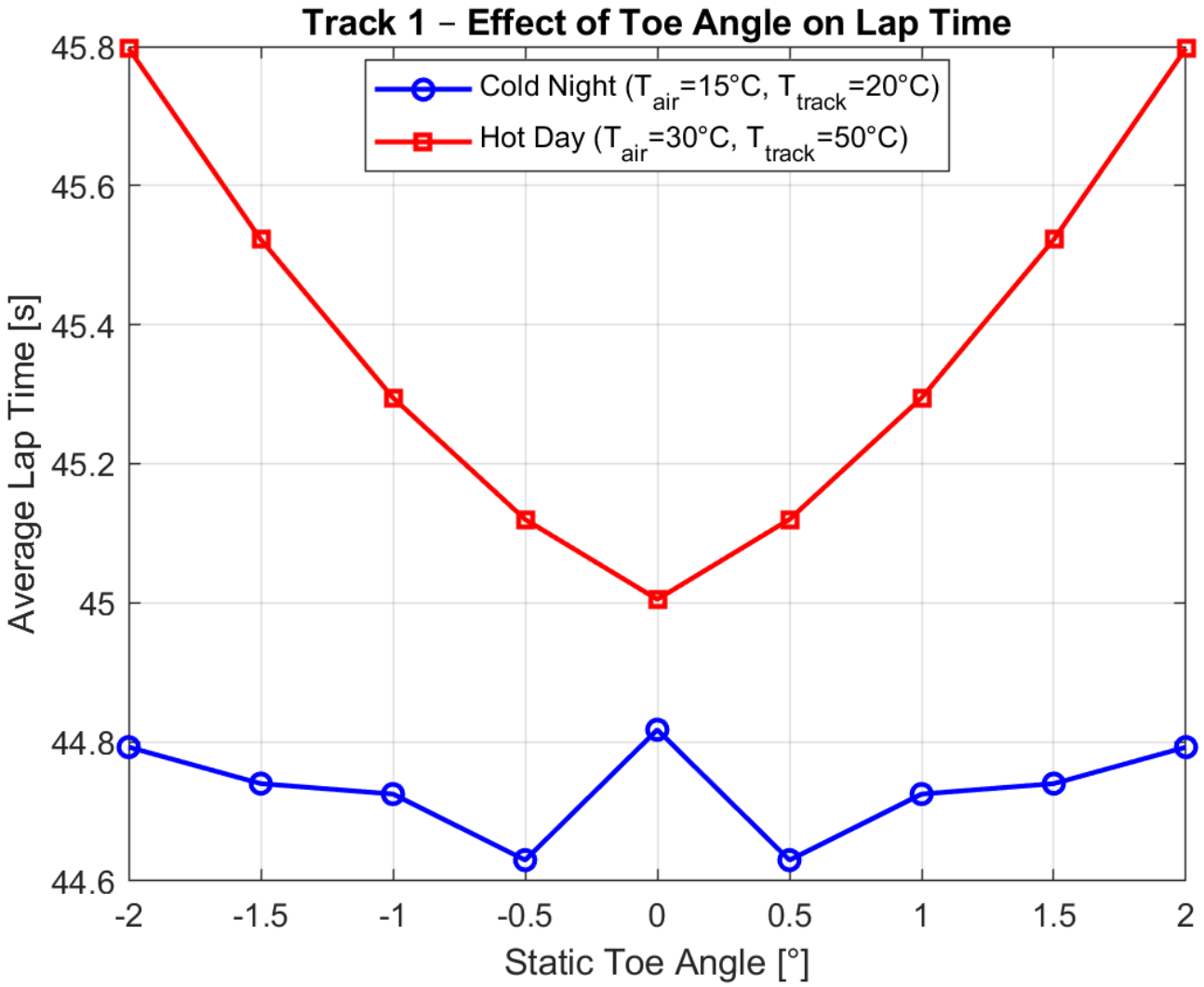


Figure 2.1: Average lap time as a function of static toe angle on Track 1 for the Cold Night (blue) and Hot Day (red) conditions.

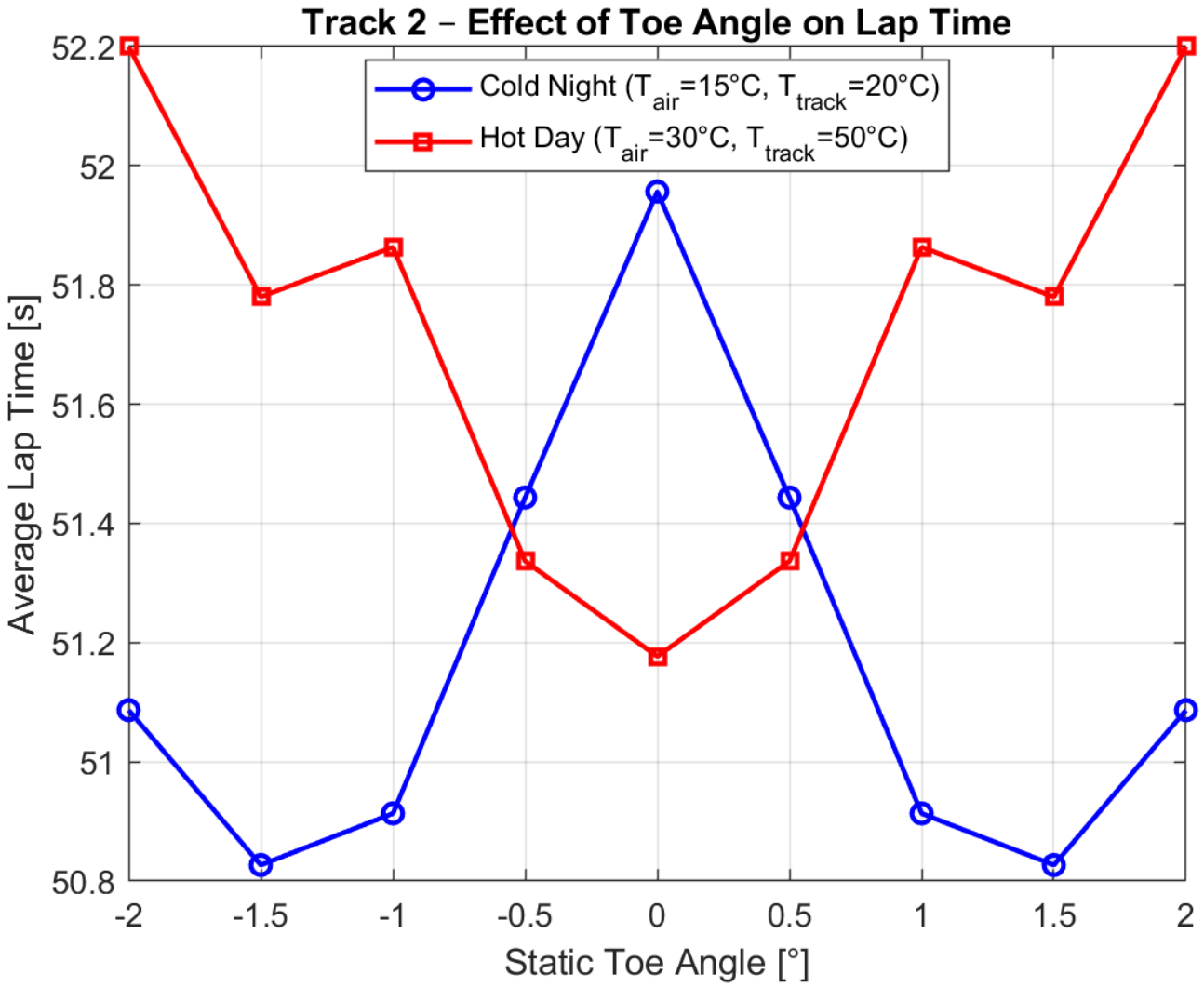


Figure 2.2: Average lap time vs. static toe angle on Track 2 (same format as Figure 2.1).

The two curves in Figure 2.1 are a good illustration of how the same set-up change can be beneficial or harmful depending on context. Under **Cold Night** conditions the tyres start well below their optimal temperature. A small toe-in of around 0.5° generates extra sliding at the contact patch, which feeds into Q_{fric} and accelerates warm-up. The tread reaches its grip peak a few laps earlier, and the lap time actually improves slightly relative to zero toe. Beyond about 1°, however, the tyre is already warm enough and the drag penalty (Eq. 2.7) dominates, so lap time rises monotonically.

Under **Hot Day** conditions the tyres are already at or above their optimal temperature from the first lap. Here any toe at all pushes the tread further into overheating while adding drag simultaneously, so the curve rises from zero with no initial benefit. The same pattern appears on Track 2 (Figure 2.2), although the exact optimal toe shifts because the different layout changes how much lateral slip and friction heat is generated on each lap.

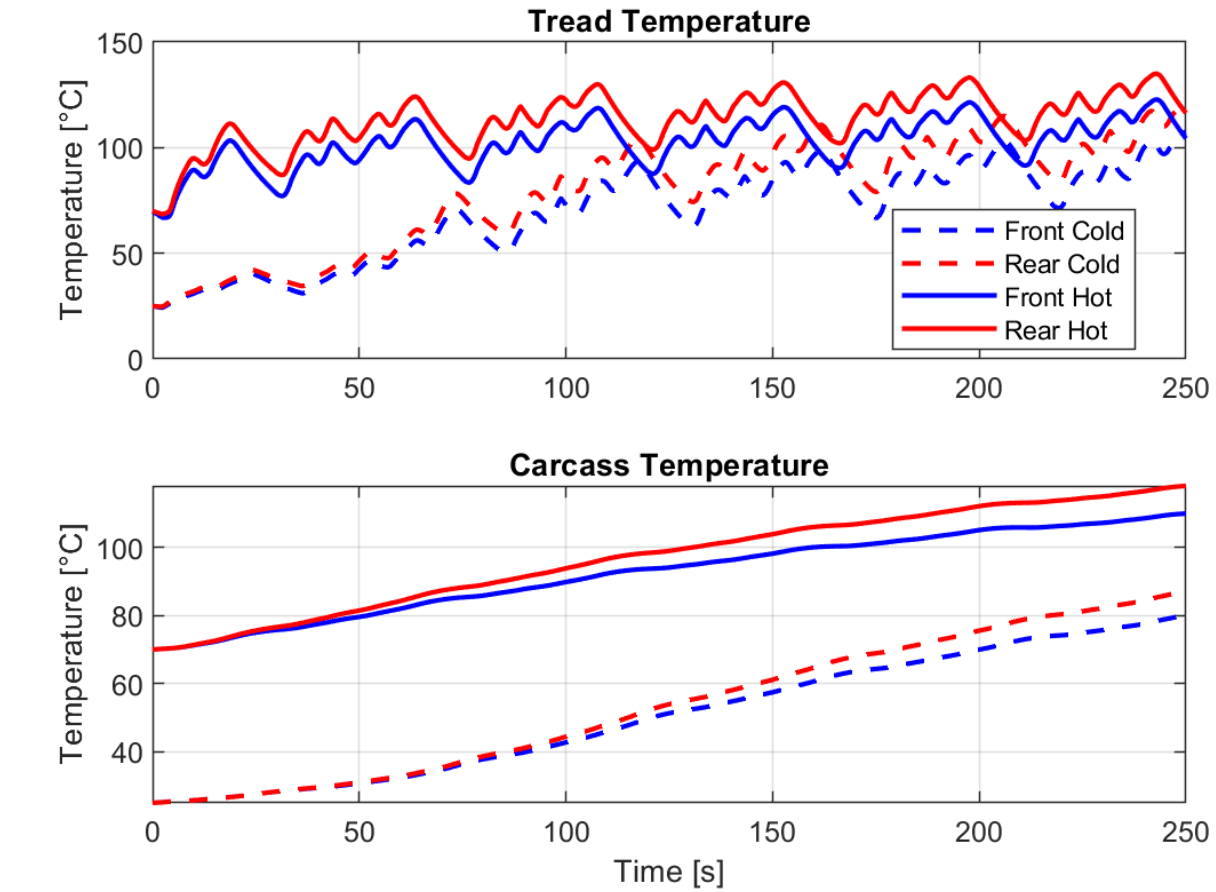


Figure 2.3: Tread and carcass temperature evolution for the reference case ($\delta_{\text{toe}} = 0$) under Cold Night (dashed) and Hot Day (solid) conditions. Front tyres in blue, rear in red.

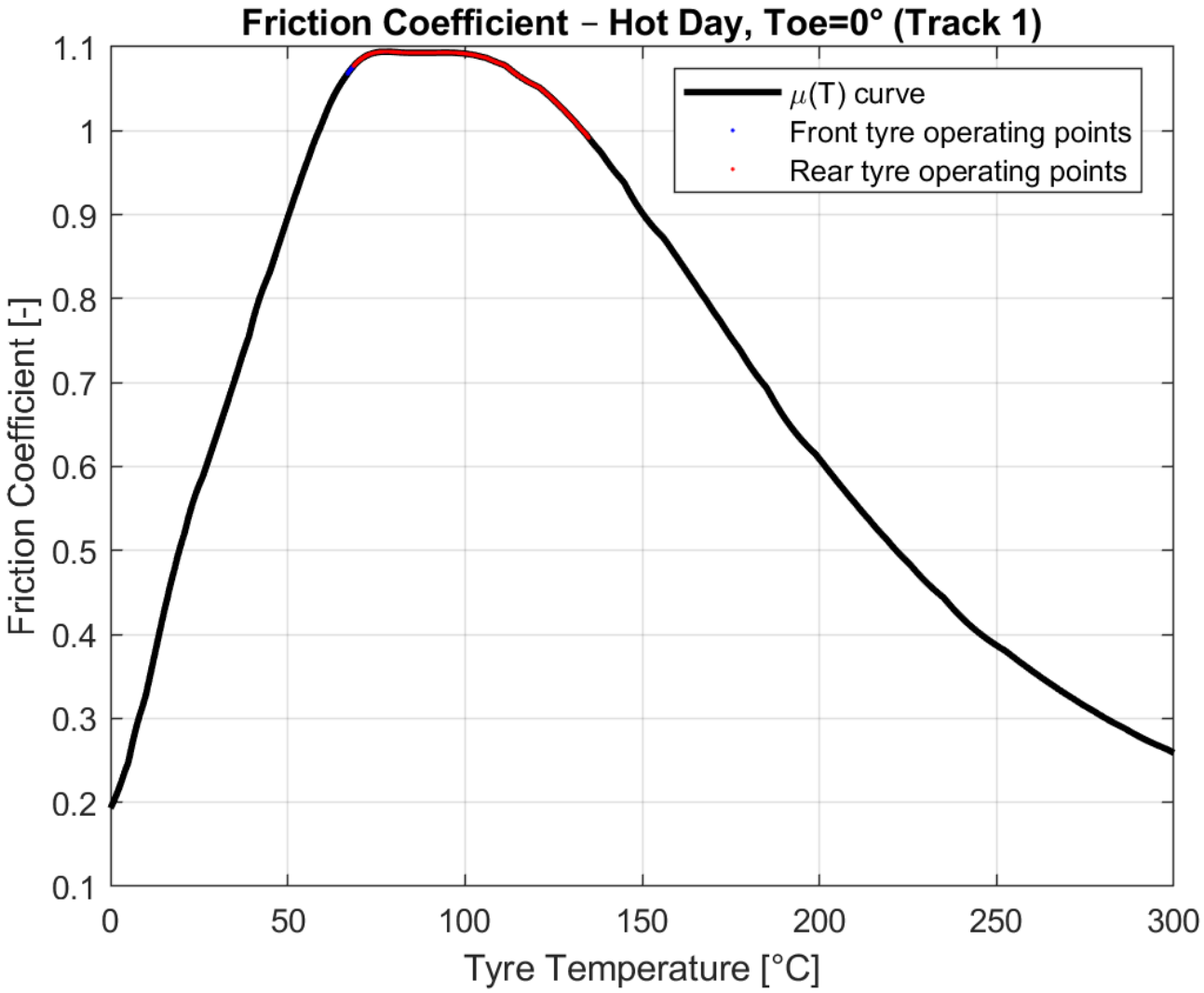


Figure 2.4: Friction coefficient $\mu(T)$ curve (black) with operating points of front (blue) and rear (red) tyres for the Hot Day reference case.

Figure 2.3 makes the warm-up difference concrete: the Cold Night tread starts at 25 °C and climbs gradually over two to three laps before settling at a steady-state, while the Hot Day tyre is already at operating temperature on lap one. Notice also that the rear tyre consistently runs hotter than the front – it carries more load, generates more traction and braking force, and therefore produces more frictional heat. Figure 2.4 shows where on the $\mu(T)$ curve each tyre is operating in the Hot Day reference case; the rear is noticeably further along the high-temperature side of the peak.

2.1.4 Effect of Weight Distribution and Aerodynamic Balance

With toe at zero and Hot Day conditions, a full factorial sweep is run:

$$WD \in \{30,\ 40,\ 50,\ 60,\ 70\}\%, \qquad AB \in \{20,\ 30,\ 40,\ 50,\ 60,\ 70,\ 80\}\%,$$

giving 35 simulation cases per track.

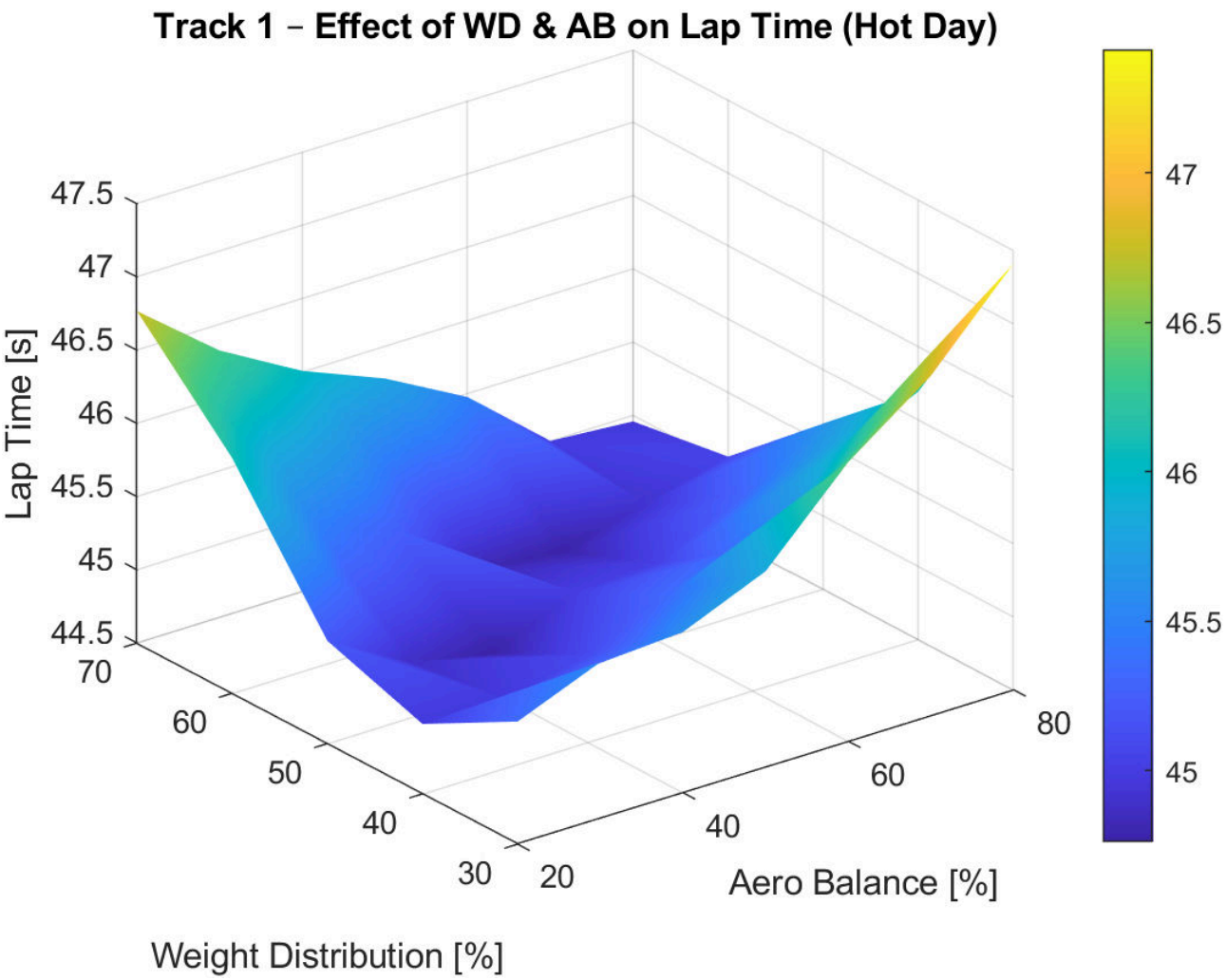


Figure 2.5: Average lap time surface as a function of weight distribution (WD) and aerodynamic balance (AB) on Track 1, Hot Day conditions.

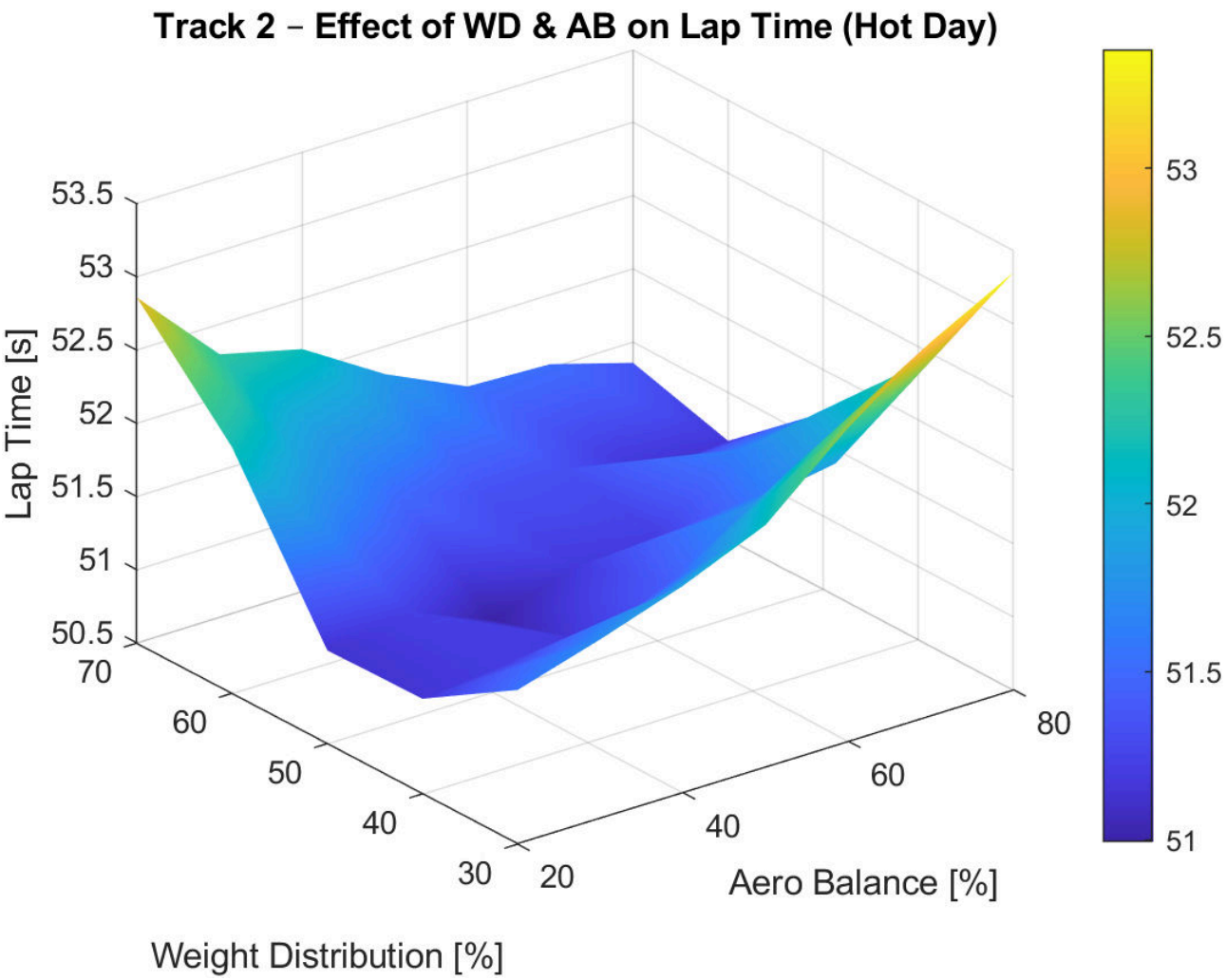


Figure 2.6: Average lap time surface as a function of WD and AB on Track 2, Hot Day conditions.

Both surfaces (Figures 2.5 and 2.6) show a clear minimum in the interior of the swept domain, which means the optimum is not at an extreme – it is genuinely a compromise. The coupling between the two parameters can be understood through Eq. (2.6): shifting weight forward loads the front axle and starves the rear of normal force, reducing rear grip and driving up rear tyre temperature. Pushing the aerodynamic balance forward compensates by adding downforce to the front axle at speed, restoring some of the thermal balance. The optimal WD/AB combination is therefore the one where both axles run close to their friction peak simultaneously.

The surface on Track 2 (Figure 2.6) shows the optimum at a different WD/AB pair, a consequence of the different speed profile and lateral load distribution that the track geometry imposes on the front and rear axles.

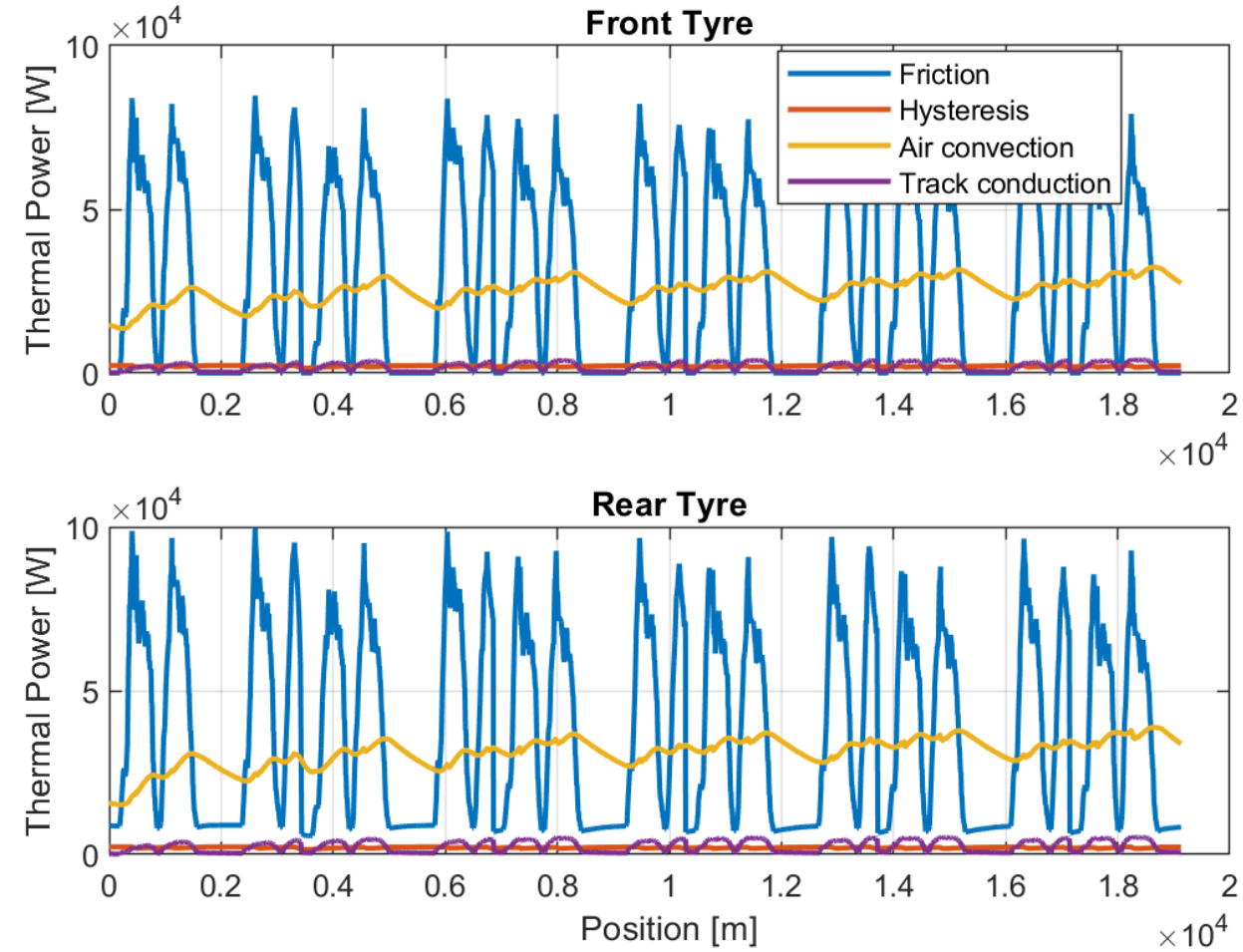


Figure 2.7: Thermal power breakdown for front (top) and rear (bottom) tyres at the reference set-up ($WD = 50\%$, $AB = 50\%$, Hot Day).

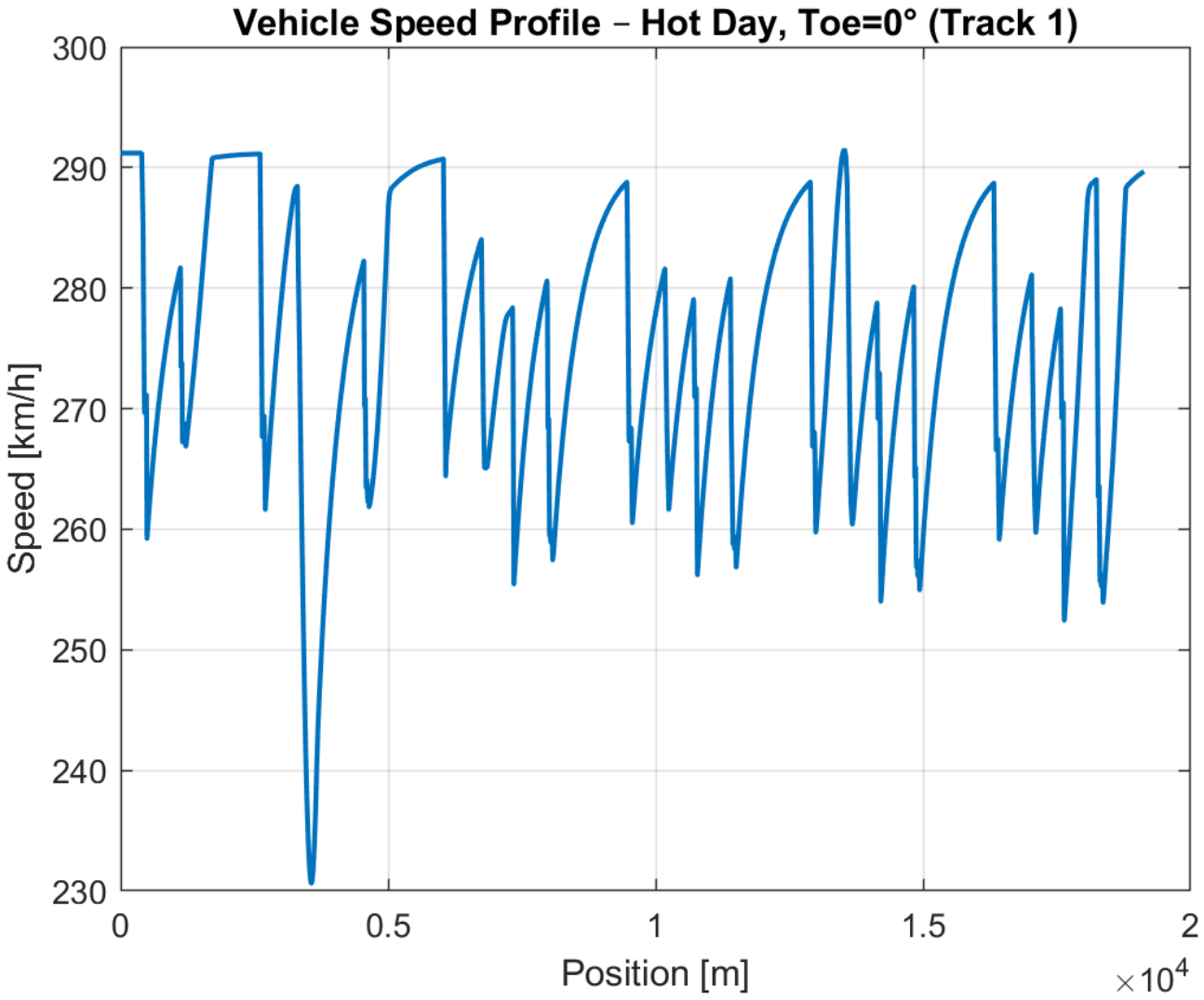


Figure 2.8: Vehicle speed profile as a function of track position for the Hot Day reference case ($\delta_{\text{toe}} = 0$, $WD = 50\%$, $AB = 50\%$).

Figure 2.7 shows the thermal power balance in detail. Frictional heating dominates at the rear, where longitudinal forces are highest, while tread-to-track conduction ($Q_{\text{cond,TT}}$) is the main cooling path on a hot surface. Air convection ($Q_{\text{conv,TA}}$) is only significant at high speed – at low speed the boundary layer thickens and convective heat transfer drops sharply.

The speed profile in Figure 2.8 reflects the interaction between the cornering limit and the power limit: peak speed on the straights rises slightly from lap to lap as the tyres warm up and μ increases, allowing the car to carry slightly more speed into each corner.

2.1.5 Tyre Warm-up Without Toe Angle

For cold starting conditions there are several ways to reach operating temperature without resorting to static toe. The most direct is aggressive driving on the out-lap – using large slip angles deliberately in corners to maximise Q_{fric} without any set-up penalty once the tyres are warm. Tyre blankets achieve a similar result before the car leaves the garage, essentially replicating the Hot Day initial condition. A less obvious but free source of heat is the track itself: when the track surface is warmer than the tyre, the sign of $Q_{\text{cond,TT}}$ reverses and the road heats the rubber rather than cooling it. The simulation shows that regardless of strategy, tread temperature stabilises within two to three laps; any warm-up intervention beyond that point is wasted effort.

2.1.6 Conclusions – Exercise 1

- **Static toe angle** plays opposite roles depending on conditions: a small toe-in can marginally improve cold-tyre lap times by accelerating warm-up, but always increases drag and tends to overheat an already-warm tyre.
- **Weight distribution and aerodynamic balance** interact to set the thermal balance between axles. The optimum combination keeps both axles near their friction peak simultaneously and must be found for each specific track.
- **Track dependency:** because the optimal set-up is driven by the speed and lateral load profile, the quantitative optimum changes with every circuit layout.
- **Tyre warm-up** can be achieved through driving style or pre-heating instead of static toe, avoiding a permanent speed penalty for the rest of the stint.

2.2 Exercise 2 – Tyre Friction Law Identification

2.2.1 Theoretical Background

Rather than assuming a friction model, this exercise lets the data speak. The goal is to reconstruct the surface $\mu(v_{\text{slip}}, T_{\text{tread}})$ purely from contact-patch measurements, treating the tyre as a black box. The effective friction coefficient at each measurement instant is:

$$\mu = \frac{F_t}{|F_z|} = \frac{\sqrt{F_x^2 + F_y^2}}{|F_z|}, \tag{2.8}$$

which is the length of the combined force vector on the friction circle, normalised by the vertical load. The two independent variables are the contact-patch sliding speed v_{slip} – combining the contributions of longitudinal and lateral slip – and the tread surface temperature T_{tread}.

2.2.2 Dataset Description

The dataset (`ContactData.mat`) contains 501 time-synchronous samples recorded across all four tyres during a representative lap. Table 2.4 summarises the range of each variable.

Table 2.4: Dataset summary: variable ranges across all four tyres.

Variable	Symbol	Min	Max	Unit
Normal force	$\lvert F_z \rvert$	696	7 174	N
Longitudinal force	F_x	−1 380	783	N
Lateral force	F_y	−8 698	7 943	N
Slip speed	v_{slip}	3.40	8.75	m/s
Tread temperature	T_{tread}	55.2	140.3	°C

The wide F_y range – nearly ±9 kN – reflects the high-speed cornering demands of the circuit. The tenfold span in F_z (from around 700 N to over 7 kN) already hints at the load sensitivity that will become visible in the results.

2.2.3 Methodology

The $(v_{\text{slip}}, T_{\text{tread}})$ domain is divided into a 35×30 uniform grid. Within each cell all valid measurements are averaged to produce a representative μ value; cells with fewer than 6 samples are discarded. The resulting binned map has some empty cells at the boundaries, which are filled using a `scatteredInterpolant` with natural-neighbour interpolation. To assess load sensitivity, the dataset is split into three equal-population F_z bands using the 1/3 and 2/3 quantiles, and a separate map is fitted to each band.

Finally, a compact quadratic polynomial in two variables is fitted to the full dataset:

$$\hat{\mu}(v, T) = a_0 + a_1 v + a_2 T + a_3 v^2 + a_4 T^2 + a_5 v T, \tag{2.9}$$

with the six coefficients found by ordinary least squares: $\mathbf{a} = \mathbf{X}^{\dagger}\boldsymbol{\mu}$, with $\mathbf{X}_{i,:} = \left[1,\ v_i,\ T_i,\ v_i^2,\ T_i^2,\ v_i T_i\right]$.

2.2.4 Results

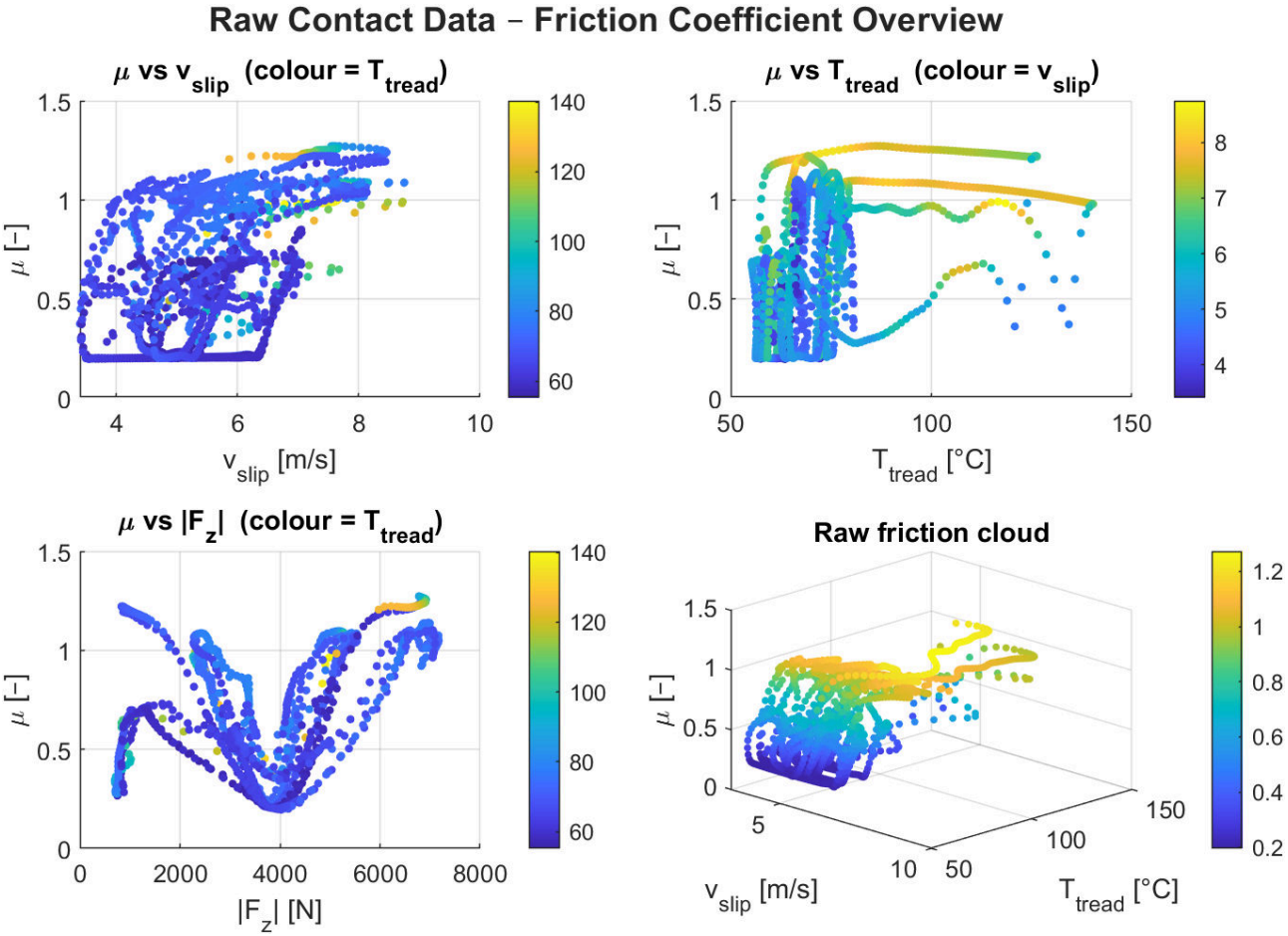


Figure 2.9: Raw friction data overview. Top-left: μ vs. slip speed (colour: T_{tread}). Top-right: μ vs. temperature (colour: v_{slip}). Bottom-left: μ vs. normal load. Bottom-right: 3-D scatter cloud.

The raw scatter in Figure 2.9 immediately shows that temperature is the dominant variable: there is a clear arch in μ vs. T, with a peak somewhere around 90 °C and a drop on both sides. The influence of slip speed is more subtle but real – at a fixed temperature, higher sliding speeds tend to produce slightly lower friction, which is consistent with the visco-elastic heating of rubber creating a thin molten layer that reduces adhesion. The μ vs. F_z scatter in

the bottom-left panel already hints at load sensitivity: the cloud slopes downward, meaning the tyre is less efficient at high loads.

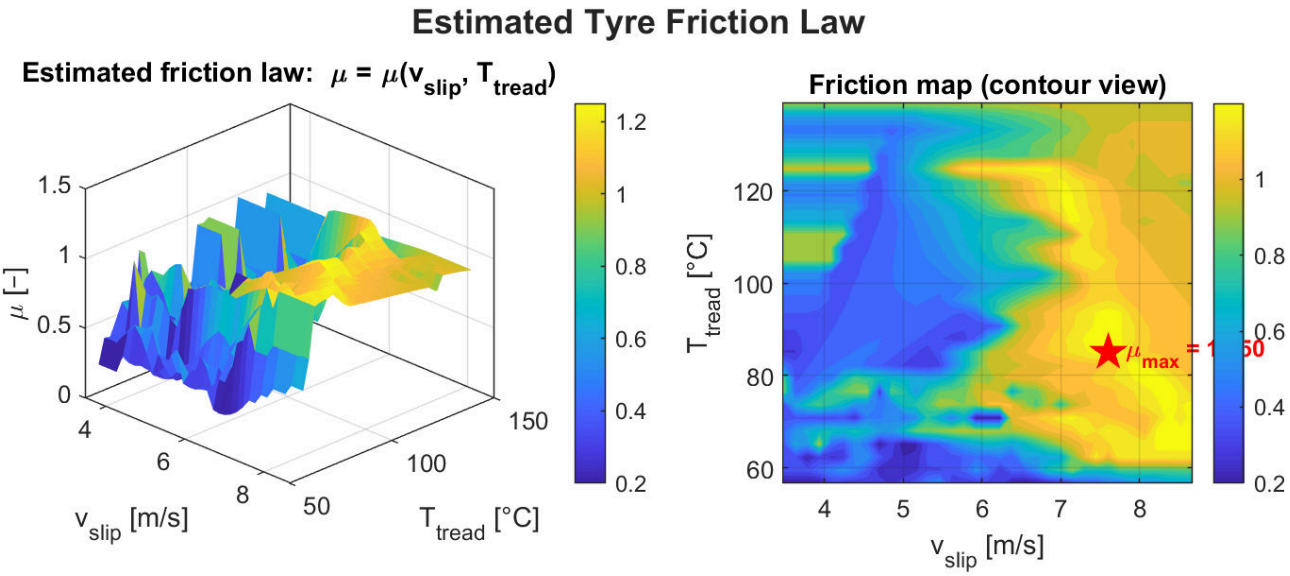


Figure 2.10: Estimated friction law surface (left) and contour map (right). The red star marks the maximum friction point.

The binned map in Figure 2.10 sharpens this picture into a smooth surface. The friction peak sits in the window 80 °C–100 °C, which is consistent with typical motorsport compound operating ranges. The contours at moderate slip speeds are nearly horizontal, confirming that temperature matters more than sliding speed in this regime. At higher temperatures the contours tilt, indicating that the two variables interact – rubber that has already softened thermally is more sensitive to the additional heat generated by sliding.

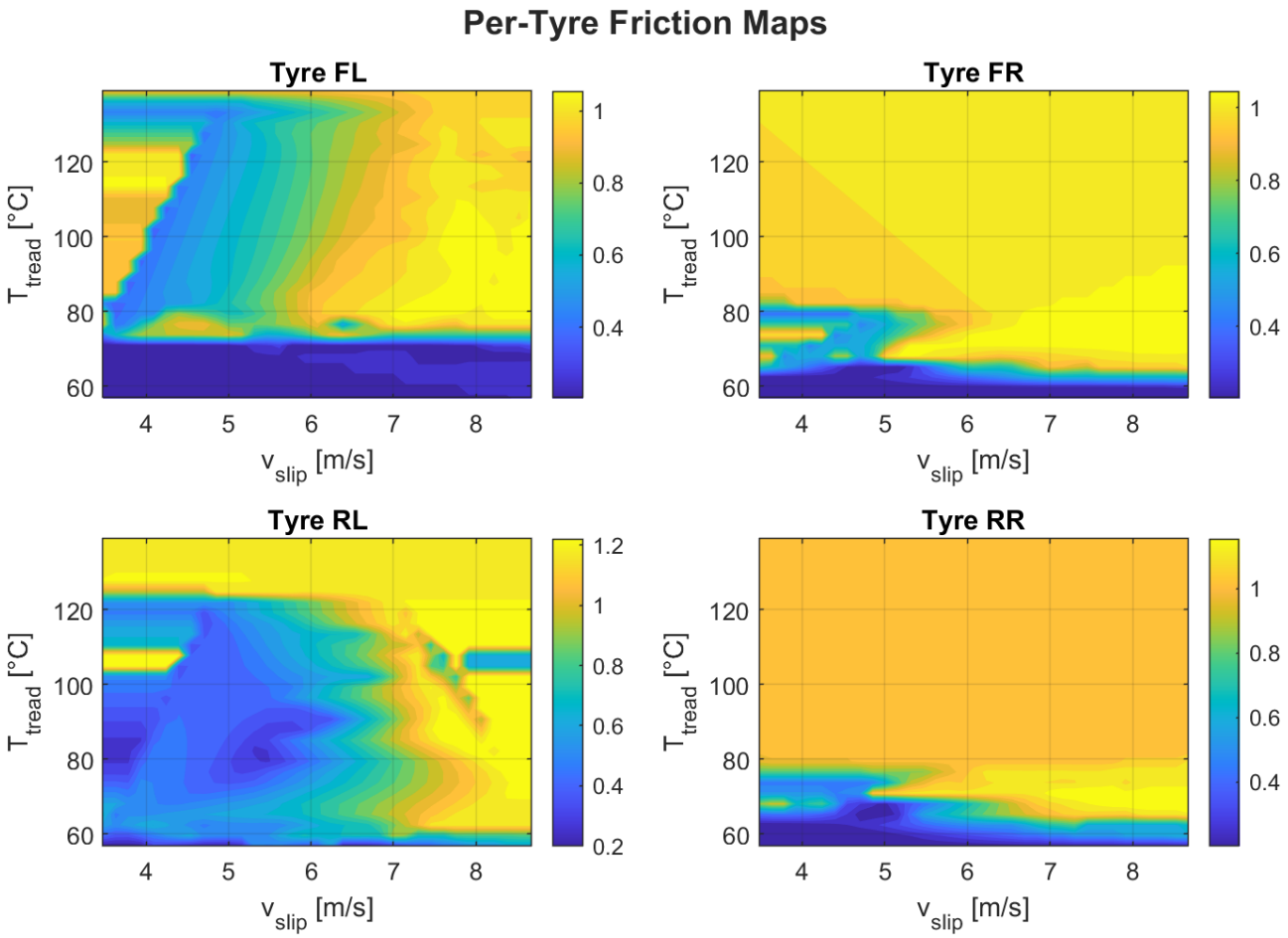


Figure 2.11: Individual friction maps for the four tyres (FL, FR, RL, RR). Differences reflect asymmetric loading due to the specific lap trajectory.

Looking at the four individual tyre maps in Figure 2.11, the underlying friction law is consistent across all four, but each tyre explores a different region of the (v, T) space. The rear tyres run hotter because they handle larger traction and braking forces. The left and right tyres sample different parts of the friction circle because the circuit has a directional bias – a predominantly clockwise or counter-clockwise layout means one side is loaded harder in fast corners.

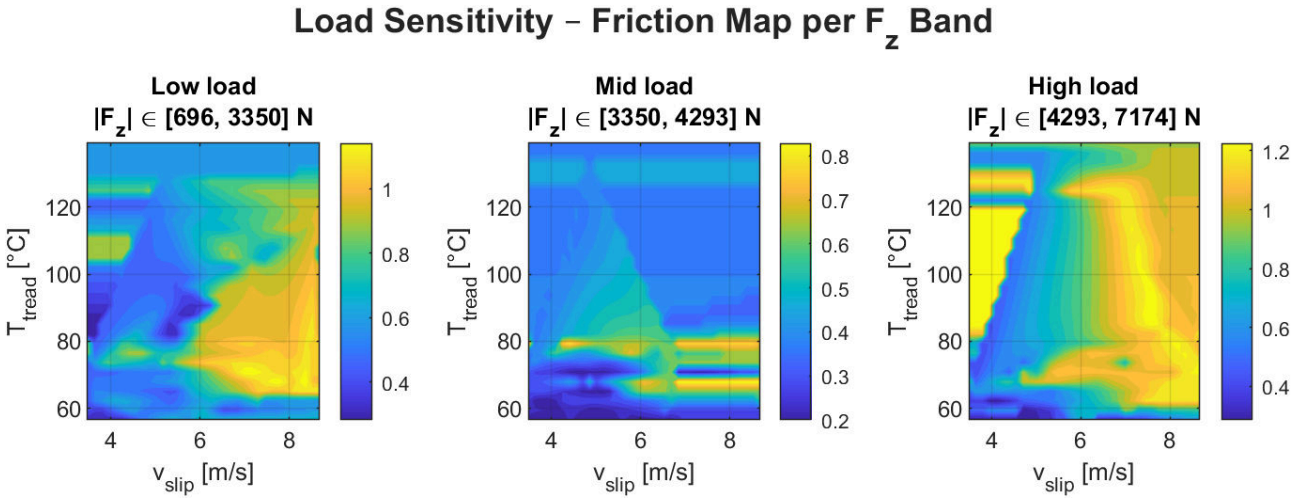


Figure 2.12: Friction maps for three normal-load bands: low (left), mid (centre) and high (right).

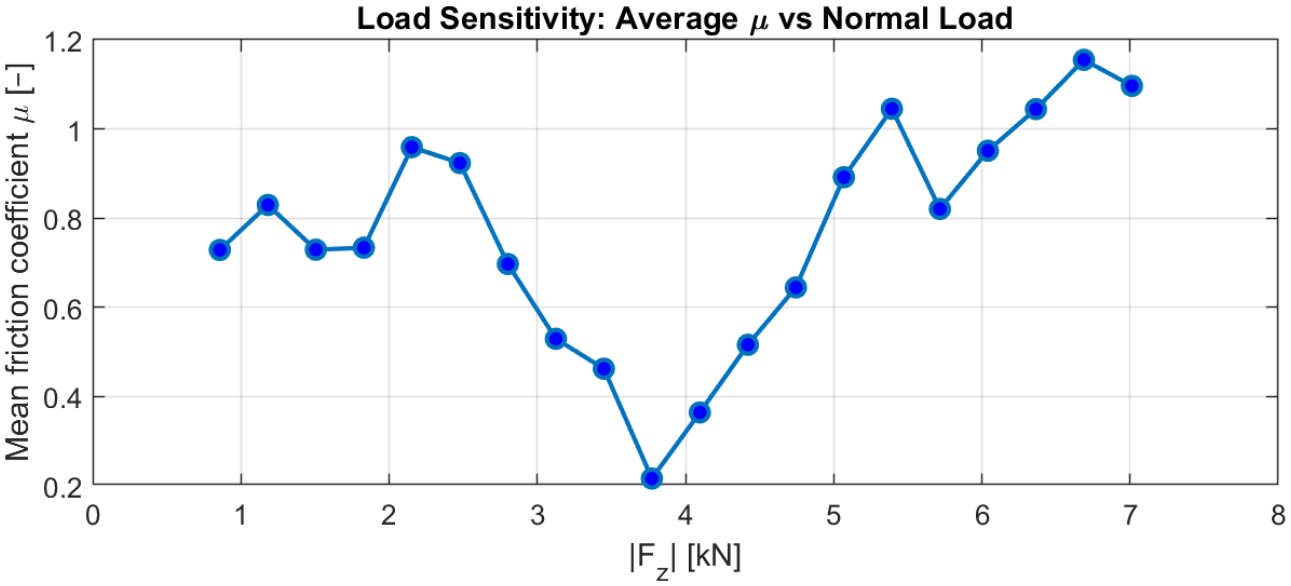


Figure 2.13: Mean friction coefficient as a function of normal load $|F_z|$.

Figures 2.12 and 2.13 make the load sensitivity explicit. The friction map shifts downward as the F_z band increases: a lightly loaded tyre grips harder per unit of normal force than a heavily loaded one. The physical reason is that grip is governed by contact pressure rather than total load; increasing F_z raises the pressure into a less adhesive deformation regime of the rubber compound. For the aerodynamicist, this means that adding downforce does increase total lateral force ($F_t = \mu F_z$) but with diminishing returns – each extra kilonewton of downforce buys progressively less grip increment.

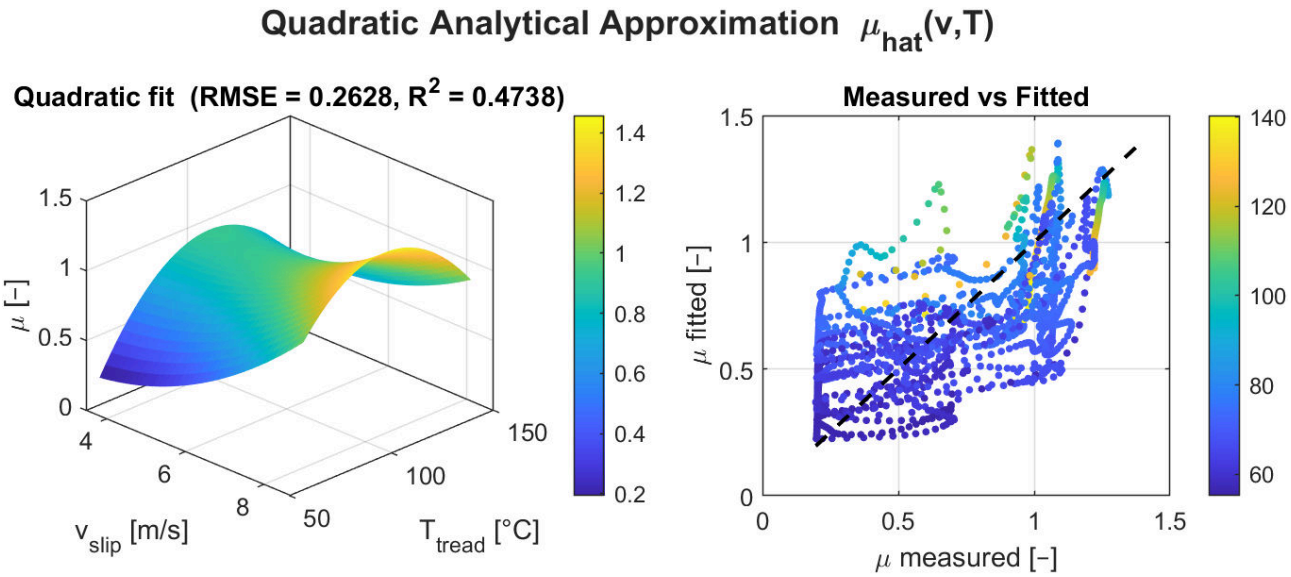


Figure 2.14: Quadratic analytical fit $\hat{\mu}(v, T)$ (Eq. 2.9). Left: fitted surface. Right: measured vs. fitted values (identity line in black dashed).

The quality metrics for the polynomial fit are shown in Table 2.5.

Table 2.5: Analytical fit quality metrics on the training dataset.

Metric	Symbol	Value
Root-mean-square error	RMSE	0.263
Coefficient of determination	R^2	0.474

An R^2 of 0.47 might look poor at first glance, but it is actually expected here and does not mean the model is wrong. The problem is that the polynomial uses only (v, T) as inputs, while the dataset contains measurements at a wide range of normal loads – and as just shown, μ drops by roughly 0.25–0.30 across the full F_z range. For any given (v, T) point, the model predicts a single value while the data contains many measurements taken at different loads, each with a different μ. That load-driven scatter cannot be captured by a two-variable model, so it lands entirely in the residuals.

Adding F_z (and F_z^2) as extra terms in the polynomial would substantially raise R^2. For the purpose of this exercise, the two-variable form remains useful: it correctly identifies the dominant temperature and slip-speed trends, is cheap to evaluate, and provides a compact analytical expression that can be plugged directly into a real-time lap-time simulator.

2.2.5 Conclusions – Exercise 2

- The friction coefficient peaks in the range 80 °C–100 °C and falls off on both sides, making thermal management essential to extract consistent grip over a full stint.
- Slip speed has a secondary but measurable effect: higher sliding speeds reduce peak friction slightly through visco-elastic heating of the rubber.
- Load sensitivity is clearly present: μ decreases monotonically with $|F_z|$, limiting the grip returns from aerodynamic downforce at high loads.
- The quadratic model $\hat{\mu}(v, T)$ captures the dominant trends and is compact enough for use in real-time simulation; a three-variable extension including F_z would improve fit accuracy at the cost of additional complexity.

General Conclusions

Taken together, the two exercises tell a coherent story. Exercise 1 shows that the lap time is sensitive to tyre temperature in a very practical way: the right set-up pushes both axles into their grip window, and the wrong set-up either leaves grip on the table (tyres too cold) or erodes it (tyres too hot). Exercise 2 then provides the quantitative evidence behind that story – a directly measured friction surface that confirms the temperature window, quantifies the slip-speed effect, and reveals the load sensitivity that limits how much downforce can help. The two results support each other: the model in Exercise 1 is more trustworthy because the friction law it relies on has been empirically verified in Exercise 2.

More broadly, the laboratory reinforces a point that is easy to overlook in simulation work: the tyre is not a static friction coefficient but a dynamic system with its own thermal state, and any model that ignores that state will make systematically wrong predictions about how set-up changes affect performance.

3 LAB 2 - Lap time simulation

3.1 Introduction

The objective of this laboratory is to explore the fundamental principles and applications of lap time simulation in the context of vehicle dynamics and performance engineering. Lap time simulation is a critical tool in modern motorsport, as it allows engineers to understand the sensitivity of a vehicle's performance to specific design parameters and setup configurations. It is very useful during the design phase of the car but also for the development of race strategies and the driving style optimization.

A robust lap time simulation requires the integration of three primary elements: a vehicle model (including handling, engine, tire, and aerodynamic sub-models), a driver model, and a track model. This laboratory focuses specifically on *"Quasi-Steady-State"* simulations, which utilize the concept of a performance envelope (GGV map) to determine the vehicle's limits across various track sections.

3.2 Exercise

The exercise conducted during this session involve the use of the OpenLap software, a MATLAB-based open-source project. Using a Tatuus FA010 vehicle model, the analysis includes:

- Evaluating the impact of fuel load on lap times across different circuits (Spa-Francorchamps, Monza, and Donington).
- Generating isochrone maps to analyze the trade-off between downforce (cl) and aerodynamic drag (cd).
- Selecting the optimal aerodynamic setup between the possible vehicle configurations given, according to the results obtained.

3.2.1 Results

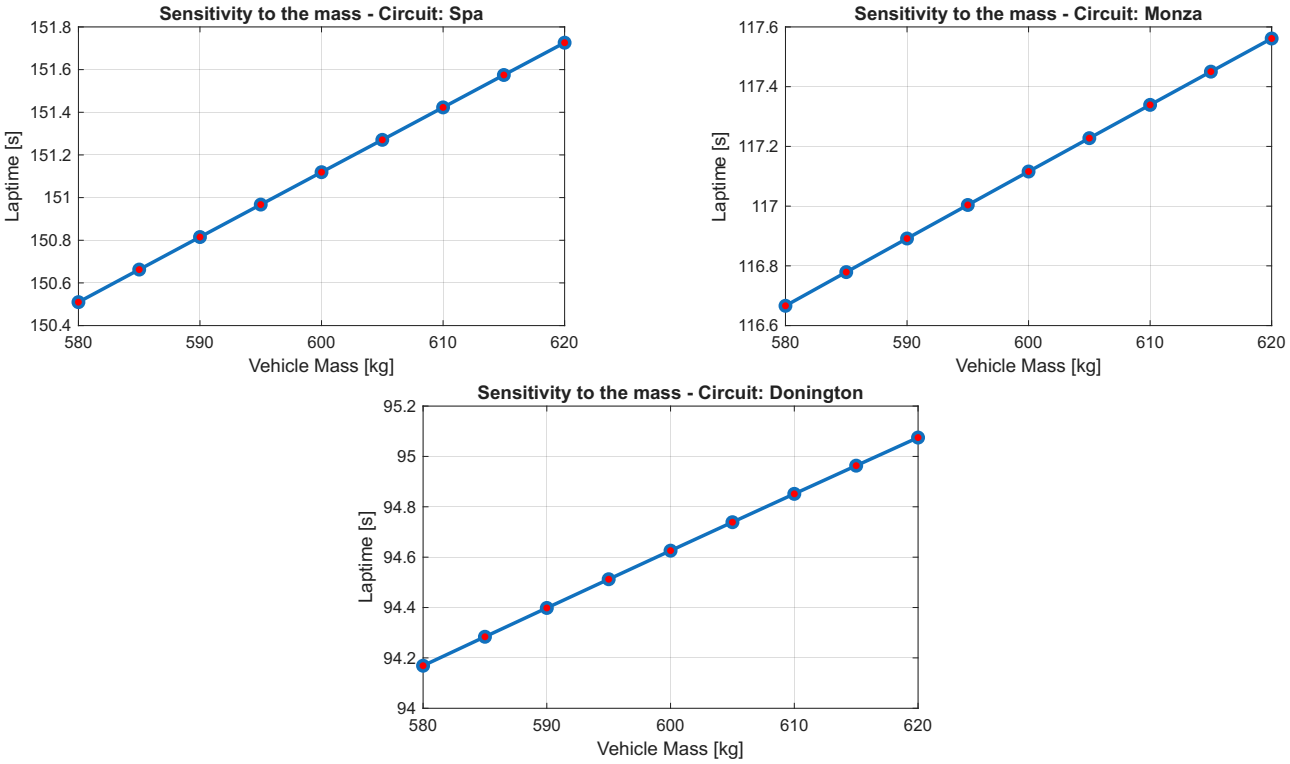
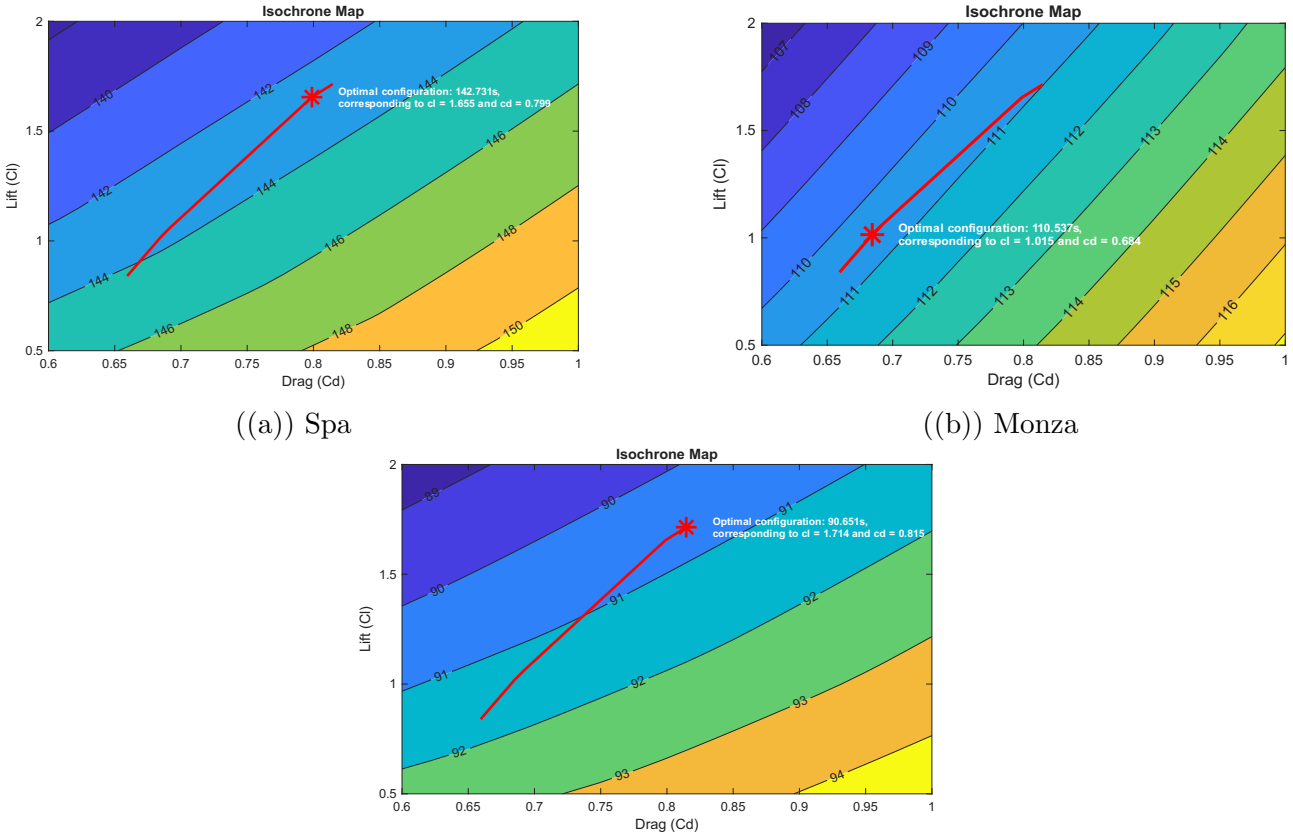


Figure 3.1: Sensitivity to the vehicle total mass (car+fuel) with cd = 1 and cl = 0.6

Quantity	Spa	Monza	Donington
Laptime Fuel Sensitivity (s/kg)	0.03040636	0.022376365	0.022648723
Fuel Sensitivity in % to Laptime (%/kg)	0.020120839	0.019106314	0.023935446

Table 3.1: Fuel effect for the three tracks



((a)) Spa

((b)) Monza

((c)) Donington

Figure 3.2: Isochrone maps of the race tracks for *cl* (0.5;2) and *cd* (0.6;1)

3.2.2 Discussion

We can observe that for all the tracks there is a linear relationship between the increase of the mass and the increase of lap time, we divided the fuel sensitivity for the lap length in seconds, in order to remove the effect of the length of the circuit that obviously affects the first data. We noticed that the fuel sensitivity percentage is very similar in all the three tracks but is a bit higher in Donington, probably due to the high speed turns that feature the english circuit.

Looking at the isochrone maps, the characteristics of the three circuits are well highlighted by the maps: Monza represents the classic "low-downforce" scenario. The simulation correctly identifies that on a track characterized by long straights and high-speed sections, minimizing drag is the priority. The optimal point is shifted towards the bottom-left of the map, indicating that the time gained on straights by reducing resistance outweighs the time lost in the chicanes and the Parabolica. Spa requires a "medium-high downforce" setup. Although sectors 1 and 3 are very fast, the long and technical Sector 2 demands significant vertical load to maintain high cornering speeds. The optimal configuration shows a balanced increase in both lift and drag compared to Monza, reflecting the need for a versatile car capable of handling both the Kemmel straight and the high-speed corners of the middle sector. Donington is the track that demands the highest level of downforce among the three. Being a shorter and more technical circuit with fewer long straights, the simulation favors aerodynamic grip over top speed. The optimal point moves further towards the top-right of the isochrone map (cl>1.7), demonstrating that maximizing cornering performance is the most effective strategy to reduce lap time here.

In the end we can reassume the results that clearly illustrate the **sensitivity of the vehicle to aerodynamic setup** based on track layout. We observe a clear progression:

- **Monza** (High Speed): Low *cl*, Low *cd*.
- **Spa** (Mixed): Medium-High *cl*, Medium-High *cd*.
- **Donington** (Technical): High *cl*, High *cd*.

In the practical application the lap time alone cannot be the only factor considered to choose a certain setup. **Tire wear** is a fundamental factor that must be taken into account alongside, for example, **handling**, **fuel consumption**, and the **ability to overtake and defend** from overtakes

4 LAB 3 - Braking systems

4.1 Introduction

The goal of this laboratory is to size the braking system of a racing car dividing this activity in three exercises

- **Exercise 1 - Disc sizing.** The temperature of braking discs needs to stay in a certain range (200 °C - 600 °C) to ensure the correct working conditions and to avoid safety problems for the entire duration of the race.
- **Exercise 2 - Braking system sizing.** The maximum tire forces are different between the front and rear axle. The ***brake balance***is the percentage of the total braking force acting on the front axle; our goal was to design the component of the braking system that are master cylinders, balance bar and calipers (number of pistons and pistons' diameter), implementing the best braking balance.
- **Exercise 3 - Bicycle model simulation.** Configure other 3 sets of components and run bicycle model simulation for comparison regarding lap time, rear and front maximum pressure, maximum driver force and brake balance.

4.2 Exercise 1 - Discs

4.2.1 Theoretical background

- **Braking power increases disc temperature**
  1. Power depends on brake torque & speed
  2. Given a Power input, temperature increase during braking depends mainly on disc thermal capacity
- **Heat dissipation reduces disc temperature**
  1. Main contribution: convection → depends on speed, dT, venting design, etc
  2. Second contribution (specially on carbon discs) → radiation

A disc works better in a specific temperature range. Outside this window, the braking performances are not optimal

Goal: Choose the best discs from the given set for the Tatuus car
Best target:

- Temperature should always stay in the range 200°C-600°C.
- A race should last (at least) 10 laps.

We neglected the torque limit reduction due to lateral dynamics and assumed all-longitudinal dynamics.

4.2.2 Methodology

1. Compute maximum longitudinal forces and torques (for each axle and total)

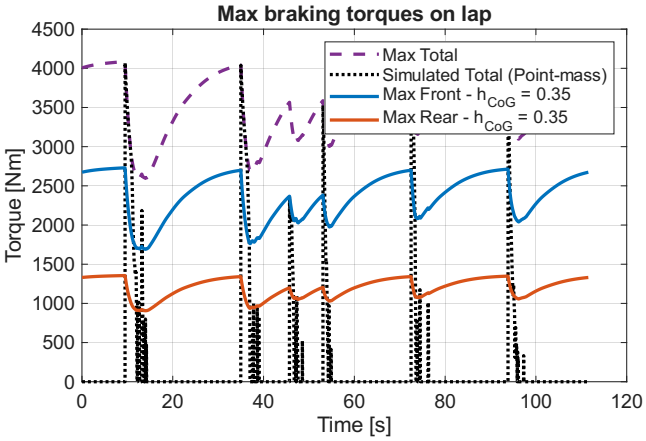


Figure 4.1: Braking torques computed with the bicycle model compared with the poin-mass model

2. Identify brakings: from torque limit profiles, the sections related to braking only are extracted
3. Maximum admissible braking power on each disc is computed
4. Select discs from the following set: **diameter 1 = 250 mm, diameter 2 = 270 mm, diameter 3 =290 mm**
5. Implement a thermal model to get temperature profiles for each axle
6. Run thermal analysis and check if the temperature remains in the desired range

4.2.3 Results

For the selected discs: (270 mm for the front axle and 250 mm for the rear axle)

Regime temperatures (excluding the transient in lap1):
T front: min = 204 °C, max = 416 °C
T rear: min = 225 °C, max = 378 °C

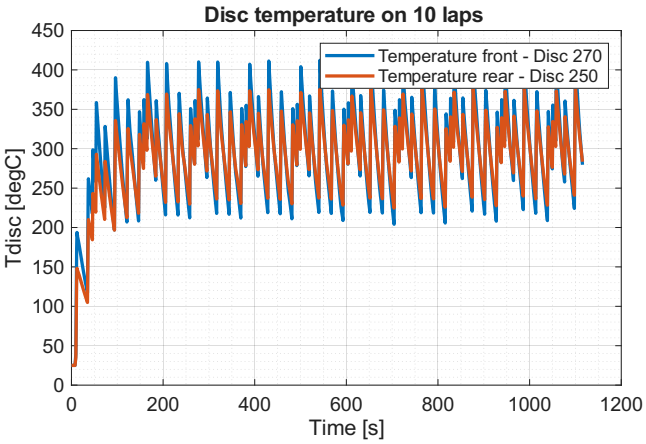


Figure 4.2: Thermal analysis on a 10 lap segment for the selected pair of discs

4.2.4 Discussion

The chosen configuration (Front: 270mm / Rear: 250mm) successfully meets all design objectives.

1. **Constraint Satisfaction:** Both axles operate well within the **200-600°C** safety window. This range ensures that the friction material (pads/discs) operates at an optimal coefficient of friction (bite) without reaching the threshold for thermal fading or structural damage (typically above 650-700°C for steel/iron discs).

2. **Asymmetric Sizing:** The requirement for a larger disc on the front (270mm vs 250mm) is a direct consequence of longitudinal load transfer. Because the front tires have higher vertical load during deceleration, they can generate more longitudinal force, requiring higher braking torque and thus dissipating more energy.

3. **Stability:** The 10-lap simulation demonstrates that the cooling capacity of the discs (convection) is sufficient to balance the heat input at Monza, as the temperature peaks do not continue to rise indefinitely but reach a steady-state "sawtooth" profile.

4. **Optimization:** While 290mm discs were available, they were not selected for the front to avoid unnecessary unsprung mass and rotational inertia, as the 270mm disc provided sufficient thermal headroom. Similarly, the 250mm rear disc is adequate for the lower energy demands of the rear axle.

4.3 Exercise 2 - Braking system

4.3.1 Theoretical background

Goal: Choose the braking system for the vehicle, implementing the best brake balance

Best target:

- As close as possible to the ideal balance for the entire speed range
- Never lock rear wheels

Constraints:

- Since the brake system can tolerate a limited amount of pressure, guarantee that the maximum pressure in each system does not exceed 120 bar
- Since there is no brake booster, ensure that the driver can full-brake with no more than 100 kg (981N). It is better if the driver force could be less than 80kg for the entire lap.

4.3.2 Methodology

- From the maximum axle braking forces of the previous exercise, calculate the ideal brake balance:

$$BB = \frac{F_{xFrontMax}}{F_{xFrontMax} + F_{xRearMax}} \tag{4.1}$$

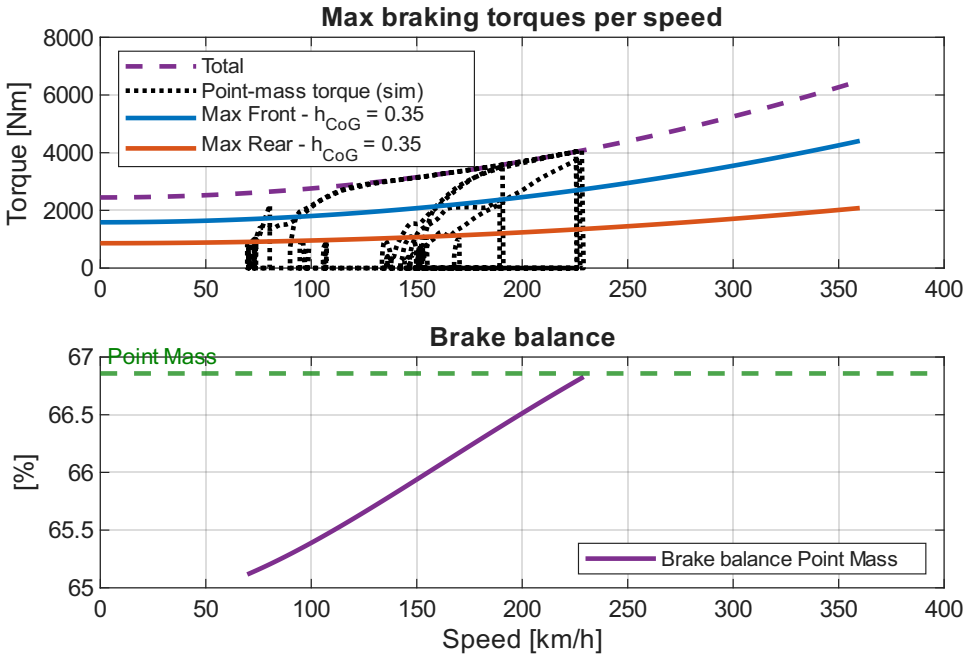


Figure 4.3: Max longitudinal torques compared with simulation and Ideal brake balance depending on speed

- choose braking components between the following (the brake balance bar can be set between 40% and 60%)

Master cylinder:

Diameter $[mm]$	Applicable on	
	Front	Rear
17	X	X
18	X	X
19	X	X
20	X	X

Calipers:

Number of pistons	D piston 1 $[mm]$	D piston 2 $[mm]$	D piston 3 $[mm]$	Applicable on	
				Front	Rear
2	28				X
2	30				X
2	32				X
4	26	26		X	X
4	26	28		X	X
4	28	28		X	X
4	28	30		X	X
4	30	30		X	X
4	30	32		X	X
6	26	26	26	X	
6	26	28	28	X	
6	26	28	30	X	
6	28	30	30	X	
6	28	30	32	X	
6	30	32	32	X	

Figure 4.4: Braking system's available components

- compute the real brake balance for this set of calipers, master cylinder and brake balance bar and compare it with the ideal brake balance, verify the constraints (p<120 bar for both axes and driver force necessary < 80kg for the entire lap).

4.3.3 Results

Ideal Braking Balance computed: 66,86%

The chosen racing system configuration was:

	Front Master Cyl.	Rear Master Cyl.	front calipers	rear calipers	BB Bar
Base BS	18 mm	17 mm	28-30-32	28-28	45%

This braking system configuration has a braking balance very close to the ideal one, and at the same time matches the requirements of maximum pressure below 120 bar and of maximum driver force of 80 kg.

```
Pressures:
    -> p front max = 63 bar
    -> p rear max  = 58 bar
-------------------------------------------------
F driver:
    -> max [N]  = 731 N = 75 kg
-------------------------------------------------
Brake balance:
    -> Front = 66.8577 %
    -> Rear  = 33.1423 %
-------------------------------------------------
```

Figure 4.5: presssures, driver force and Brake Balance computed for the chosen braking system

4.3.4 Discussion

We reached this configuration by changing the parameters one at time and we noticed that they have the following effect on the brake balance:

- increasing the diameter of the front master cylinder causes a decrease in the Braking Balance (on the other hand increasing the rear master cylinder's make the braking balance increasing).
- increasing number and diameter of the calipers' pads causes an increase in the Braking Balance (on the other hand decreasing them make the braking balance decreasing).
- increasing the braking balance bar % causes a decrease in the Braking Balance (on the other hand decreasing it make the braking balance increasing).

4.4 Exercise 3 - Bicycle model and comparison

4.4.1 Methodology

We compared the configuration of the previous exercise with three different configurations.

	Front Master Cyl.	Rear Master Cyl.	front calipers	rear calipers	BB Bar
Base BS	18 mm	17 mm	28-30-32	28-28	45%
Altern. 1	20 mm	19 mm	28-30-32	28-28	45%
Altern. 2	19 mm	20 mm	30-32-32	26-26	55%
Altern. 3	18 mm	17 mm	28-30-32	28-28	45%

We choose to stay as close as possible to the ideal brake balance by varying the parameters: in the first alternative we increased the diameter of both master cylinders, in the second one we increase the brake balance bar percentage by ten points, in the third we reduced the number of pads in the calipers.

4.4.2 Results

Simulation	LapTime_s	DeltaVsPointMass_s	Max pressure front [bar]	Max pressure rear [bar]	Max driver force [kg]	Brake Balance
"Point Mass"	111.54	0	63	58	74.516	0.66858
"SingleTrack Baseline"	112.77	1.2225	63	58	74.618	0.66858
"SingleTrack alternative 1"	112.77	1.2264	63	57	92.049	0.67117
"SingleTrack alternative 2"	112.79	1.2423	58	64	93.272	0.67928
"SingleTrack alternative 3"	112.77	1.2275	109	100	128.75	0.67051

Figure 4.6: Laptimes, pressures, driver forces and brake balances of the three alternative braking systems compared to chosen one and to the single point mass model

Even if all the alternatives have an acceptable final brake balance we have to notice that they do not match the constraints, they all require a force of the driver greater than 80 kg and the third one has also a maximum front pressure very close to the limit of 120 bar

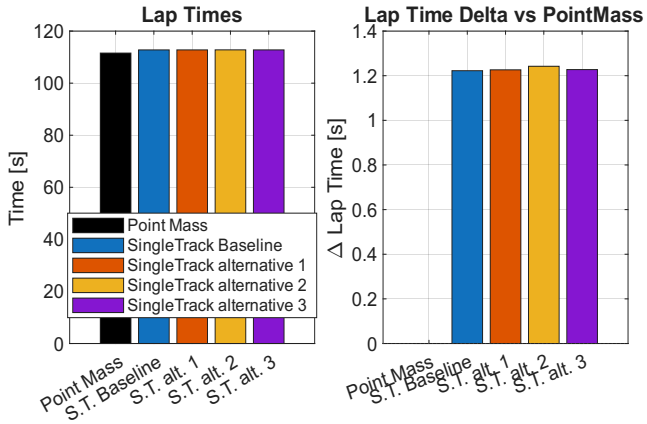


Figure 4.7: Lap times of the bicycle model with the four different braking systems compared to the single point mass simulation

The differences in terms of lap time between the four braking systems are almost negligible, since the brake balances are similar. We can instead notice a difference of 1.2 seconds with the point mass simulation, we can explain it with the major complexity of the bicycle model.

4.4.3 Discussion

- **Master Cylinder seizing**: If an alternative setup uses master cylinders with a larger bore diameter, the driver must push harder. Pressure is defined as $P = F_{push}/A_{cylinder}$. A larger area means that for every Newton of force applied by the driver, less hydraulic pressure is generated. Therefore, to get the pressure needed for high deceleration, the driver must increase their physical effort.

- **Caliper Piston Area:** The clamping force on the disc is $F_{clamp} = P \times A_{pistons}$. If the front calipers have smaller total piston areas compared to the rear (or compared to another configuration), the system must compensate by increasing the hydraulic pressure to achieve the required clamping force.

5 LAB 4 - Lap time simulation 2

5.1 Introduction

The aim of this laboratory is to study how two different setup and race-management levers act on the lap time of a single-seater at the Monza circuit, by means of the OpenLap quasi-steady lap simulator. The lab is split into two independent exercises:

- **Exercise 1 – Ride heights.** The aerodynamic coefficients of the car (c_l, c_d and the aerodynamic balance Bal) are not constant: they depend on the front and rear ride heights (FRH and RRH), which in turn depend on the downforce produced at the current speed. We are asked to estimate speed-dependent functions $c_l(v)$, $c_d(v)$ and $Bal(v)$ from the supplied aero map and the static ride heights and ground stiffnesses, then to choose the setup that minimises the lap time at Monza without making the car touch the ground.
- **Exercise 2 – Lift and coast.** Using a modified version of OpenLap in which the throttle can be cut over a configurable window in the Prima Variante and/or Lesmo 1 corners, we have to (i) quantify the impact of lift & coast on lap time, brake energy and fuel consumption, (ii) decide which of the two corners is the most convenient for it, (iii) evaluate whether it is worth doing it on every lap of a 40-lap race in order to start with less fuel, and (iv) discuss the limitations of the model.

5.2 Exercise 1 – Ride heights

5.2.1 Theoretical background

The underfloor of a modern racing car generates a large fraction of the total downforce. Even small changes in front or rear ride height can therefore substantially modify c_l, c_d and the aerodynamic balance Bal (the fraction of total downforce on the front axle). The ride height is also limited by:

- plank wear and minimum ground clearance regulations,
- the operating window of diffusers (downforce drops sharply outside it),
- the need to avoid bottoming on bumps and high-speed kerbs,
- its (small) effect on the centre of mass and load transfer.

The aero map is normally measured by a combination of *wind-tunnel tests*, *CFD simulations* and *constant-speed track tests* (where pushrod loads give the tire forces, hence the downforce, while the ride heights are measured by laser ride-height sensors).

Because the ride height changes with speed (the heavier the car is pushed down by aero, the lower the body sits), the coefficients seen by a lap-time simulator must be expressed as a function of speed, not as a function of $(\mathrm{FRH}, \mathrm{RRH})$. This is the link the present exercise is asking to build.

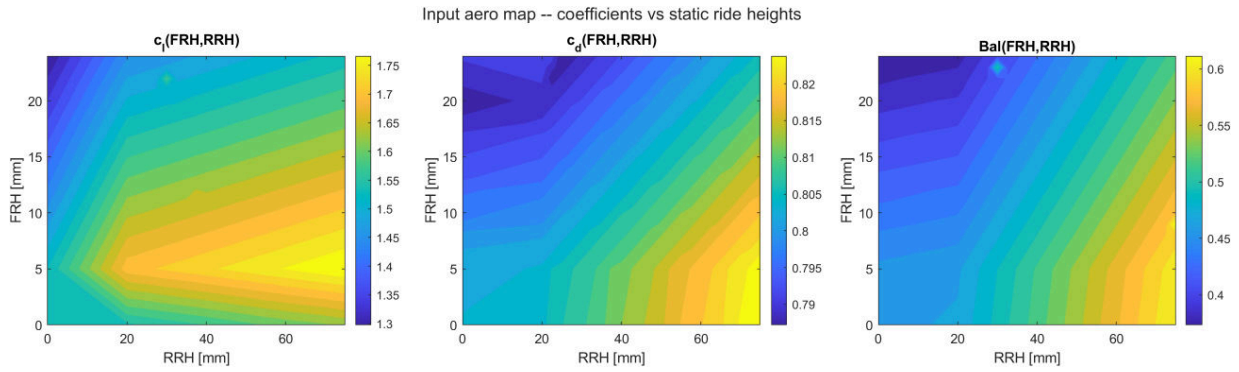


Figure 5.1: Input aero map: contours of c_l, c_d and aerodynamic balance Bal as a function of front and rear static ride height.

5.2.2 Methodology

Speed-dependent aero functions. Given the static ride heights $\mathrm{FRH}_{\mathrm{stat}}$ and $\mathrm{RRH}_{\mathrm{stat}}$ and the front/rear ground stiffnesses k_f and k_r, a Picard-style march is performed from 0 to 360 km/h in 1 km/h steps:

1. At the current ride heights, look up c_l, c_d, Bal from the aero map (linear bivariate interpolation).
2. Compute the total downforce $F_z = \frac{1}{2}\,\rho\, c_l\, A\, V^2$, split it according to Bal between front ($F_{z,f}$) and rear ($F_{z,r}$) axles.
3. Update the new ride heights with the linear ground-stiffness model: $\mathrm{FRH} = \mathrm{FRH}_{\mathrm{stat}} - F_{z,f}/k_f$ and $\mathrm{RRH} = \mathrm{RRH}_{\mathrm{stat}} - F_{z,r}/k_r$.
4. Step to the next speed.

The resulting $c_l(v)$, $c_d(v)$ and $Bal(v)$ are saved and used inside OpenLap.

Setup sweep. Four design variables are explored: $\mathrm{FRH}_{\mathrm{stat}} \in \{10, 12, 14, 16\}$ mm, $\mathrm{RRH}_{\mathrm{stat}} \in \{35, 40, 45, 50\}$ mm, and $k_f, k_r \in \{206, 264, 294, 324\}$ N/mm. This gives $4 \times 4 \times 4 \times 4 = 256$ combinations.

On-track validity check. A naive criterion would discard every combination that drives the ride height to zero anywhere in the 0–360 km/h build sweep. This is too strict, because the car does not actually reach 360 km/h on Monza. The implementation adopted here therefore:

1. Always runs the OpenLap simulation, even if the speed-march reaches a bottoming condition at high speed (the simulator clamps the ride height to a small positive value so that aero coefficients remain defined).
2. After the lap, computes the maximum on-track speed $V_{\mathrm{max,lap}} = \max(V(s)) \cdot 3.6$.
3. Validates the setup by checking that the minimum FRH and RRH for $v \le V_{\mathrm{max,lap}}$ are strictly positive.

A setup that bottoms out at 360 km/h but is fully clear at the maximum speed reached on Monza is therefore retained.

5.2.3 Results

The 256-combination sweep produced the laptime distribution of Fig. 5.2 (best k_f, k_r pair shown for each static ride-height pair). The winning setup is summarised in Table 5.1, and Fig. 5.3 shows the speed-dependent quantities of the corresponding lap.

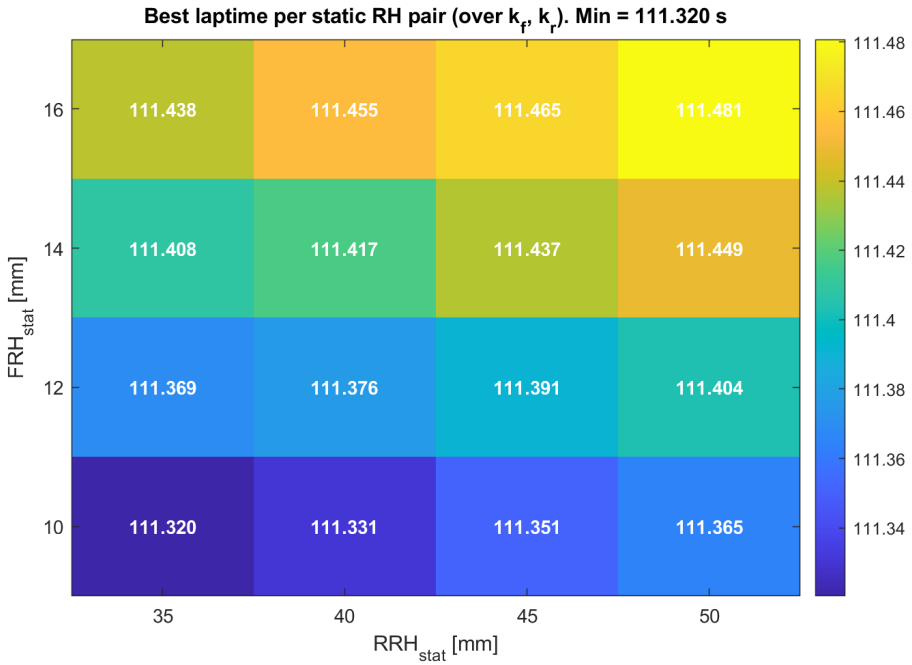


Figure 5.2: Best laptime per $(\mathrm{FRH_{stat}}, \mathrm{RRH_{stat}})$ pair (i.e. minimum over the 4×4 stiffness combinations).

Quantity Unit	$\mathrm{FRH_{stat}}$ mm	$\mathrm{RRH_{stat}}$ mm	k_f N/mm	k_r N/mm	Lap time s
Best setup	10	35	206	206	111.32

Table 5.1: Best static ride heights / ground stiffnesses for Monza, from the 256-combination sweep with on-track validation.

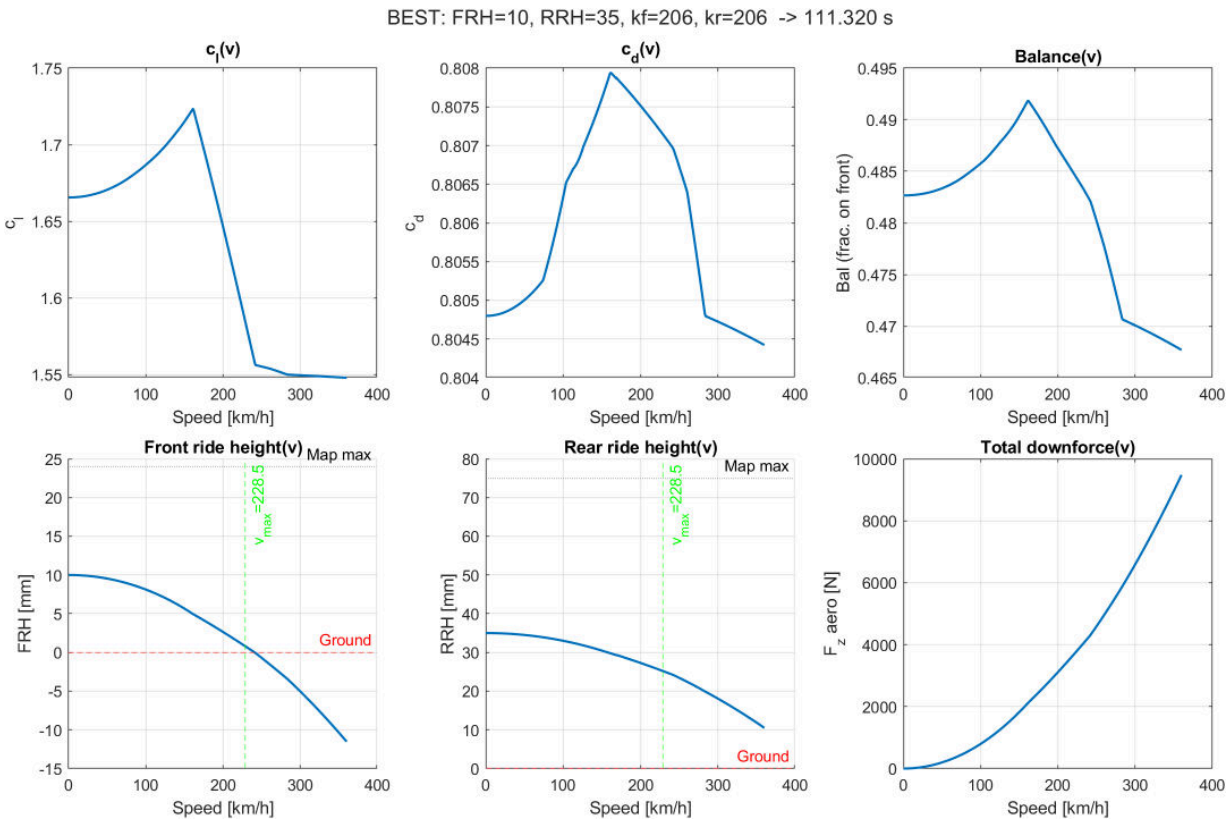


Figure 5.3: Aero quantities of the winning setup as a function of speed: $c_l(v)$, $c_d(v)$, $Bal(v)$, FRH(v), RRH(v) and total downforce. The green dashed line marks the maximum speed reached on the lap, the red dashed line marks the ground.

5.2.4 Discussion

The heatmap in Figure 5.2 shows a clear trend: lower static ride heights consistently produce faster lap times because they keep the car closer to the ground, maximising the underfloor downforce. The penalty for going too low is bottoming, which causes a sudden loss of downforce and risks mechanical damage. The winning setup of FRH = 10 mm and RRH = 35 mm sits right at the edge of what is admissible at Monza's maximum speed.

Suspension stiffness plays a secondary but real role. Softer springs allow the car to squat deeper at high speed, which increases downforce but reduces the ride-height margin. Stiffer springs keep the ride height closer to the static value and protect against bottoming on the straights, but at the cost of reduced compliance over kerbs, where the instantaneous load spike can momentarily compress the suspension beyond the steady-state prediction.

The on-track validation criterion was crucial for getting the right answer. A naive check that discards any setup bottoming anywhere in the 0–360 km/h sweep would have eliminated setups that only bottom beyond the actual maximum speed reached on Monza. Several of the fastest valid combinations would have been wrongly rejected, and the reported optimum would have been unnecessarily conservative.

5.3 Exercise 2 – Lift and coast

5.3.1 Theoretical background

Fuel saving is a recurring need in long-distance motorsport (Le Mans, F1 with limited fuel-flow regulations, etc.) and in tracks where the power unit cannot be run at full output (e.g. Mexico

GP at 2238 m altitude, where the air is ∼25 % less dense and cooling capacity drops). Two main strategies are available:

- Specific engine maps that lower the power output (and hence fuel flow);
- *Lift and coast*: the driver releases the throttle before braking, so that the car coasts to the corner.

Lift and coast is most efficient at the *end of a straight*, just before the braking zone, because the time loss is concentrated in a short section of track where the car is anyway about to slow down.

The single-lap time difference between the lift and the no-lift strategies is, to first order,

$$\Delta T \approx \int_0^L -\frac{\Delta v(s)}{v(s)^2}\, ds, \tag{5.1}$$

where $\Delta v(s)$ is the speed drop introduced by the lift along the track.

5.3.2 Methodology

The lift-and-coast logic is governed by two pairs of parameters, one per corner: a *lift-start position* (an index along the track mesh, from which the throttle is cut to zero up to the corresponding corner-entry braking zone) and a binary *enable* flag. The smaller the lift-start position, the earlier the driver lifts off the throttle. The maximum (latest) lift-start positions are 600 for Prima Variante and 2220 for Lesmo 1.

The analysis is built on the following sequence:

1. A baseline run with both lifts disabled, used as reference for all subsequent deltas.
2. A *mass sensitivity* sweep: the same baseline lap is re-simulated at several values of vehicle mass to fit a linear coefficient $\partial T/\partial m$ in s/kg.
3. A fine sweep of the Prima Variante lift-start position from 0 to 600 (step 25), with the Lesmo lift disabled.
4. A fine sweep of the Lesmo lift-start position from 0 to 2220 (step 50), with the Prima Variante lift disabled.
5. A 40-lap race analysis at every sweep point: the time gained on each lap due to the lighter average mass is added to the time lost by lifting, yielding a net per-lap and net 40-lap delta.

40-lap race analysis. If the car saves ΔF kg of fuel per lap, then over 40 laps the total saving is $40\,\Delta F$. In a real race this saving allows the team to start with a lighter fuel load. Because fuel burns approximately linearly along the race, the *average* mass reduction across the 40 laps is half of the total saving, $\overline{\Delta m} = \frac{N_{\text{laps}}}{2}\,\Delta F$. The corresponding lap-time gain per lap is

$$\Delta T_{\text{mass}} = \overline{\Delta m}\,\frac{\partial T}{\partial m} = \frac{N_{\text{laps}}}{2}\,\Delta F\,\frac{\partial T}{\partial m}. \tag{5.2}$$

The net effect per lap is therefore

$$\Delta T_{\text{net}} = \Delta T_{\text{lift}} + \Delta T_{\text{mass}}, \tag{5.3}$$

where $\Delta T_{\text{lift}} > 0$ (time lost by lifting) and $\Delta T_{\text{mass}} < 0$ (time gained by lighter car). If $\Delta T_{\text{net}} > 0$ on every sweep point, lift and coast is *never* convenient for raw lap time over the race.

5.3.3 Results

Mass sensitivity. A linear fit of lap time versus mass perturbation (Fig. 5.4) gives $\partial T/\partial m = 0.0285\,\mathrm{s/kg}$ at the $M_0 = 605\,\mathrm{kg}$ baseline. The fit is essentially linear over the $\pm 20\,\mathrm{kg}$ range explored.

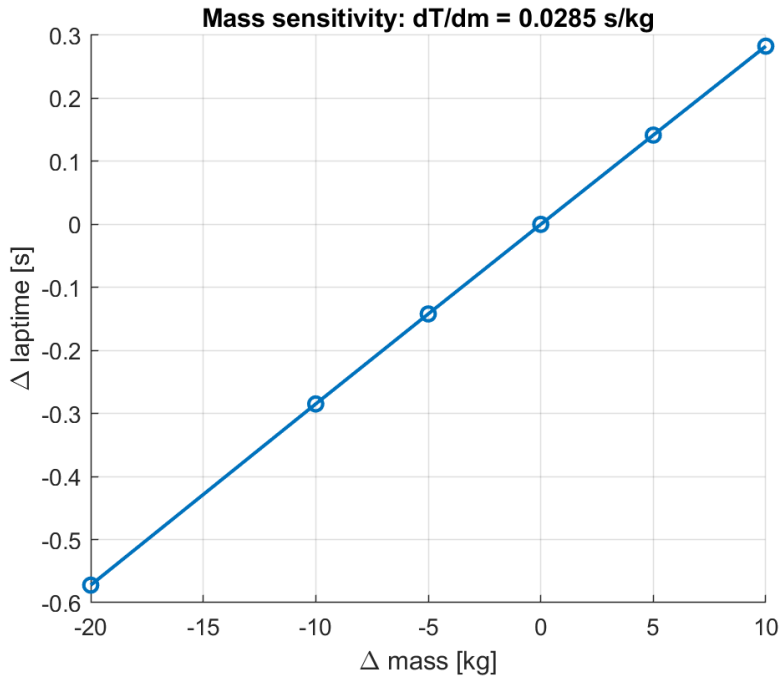


Figure 5.4: Lap time delta as a function of vehicle mass perturbation around the 605 kg baseline. The slope is $\partial T/\partial m = 0.0285\,\mathrm{s/kg}$.

Per-lap trade-off. Table 5.2 summarises four representative cases: baseline, PV-only at `pos_PV` = 300, Lesmo-only at `pos_L` = 2100, and both corners enabled. The full sweep is reported in Fig. 5.6.

Case	Lap time [s]	Fuel [kg]	Brake energy [MJ]	ΔFuel [g]
Baseline (no lift)	111.461	0.8384	0.191	—
PV only (pos_PV=300)	112.197	0.7972	—	−41.1
Lesmo only (pos_L=2100)	111.905	0.8135	—	−24.8
Both	112.640	0.7723	—	−66.0

Table 5.2: Per-lap impact of lift & coast, representative sweep points. Brake-energy entries are taken from `simulate_lc.m` (`out.brake_energy`).

Speed and throttle traces. Fig. 5.5 compares the speed and throttle traces of the four representative cases over the lap, making the cut-throttle windows in PV and Lesmo 1 visible.

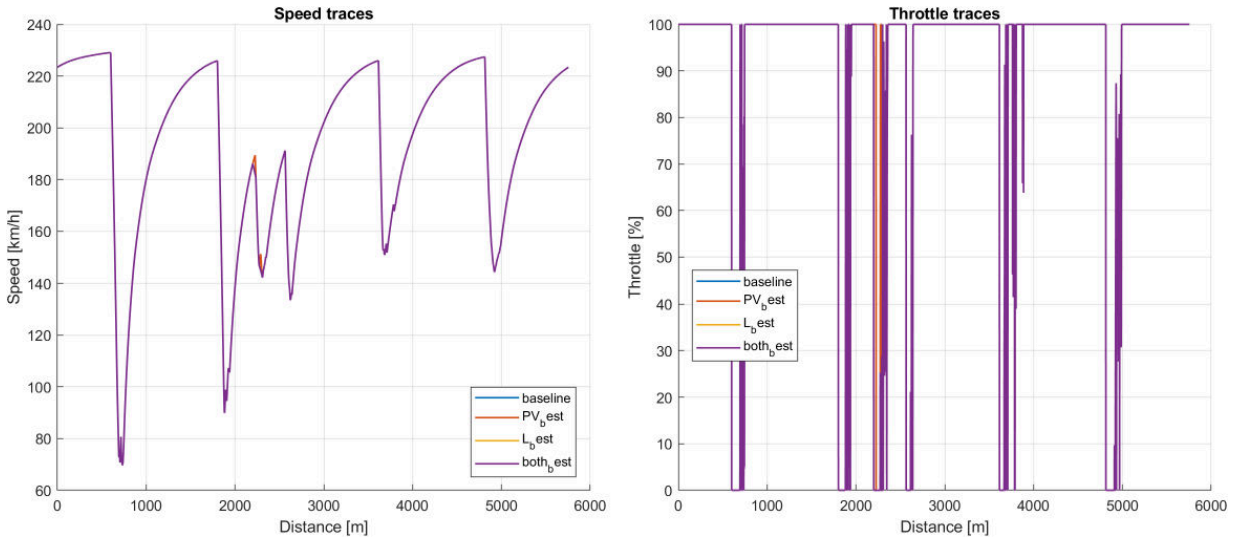


Figure 5.5: Speed (left) and throttle (right) traces along the lap for the four representative cases. The cuts in throttle correspond to the lift windows in Prima Variante and Lesmo 1.

Which corner is the most convenient? The Pareto front in Figure 5.6 gives a direct answer: at any chosen time-loss budget, Prima Variante always saves more fuel than Lesmo 1. At a modest penalty of 0.15 s/lap, PV saves 20.6 g/lap against Lesmo's 15.8 g/lap. Push the lift window out to 0.5 s/lap and the gap widens to 34.3 vs 24.8 g/lap. Beyond 1 s/lap, PV reaches savings of nearly 48 g/lap while Lesmo cannot sensibly extend that far.

The reason is circuit geometry. Prima Variante is approached along the long start/finish straight, where the car spends several seconds at full throttle and high speed. Cutting the throttle earlier there removes a large block of fuel consumption at a relatively modest speed penalty. Lesmo 1 is approached after Curva Grande, a much shorter run, so the useful lift window is physically narrower and the fuel saving per second of time lost is smaller. In short, **Prima Variante is the more efficient corner for lift & coast**.

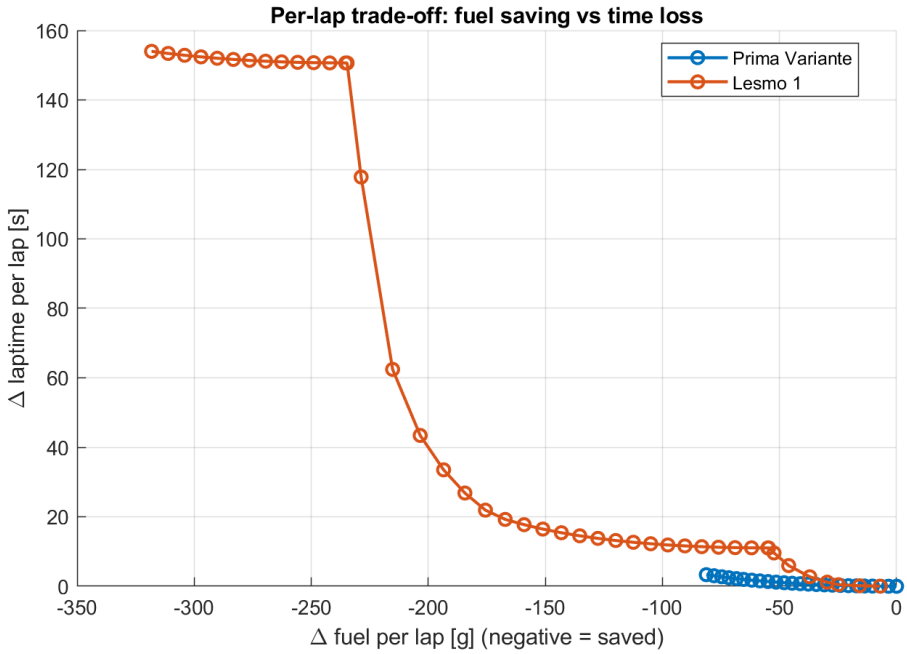


Figure 5.6: Per-lap Pareto front: Δlap time vs Δfuel mass for Prima Variante (blue) and Lesmo 1 (orange) sweeps. For every ΔT value, the PV curve sits to the left of Lesmo, i.e. saves more fuel for the same time loss.

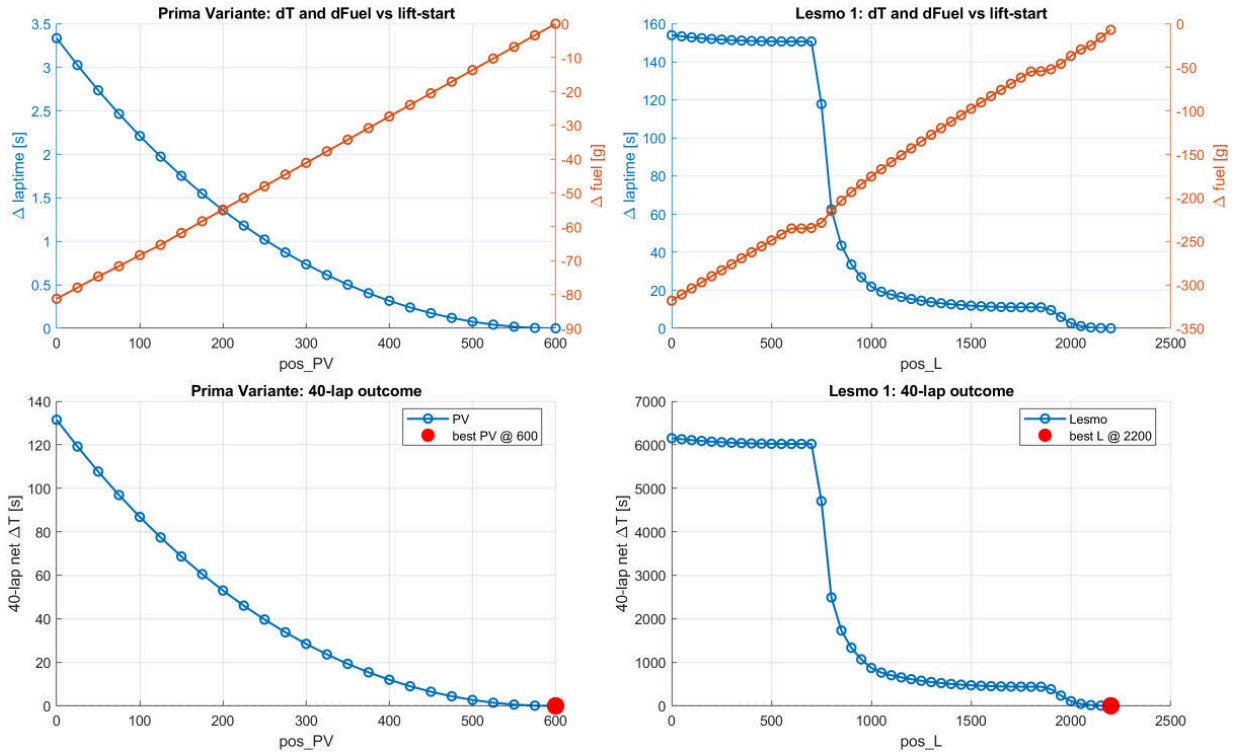


Figure 5.7: Per-corner sweep summary: top row shows ΔT and Δfuel as a function of lift-start position; bottom row shows the 40-lap net ΔT outcome. The red marker highlights the best (least bad) lift-start position from a pure laptime perspective.

Is it worth lifting on every lap of a 40-lap race? Combining the time lost by lifting with the time gained from a lighter average mass over 40 laps (Eq. 5.3) yields the net curves of Fig. 5.8.

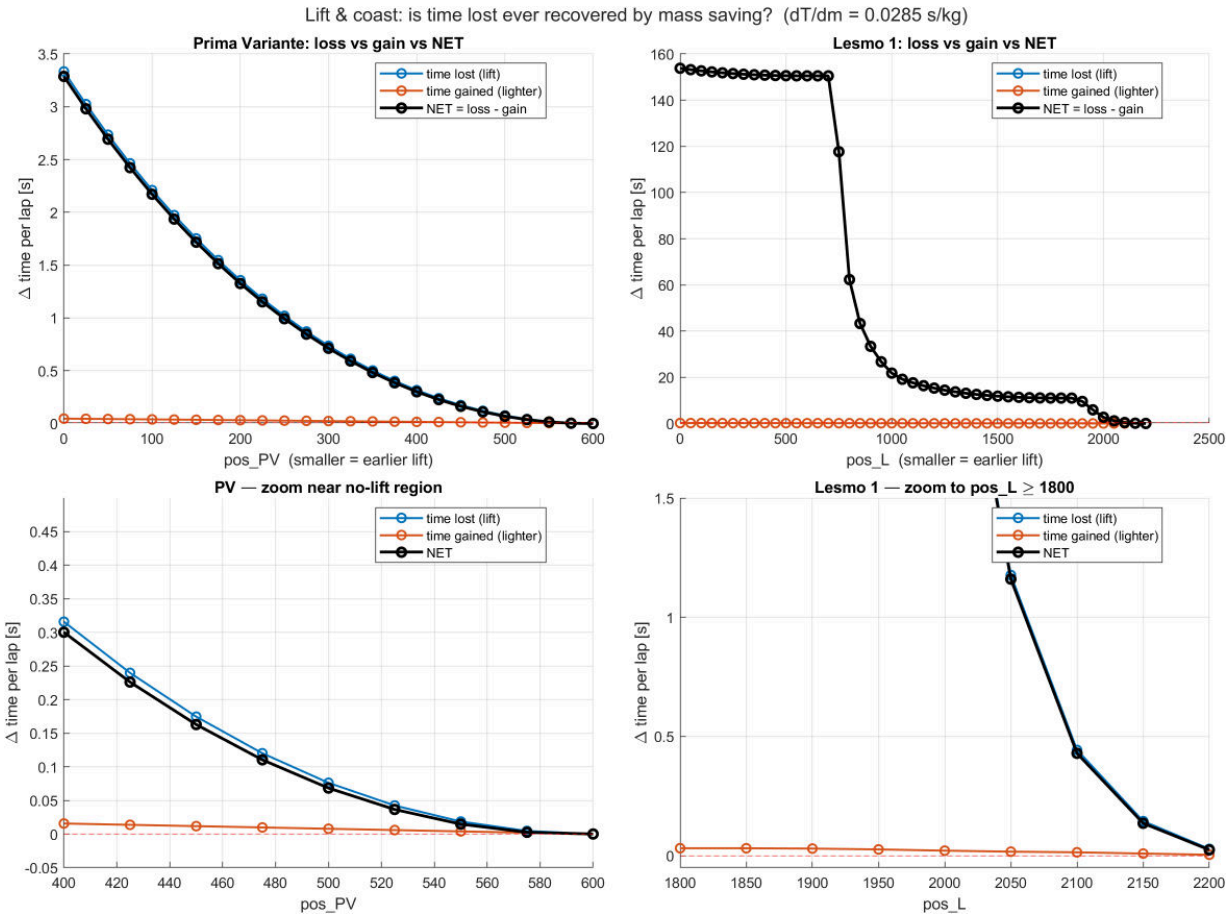


Figure 5.8: Per-corner decomposition: time lost by lifting (blue), time gained from the lighter average car (orange), and net per-lap delta (black). The red dashed line marks the break-even. **None of the explored points crosses below zero**, i.e. the mass saving never pays back the time directly lost in the lift window. Top row: full range. Bottom row: zoom around the no-lift region.

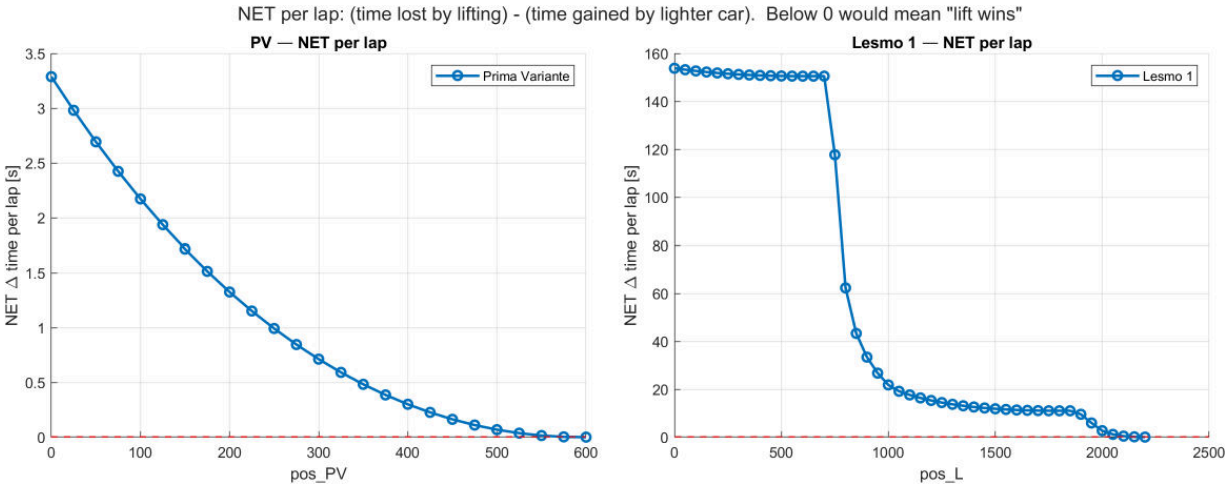


Figure 5.9: Net per-lap delta versus lift-start position for the two corners (separate axes for readability, since the Lesmo NET ranges up to several seconds while the PV NET stays well below). Both curves are non-negative everywhere.

The smallest non-trivial net found is $+0.07\,\mathrm{s}$/lap at `pos_PV` $= 500$, which over 40 laps amounts to $+2.7\,\mathrm{s}$ – still strictly positive. **In pure lap-time terms, the model says it is never convenient to lift on every lap of the race**.

The reason is purely numerical: $\partial T/\partial m \approx 0.0285\,\mathrm{s/kg}$ is small, and the average mass saving in a 40-lap race is only half of the total saving (the fuel burns linearly), so each gram saved per lap converts into a tiny fraction of a millisecond per lap. By contrast, the throttle cut directly costs hundreds of milliseconds per lap as soon as the lift window covers a non-trivial portion of the throttle-on region.

5.3.4 Model limitations – what does it neglect?

The lap simulator used here is quasi-steady and does not capture several phenomena that, in reality, justify the use of lift & coast in a race despite the negative lap-time prediction:

- **Brake temperatures.** Lifting at the end of a straight reduces the kinetic energy that the brakes have to dissipate. In races where brake cooling is marginal (Singapore, Mexico, Monaco) this alone is often the dominant reason to lift.
- **Engine and powertrain temperatures.** At high altitude (e.g. Mexico) or in long stints, lift & coast keeps the power unit within thermal limits.
- **Fuel-flow regulations.** F1 limits maximum fuel flow; if a track demands more energy per lap than the regulation allows over a stint, lift is mandatory. The simulator does not enforce any flow limit.
- **Tire wear and degradation.** Arriving at the corner with less energy reduces the longitudinal slip during braking and the lateral load on entry, lengthening tire life. Tire degradation is not modelled.
- **Battery state of charge (hybrid cars).** Lift and coast frees up combustion energy that would otherwise be wasted into heat; that energy can be partially harvested by the MGU-K. Not modelled.
- **Throttle dynamics.** The model treats the throttle command as a binary on/off step; in reality drivers lift partially and modulate the pedal, with non-trivial inertia of the engine response.
- **Driver consistency and traffic.** Real drivers cannot exactly repeat a lift point each lap; in traffic, lift may also be used for tactical reasons.

5.4 Conclusion

The two exercises illustrate the two ways in which a quasi-steady lap simulator is used in motorsport engineering: as a *setup optimiser*, when the design variables (ride heights, stiffnesses, aero) are explored systematically against a lap-time cost function (Exercise 1), and as a *race-strategy tool*, when the question is not the absolute fastest lap but the trade-off between lap time and other metrics, such as fuel consumption (Exercise 2).

The most useful lessons are:

1. Physical validity criteria must be applied at the actual operating point (here, the on-track speed envelope), not at the boundary of the look-up tables – otherwise valid setups are discarded.
2. For Monza, Prima Variante is more efficient than Lesmo 1 for lift and coast: it saves more fuel per second of lap time lost.

3. According to a quasi-steady, single-lap-mass-balance model, lifting on every lap of a 40-lap race is *not* convenient for raw lap time. The reason is the small lap-time sensitivity to mass (0.0285 s/kg) compared with the direct time loss of the throttle cut.

4. The fact that real teams *do* use lift and coast in races is therefore not a contradiction but a sign that the dominant motivations are thermal management, fuel-flow regulations, tire degradation and energy harvesting – all of which lie outside the scope of a quasi-steady lap-time simulator.

6 LAB 5 – Racing Car Data aquisition in Motorsport with Marelli (Wintax lab)

6.1 Introduction

Two panellists of *Marelli Motorsport*, Alberto Palma and Donia Youssef, took a lesson about telemetry and data in motorsport.
Data are widely used in Motorsport for all almost all aspects of this sport.

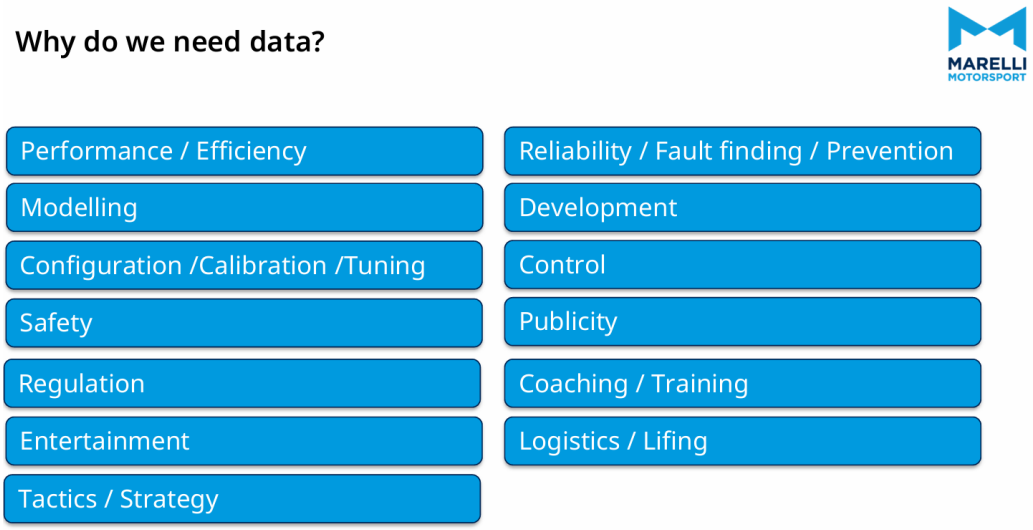


Figure 6.1: Data are needed for every aspect of the motorsport

Telemetry is the wireless transmission and reception of data between a vehicle and a remote location (typically pit crew or data center). Telemetry is a **real time** monitoring.
It's important to distinguish between live telemtery and logged data.

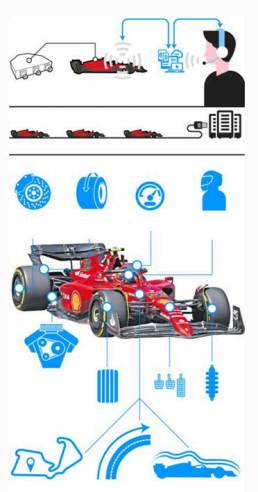


Figure 6.2: telemtry's architecture

The architecture of telemetry:

- sensors measure parameters (engine, brakes, GPS,...)
- Data logger collects and formats data
- Data are transmitted via fiber ring
- Data are received and displayed in real time for analysis

In a second moment, they presented us the Wintax program, the top ranking of PC applications for data acquisition and analysis in Motorsport competitions.

The exercise proposed consisted in computing longitudinal and lateral acceleration of a motorbike in the inertial reference system starting from signals in the vehicle (IMU) reference system, in two different cases:

- using raw acceleration data
- using acceleration data filtered with fc = 10 Hz

Choose then which of the two signals is preferable. Afterwards, only for the case filtered at 10Hz, we had to use the lap report on the lateral acceleration channel to identify, for each lap, in which corners of the circuit the maximum lateral acceleration is detected.

6.2 Theoretical Background

Wintax is a program with numerous features including customization, high performance, many analysis windows (Graphs, Reports, FFT), circuit data mapping, commercial tool integration, import, export, ECU data virtualization, real time telemetry...

The vehicle (motorbike) has a completely different reference system with respect to the inertial one, so we had to use the transformation matrices to pass to one system to the other. We created some virtual channels to load the data on Wintax.

6.3 Results

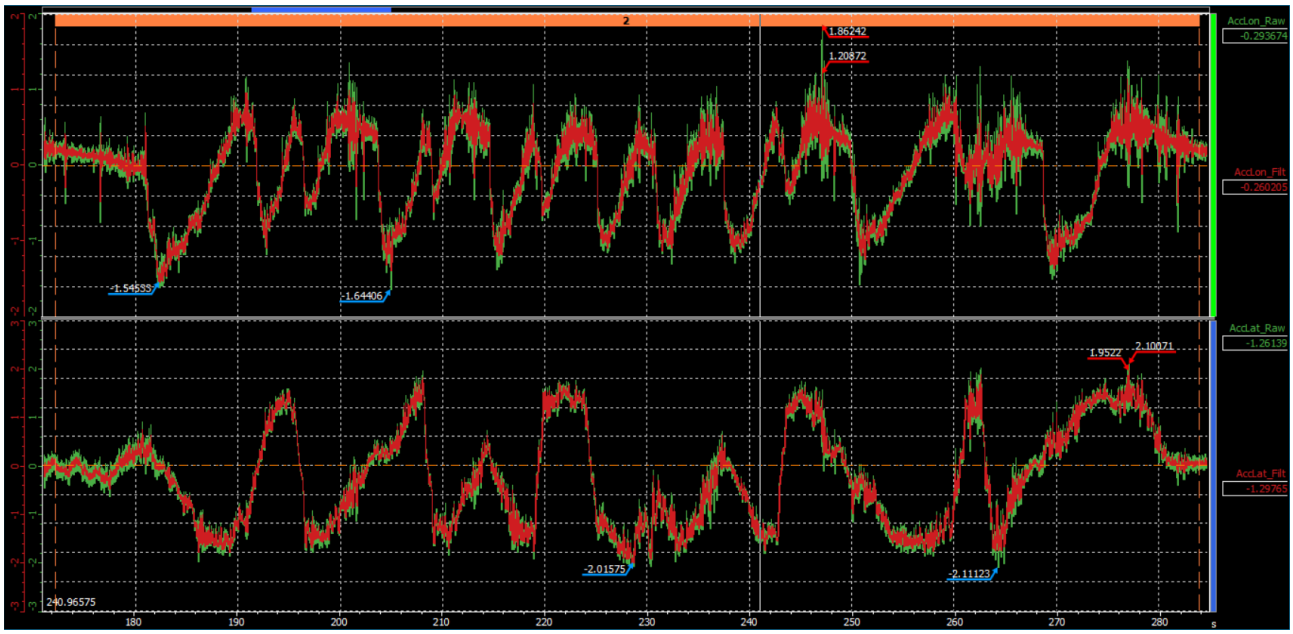


Figure 6.3: Longitudinal (upside) and Lateral (downside) acceleration data relative to lap 2, with raw signals in green and filtered signals in red

We observed that using filtered acceleration data is significantly better. Racing motorcycles are subject to intense, high-frequency vibrations from the engine, chain slap, and track surface imperfections. Raw accelerometer data is heavily corrupted by this noise, making it unreadable. The accelerometer captures every microscopic mechanical vibration. In the raw lateral acceleration, we can see a thick, chaotic band of noise. This makes it impossible to pinpoint when the rider initiates a lean or reaches the apex. The same applies to longitudinal acceleration: high-frequency noise masks the exact moments of initial braking and throttle application.

We observed three different cases regarding the max lateral acceleration registered during the eight laps:

- Lap 1, the max lateral acceleration registered is at **turn 12**, this difference with the other laps is probably due to the standing start that caused minor speeds and accelerations during the first lap, in fact the value of maximum acceleration for lap 1 (1.62g) is the lowest among the eight laps.

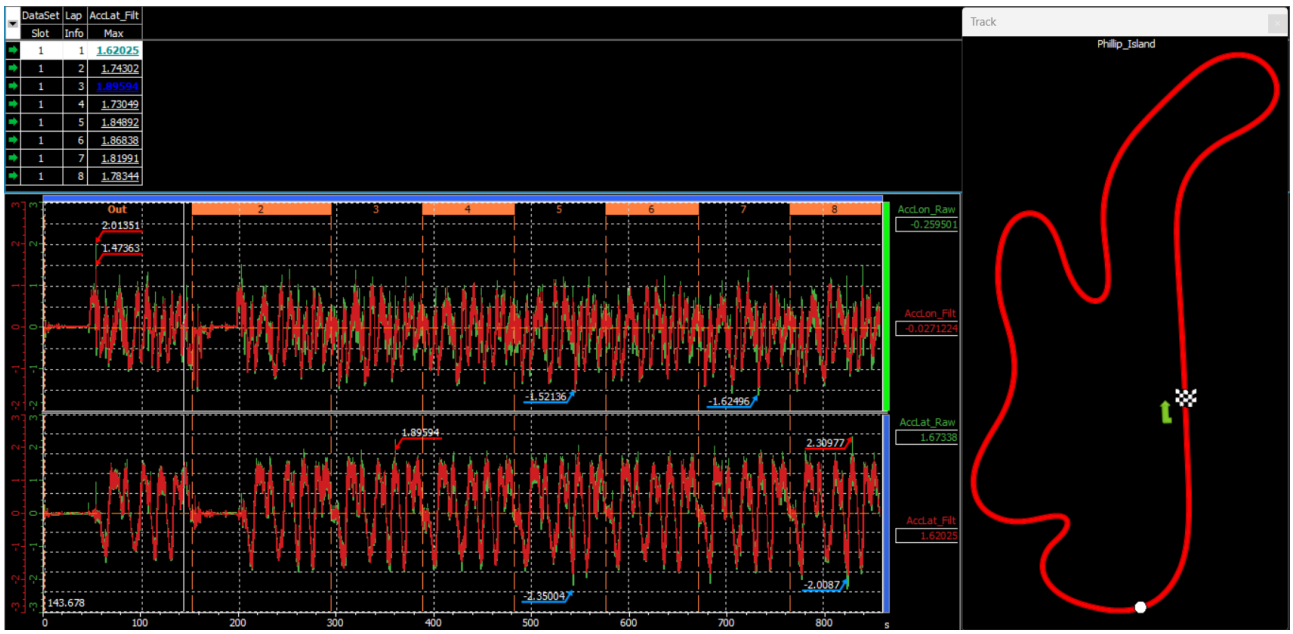


Figure 6.4: Lap report on the lateral acceleration channel for lap 1

- Lap 2,3,4,5,7,8 the max lateral acceleration registered is at **turn 9**, during lap 3 the max acceleration among all the laps was registered

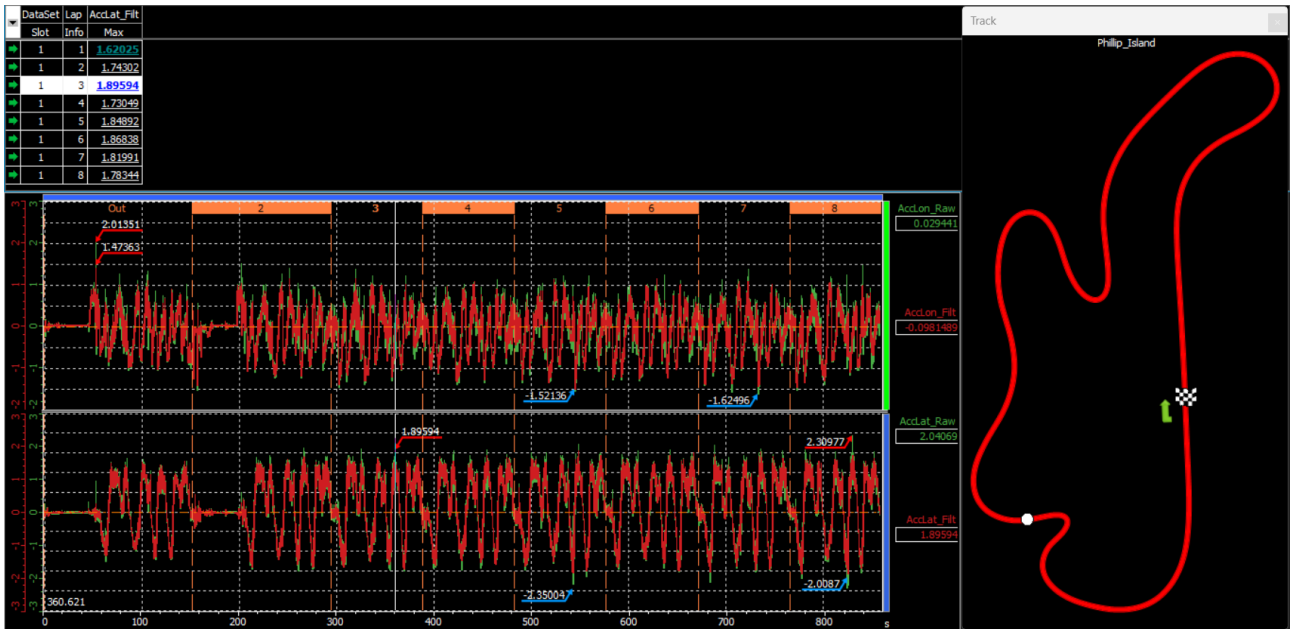


Figure 6.5: Lap report on the lateral acceleration channel for lap 3

- Lap 6, the max lateral acceleration registered is at **the beginning of the straight**

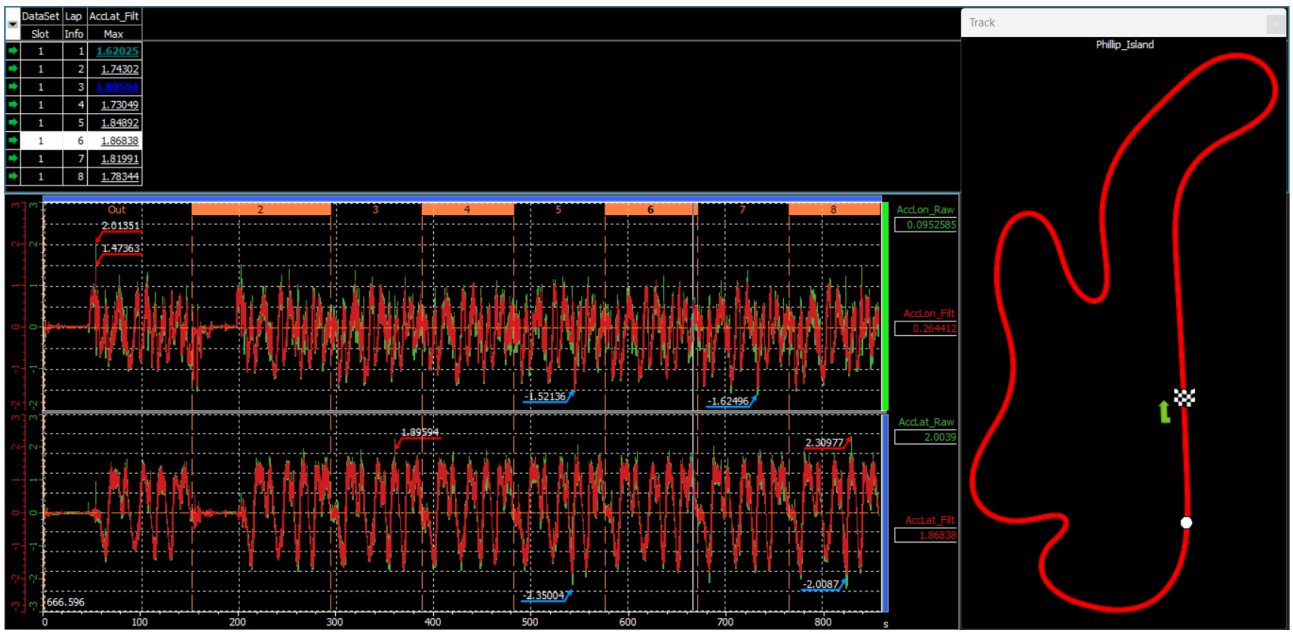


Figure 6.6: Lap report on the lateral acceleration channel for lap 6

7 LAB 6 – Advanced Lap-Time Simulation Modelling

7.1 Introduction

Modern motorsport championships increasingly regulate not only the power output of a car but also the total energy it may deploy over a single lap. Formula E is the obvious example, but even traditional single-seater formulae are moving in the same direction. In this context, lap-time simulation is no longer just a tool for finding the fastest racing line: it becomes an energy management problem, where the engineer must decide how to spread a finite energy budget around the circuit so that as little lap time as possible is lost.

This laboratory session explores exactly that question. Starting from the minimum lap-time problem (MLTP) already introduced in previous labs, an energy budget constraint is added to the optimal control formulation and three budget levels are compared on the Barcelona-Catalunya circuit: 15 MJ, 12 MJ, and 10 MJ. The 15 MJ case turns out to be effectively unconstrained – the car's free optimum consumes less – and therefore acts as the performance reference throughout. The two tighter budgets then reveal how the optimal strategy changes as the constraint bites harder.

All simulations use the direct-collocation MLTP toolbox provided for the course, with CasADi handling the symbolic algebra and IPOPT solving the resulting nonlinear programme.

7.2 Theoretical Background

7.2.1 Optimal Control Problem Formulation

The core idea of the MLTP is to recast the question "what is the fastest way to drive this car around this circuit?" as a mathematical optimisation problem. The state of the car, the driver inputs, and the vehicle equations of motion are all written down explicitly, and a numerical solver finds the input history that minimises the total lap time while satisfying every constraint. One practical subtlety is the choice of independent variable. Using time t would make the total lap duration appear in the integration limits, complicating the problem. Instead, arc-length s along the track centreline is used, which converts a free-final-time problem into a fixed-domain one. The objective becomes

$$\min_{\mathbf{u}(s)} \; T_{\text{lap}} = \int_0^{s_f} \sigma(s)\,\mathrm{d}s, \qquad \sigma = \frac{1 - n\kappa}{v_x \cos\varepsilon - v_y \sin\varepsilon}, \tag{7.1}$$

where $\sigma = \mathrm{d}t/\mathrm{d}s$ is simply the time cost per metre of track, n is the lateral deviation from the centreline, $\kappa(s)$ the local curvature, and ε the heading-angle error, all expressed in the Frenet–Serret frame. The optimisation is subject to the vehicle dynamics $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{u}, \kappa(s))$, path constraints $\mathbf{h}(\mathbf{x}, \mathbf{u}) \geq \mathbf{0}$, periodic boundary conditions that close the lap, and the global energy constraint

$$E_{\text{tot}} = \int_0^{T_{\text{lap}}} P_{\text{motor}}(t)\,\mathrm{d}t \;\leq\; E_{\max}. \tag{7.2}$$

This last constraint is what distinguishes this lab from the baseline MLTP: a single inequality linking the entire lap limits the total energy the motor may consume.

7.2.2 Vehicle Model

The car is described by a planar bicycle model with seven states:

$$\mathbf{x} = \begin{bmatrix} v_x,\ v_y,\ r,\ n,\ \varepsilon,\ \Omega_f,\ \Omega_r \end{bmatrix}^\top, \tag{7.3}$$

where v_x and v_y are the body-frame velocities, r the yaw rate, n the lateral position on track, ε the heading error, and $\Omega_{f,r}$ the wheel angular speeds. There are three control inputs – drive torque T_{drive} (rear axle only), brake torque T_{brake} (split 62.5/37.5 front/rear), and steering angle δ – collected in $\mathbf{u} = [T_{\text{drive}},\ T_{\text{brake}},\ \delta]^\top$. Tyre forces are generated by the Pacejka Magic Formula:

$$F = D \sin\Big(C \arctan\big(Bs - E(Bs - \arctan(Bs)) \big) \Big), \tag{7.4}$$

with s being the slip ratio or slip angle depending on the direction. An important detail is that the peak friction coefficient is not constant but *load-dependent*:

$$\mu(F_z) = \mu_0 + p_{D2}\,\frac{F_z - F_{z0}}{F_{z0}}, \qquad \mu_0 = 1.4, \quad p_{D2} = -0.3, \quad F_{z0} = 6000\,\text{N}. \tag{7.5}$$

Because p_{D2} is negative, a lightly loaded tyre actually grips *better* than the reference value μ_0. This turns out to matter for interpreting the results: the front and rear axle loads of this 605 kg car (2374 N and 3561 N respectively) are well below the reference load of 6000 N, so the real friction limit during the lap sits around $\mu \approx 1.50$–1.58 – noticeably above the $\mu_0 = 1.4$ circle that will be drawn on the friction-circle plots.

Aerodynamic forces follow the standard quadratic model with $C_D = 0.75$, $C_L = 1.70$, and $A = 1.0\,\text{m}^2$. The full set of vehicle parameters is listed in Table 7.1.

Table 7.1: Key vehicle parameters (`Electric_formula.m`).

Parameter	**Symbol**	**Value**	**Unit**
Total mass	m	605	kg
Wheelbase	l	2.70	m
Weight bias (front)	w_B	0.40	–
CoG height	h_{cg}	0.30	m
Wheel radius	R_w	0.275	m
Wheel inertia	J_w	1.80	kg m^2
Max drive torque	$T_{\text{drive,max}}$	3125	N m
Max motor power	P_{max}	300	kW
Max brake torque	$T_{\text{brake,max}}$	12 000	N m
Brake balance	β	0.625	–
Tyre μ_0	μ_0	1.40	–
Load sensitivity	p_{D2}	−0.30	–
Ref. tyre load	F_{z0}	6000	N

7.2.3 Direct Legendre Collocation

Solving the OCP analytically is not feasible for a real circuit, so it is transcribed into a large but finite-dimensional nonlinear programme (NLP) using *direct Legendre collocation*. The arc-length domain is split into $N = 1551$ intervals of $\Delta s = 3\,\text{m}$, and within each interval the state trajectory is approximated by a cubic polynomial. The polynomial coefficients are pinned by enforcing the equations of motion at three Legendre–Gauss–Radau (LGR) quadrature nodes per interval. The result is an NLP with roughly 51 000 decision variables and 56 000 constraints,

which IPOPT solves using the MUMPS sparse linear solver. The energy inequality (7.2) enters as a single additional constraint, discretised via the same LGR weights:

$$E_{\text{tot}} = \sum_{k=0}^{N-1} P_{k+1} \left(\mathbf{Q}_k \, \mathbf{B}\right) \Delta s_k \;\leq\; E_{\max}, \tag{7.6}$$

where $\mathbf{Q}_k$ evaluates power at the collocation nodes of interval k and $\mathbf{B}$ contains the LGR quadrature weights.

7.3 Simulation Setup

The circuit geometry comes from `Barcelona_circuit.mat`, which provides the arc-length s and curvature $\kappa(s)$ arrays together with Cartesian coordinates for plotting. The initial longitudinal speed is fixed to $v_{x,0} = 67\,\text{m/s}$ and the final lateral position to $n_f = 3.3\,\text{m}$ to close the lap correctly on the pit straight. The main collocation and solver settings are collected in Table 7.2.

Table 7.2: Collocation and solver options (`userOpts.m`).

Option	**Value**	**Note**
Spatial step Δs	3 m	$N = 1551$ intervals
Polynomial degree d	3	LGR quadrature
Input interpolation	linear	Between knot points
`max_iter`	1000	IPOPT iterations
`acceptable_tol`	10^{-1}	$100\,000\times$ default

One setting that deserves a comment is `acceptable_tol` = 10^{-1}. The IPOPT default is 10^{-6}, so this is five orders of magnitude more lenient. In practice it allows the solver to stop at an "Acceptable Level" solution well before a certified optimum is reached, which cuts the computation time substantially. The lap times reported in the next section are not significantly affected, but the looser tolerance does allow a handful of path constraints – including the friction circle – to be marginally violated at isolated points. The three cases are run in sequence from the loosest to the tightest budget (15 MJ → 12 MJ → 10 MJ), with each solution warm-starting the next.

7.4 Results

7.4.1 Lap Times and Energy Consumption

The headline numbers are in Table 7.3 and Figure 7.1. The free optimum uses 14.57 MJ, so the 15 MJ budget is never reached and the 15 MJ case is effectively unconstrained. From there, tightening the budget to 12 MJ costs just 0.29 s – barely a quarter of a second over a lap of more than 100 s. Going further to 10 MJ costs an additional 0.56 s, bringing the total penalty to 0.85 s. The non-linearity is clear: each megajoule removed is more expensive than the one before it, and the rate roughly doubles when moving from the 15-to-12 step to the 12-to-10 step. The reason for this acceleration will become clearer when looking at the power traces.

Table 7.3: Lap-time and energy results for the three energy scenarios.

Case	Lap time [s]	Energy used [MJ]	Energy limit [MJ]	ΔT vs 15 MJ [s]
15 MJ	100.190	14.571	15	0.000 (ref.)
12 MJ	100.479	12.053	12	+0.289
10 MJ	101.040	10.058	10	+0.850

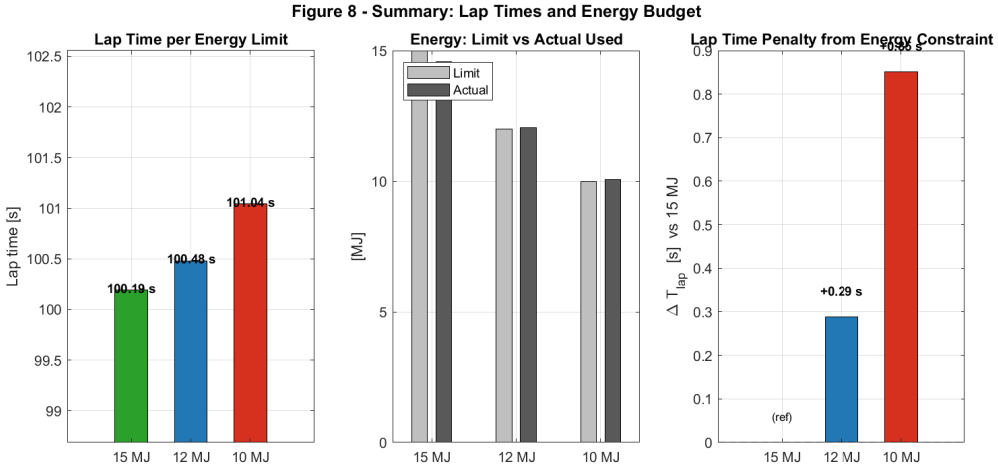


Figure 7.1: Summary of the three scenarios: lap time (left), energy limit vs. actual consumption (centre), and lap-time penalty relative to the 15 MJ reference (right).

7.4.2 Optimal Racing Lines

Looking at Figure 7.2, the three racing lines are almost on top of each other for the majority of the circuit. On the main straight and through the high-speed sections the geometry is essentially identical: reducing the energy budget changes throttle application, not where the car is on the track. Small differences emerge at the slower chicanes and hairpins, where the 10 MJ case takes a slightly tighter entry to reduce motor demand at corner exit, but these deviations are modest.

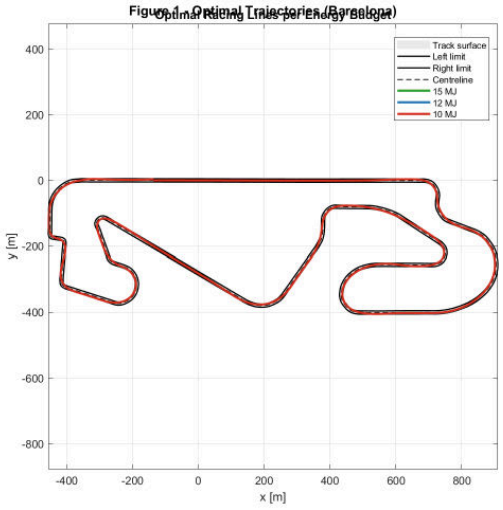


Figure 7.2: Optimal racing lines for the three energy budgets overlaid on the Barcelona-Catalunya circuit. Green: 15 MJ; blue: 12 MJ; red: 10 MJ.

7.4.3 Speed Profiles

The speed comparison in Figure 7.3 tells the same story in more detail. The 12 MJ profile is nearly indistinguishable from the reference: the speed deficit barely registers except at a few corner exits. The 10 MJ case is a different matter. Deficits of up to 12 km/h appear at slow-corner exits, and the top speed on the final straight is measurably lower. Importantly, the deficit is spread around the entire lap rather than being concentrated in one place, which already hints at the energy management strategy that will emerge in the power traces.

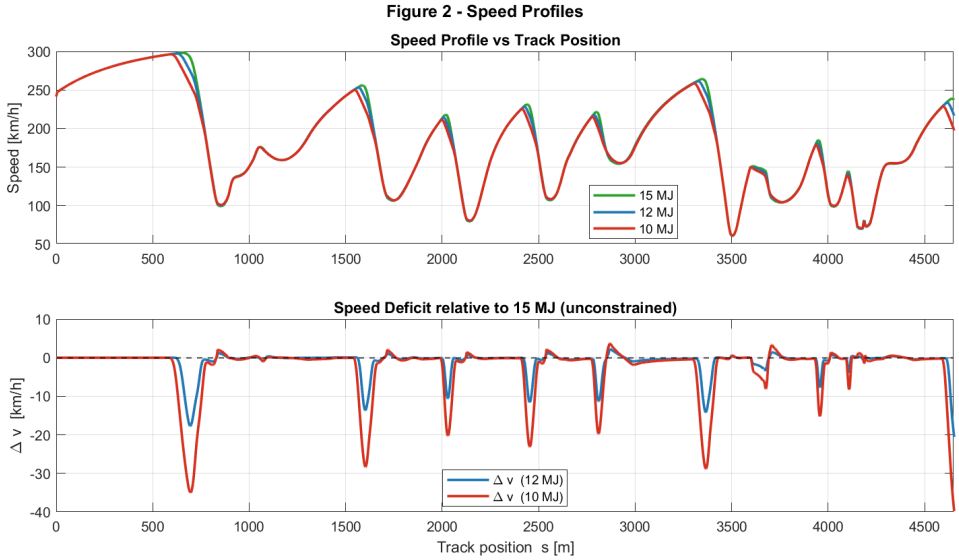


Figure 7.3: Speed profiles (top) and speed deficit relative to the 15 MJ reference (bottom) vs. track position. Negative values indicate where the energy-limited car is slower.

7.4.4 Driver Inputs

Figure 7.4 shows the throttle, brake, and steering histories. The throttle is normalised to account for the torque-to-power crossover that occurs at $v_{\text{lim}} = P_{\max} R_w / T_{\text{drive,max}} \approx 26\,\text{m/s}$: below that speed the motor is torque-limited, so the normalised value is $T_{\text{drive}}/T_{\text{drive,max}}$; above it the motor is power-limited and the normalisation becomes $T_{\text{drive}} v_x / (R_w P_{\max})$. The 10 MJ trace never fully saturates the motor for long: full-throttle bursts are shorter and partial-throttle plateaus appear at almost every corner exit.

Braking is virtually identical across all three cases, as expected – the energy constraint only touches the motor. Steering is also unchanged, consistent with the similar racing lines seen in Figure 7.2.

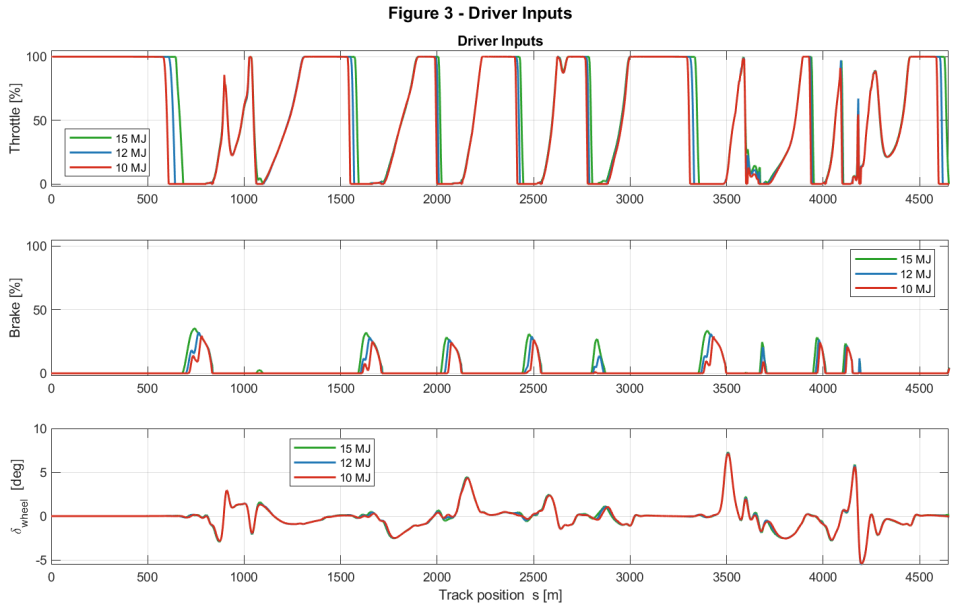


Figure 7.4: Normalised throttle (%), brake (%), and road-wheel steering angle (°) vs. track position for the three energy budgets.

7.4.5 Motor Power and Cumulative Energy

Figure 7.5 is perhaps the most informative of all the plots because it shows directly how the solver has chosen to use the available energy. In the 15 MJ case the motor hits $P_{\max} = 300\,\text{kW}$ on every significant straight – the car is driving flat-out whenever traction allows. The 12 MJ case mirrors this almost exactly, with only minor reductions at a few locations. The 10 MJ case looks qualitatively different. Rather than a few large cuts, the power is trimmed everywhere: the cumulative energy curve tracks just below the 10 MJ limit from the very first corner all the way to the finish line. This is the signature of a distributed energy management strategy, and it is optimal precisely because it spreads the performance cost as evenly as possible rather than accepting a large deficit in one sector.

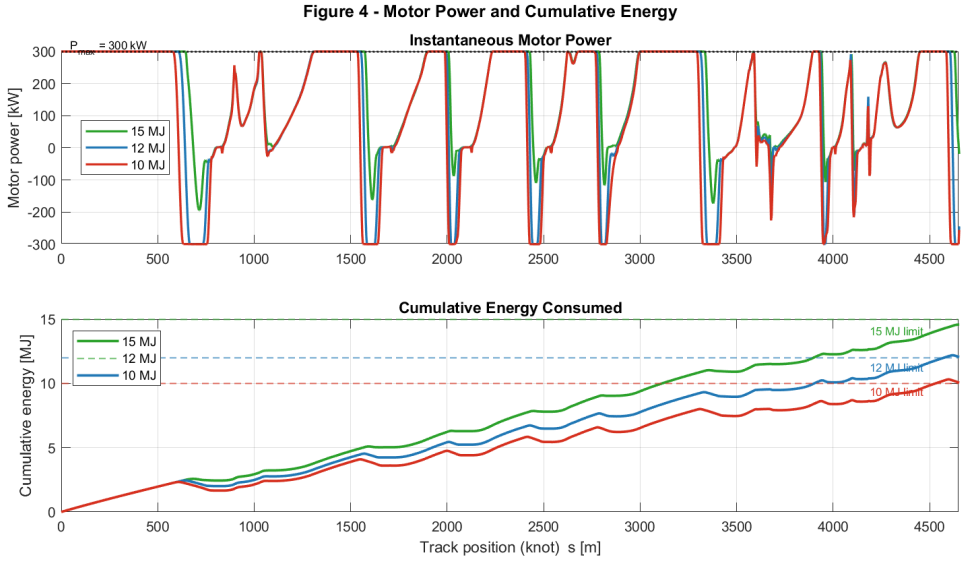


Figure 7.5: Instantaneous motor power (top) and cumulative energy consumption (bottom) vs. track position. Dashed horizontal lines mark the energy limits; the dotted line marks $P_{\max} = 300\,\text{kW}$.

7.4.6 Tyre Friction Utilisation

Figure 7.6 plots the combined lateral and longitudinal force magnitude relative to the vertical load, $|F_{xy}|/F_z$, at every knot point. At first glance it may look as though the tyres are consistently exceeding the friction limit: both axles regularly show values above 1.4. This is not a violation, however. Recall from Equation (7.5) that $\mu_0 = 1.4$ is the friction coefficient at the reference load $F_{z0} = 6000\,\text{N}$, and that the actual friction limit is higher when the tyre is lightly loaded. Since the car's axle loads sit well below 6000 N, the real limit during the lap is $\mu \approx 1.52$–1.58, and that bound is respected.

The main difference between the three cases visible here is on the rear axle: the 10 MJ case (red) shows lower utilisation on the exits of straights, where the motor is deliberately de-rated to stay within the energy budget.

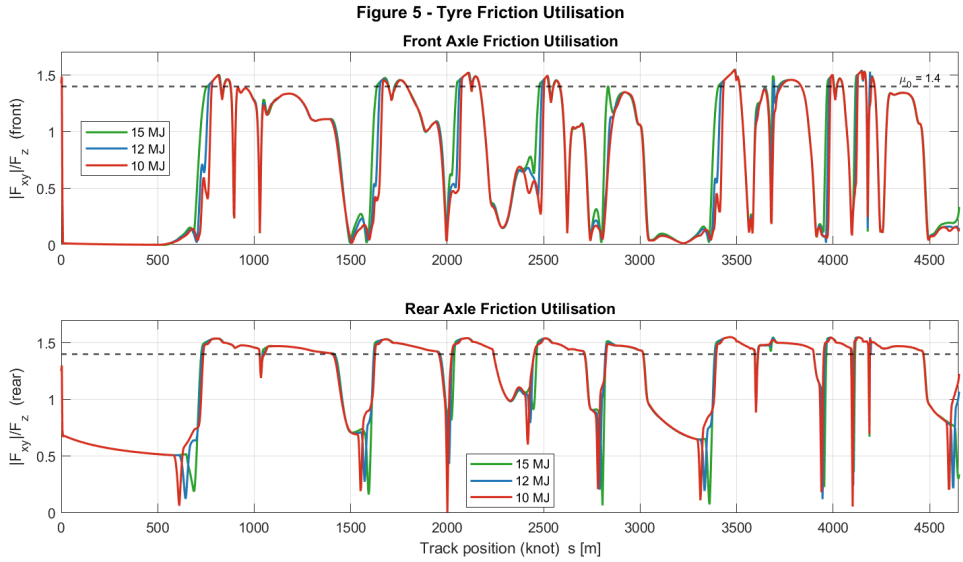


Figure 7.6: Tyre friction utilisation $|F_{xy}|/F_z$ for front (top) and rear (bottom) axles vs. track position. The dashed line marks the nominal $\mu_0 = 1.4$; values above it are physically admissible due to load sensitivity.

7.4.7 Tyre Friction Circles

The friction circles in Figure 7.7 reinforce the same point in a different representation. The front axle cluster (left panel) is almost entirely vertical – no drive torque, modest braking, so the front is working almost purely in the lateral direction. The rear axle (right panel) spans all four quadrants: it must handle combined drive, brake, and lateral demands depending on where on the circuit the car is. Points that fall outside the $\mu_0 = 1.4$ reference circle are not constraint violations; they lie within the true, load-dependent friction envelope. The three cases share essentially the same scatter shape, confirming that the energy constraint does not fundamentally change the way the tyres are being used – it just adjusts the motor output.

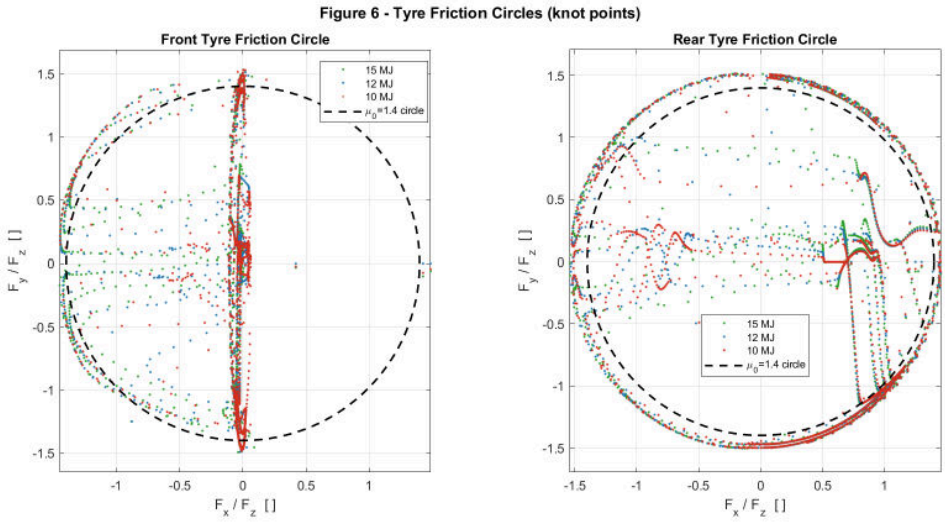


Figure 7.7: Friction-circle scatter for front (left) and rear (right) axles. The dashed circle is $\mu_0 = 1.4$; points outside it are admissible because the load-dependent friction limit is larger at this car's operating loads.

7.4.8 Tyre Forces

Figure 7.8 shows the individual force components over the lap. Peak front braking force reaches around 8 kN at the main braking zones – turn 1 near $s \approx 650$ m and the chicane around $s \approx 2000$ m. At the rear, peak longitudinal force during acceleration is roughly 7 kN. As expected, the 10 MJ case shows consistently lower rear longitudinal forces on corner exits, which is just the force-level view of what was already visible in the throttle trace. Lateral forces are nearly the same across all three cases, further confirming that the racing line geometry has not changed.

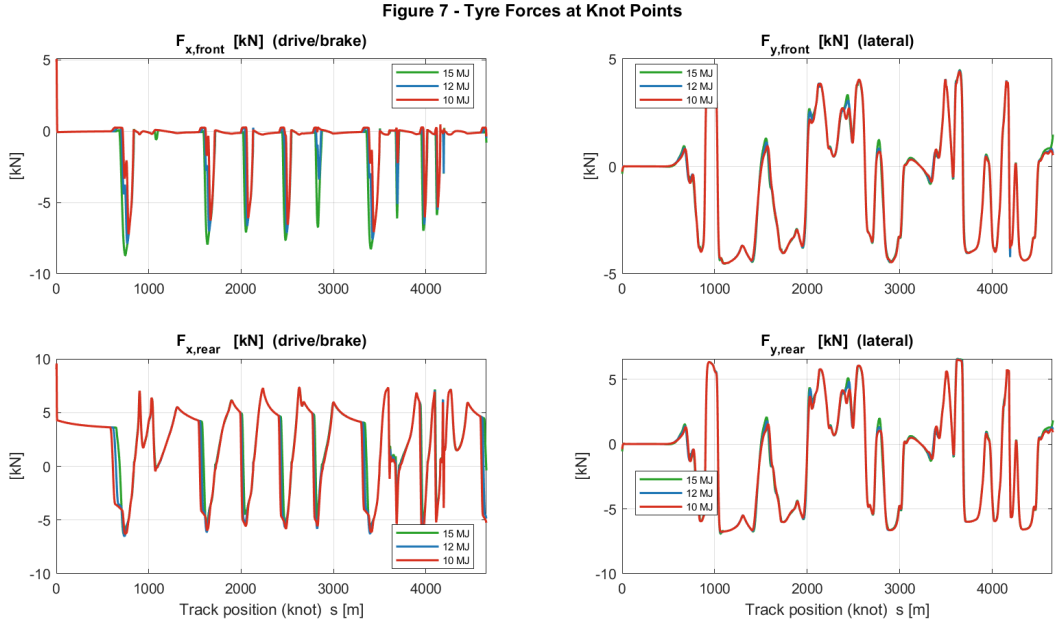


Figure 7.8: Tyre forces at knot points: longitudinal and lateral components for front and rear axles, with the three energy cases superimposed.

7.5 Discussion and Conclusions

The lap-time penalty grows faster than linearly as the budget tightens: saving 2.57 MJ (from 14.57 to 12 MJ) costs only 0.29 s, while the next 2 MJ costs 0.56 s – nearly twice as much per megajoule. The reason is that the free optimum already leaves power on the table at slow corners where traction, not motor output, is the binding constraint. Trimming the budget from 14.57 to 12 MJ targets those soft spots first, with little lap-time consequence. Once they are gone, any further cut bites into corner-exit acceleration, where the sensitivity of lap time to motor output is high, and the penalty rate doubles.

The cumulative energy curves tell the same story from a different angle: the 12 MJ case follows the unconstrained trajectory for most of the lap and only bends towards its limit in the final sector, whereas the 10 MJ curve is pressed against its ceiling from the very first corner. These are two qualitatively different strategies – selective lift-and-coast versus distributed de-rating – and the distributed one is optimal precisely because spreading the energy cut evenly avoids large, localised speed deficits that would be disproportionately expensive in lap time.

One result that can look alarming but is not is the scatter outside the $\mu_0 = 1.4$ friction circle: those points are not constraint violations but a direct consequence of the load-sensitive tyre model, which assigns a higher friction limit to a lightly loaded tyre. At this car's operating axle loads the real boundary sits around $\mu \approx 1.55$, and it is never exceeded. The handful of points that visually breach even that threshold are artefacts of the relaxed solver tolerance and would disappear with a tighter convergence criterion. Overall, the direct-collocation framework handles the energy constraint naturally – a single global inequality is all that is needed, and the solver distributes the power reduction optimally without any manual intervention.

8 Seminars

8.1 Race Track Engineering: Management, Preparation and Strategy

Speaker: Marco Fainello, an experienced mechanical engineer with 30+ years in motorsport and automotive industries.
Topic: Practical lessons on mindset, career strategy, teamwork, and real-world engineering, the skills that university rarely teaches.

8.1.1 Industry Context and Career Orientation

Motorsport and automotive are fundamentally different environments. Within motorsport, Formula 1 prioritises maximum performance and operates at an extremely fast development pace, while endurance racing demands reliability and adaptability over long stints. Choosing a direction early is useful, but the key distinction the speaker emphasised is that what works in one context may be entirely wrong in another.

8.1.2 Attitude and Competition

Technical competence is a baseline: nearly all engineers in the field are technically capable. As illustrated in Figure 8.1, the larger and less visible foundation that determines long-term success is attitude, resilience, and the capacity to keep performing under failure and pressure. *The bottleneck is rarely knowledge, it is behavior.*



Figure 8.1: The engineer iceberg: technical skills are the visible tip, while attitude, resilience, and adaptability form the dominant base.

Motorsport is inherently competitive and losing is far more frequent than winning. Two dynamics can emerge within a team (Figure 8.2): positive competition, where teammates push each other to improve collectively, and negative competition, where individuals undermine

colleagues to appear superior. The latter destroys team performance regardless of individual talent.

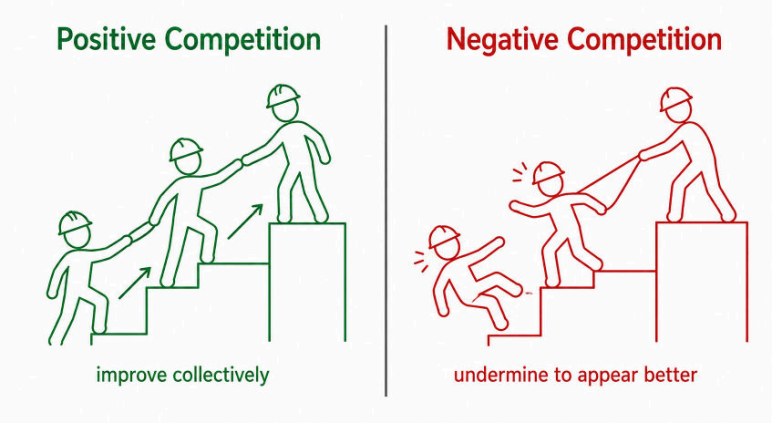


Figure 8.2: Positive competition drives collective growth; negative competition is destructive and must be actively avoided.

8.1.3 Teamwork, Communication and Leadership

No single engineer wins a race alone. Outcomes depend on the simultaneous contributions of many people, and hoarding knowledge is counterproductive. As shown in Figure 8.3, three human dimensions shape how teams actually function: culture (implicit assumptions shaped by background), communication (messages are frequently misunderstood even in the same language), and attitude (individual motivations and fears). Effective collaboration requires all three to be aligned, especially in the multicultural environments standard in international motorsport.

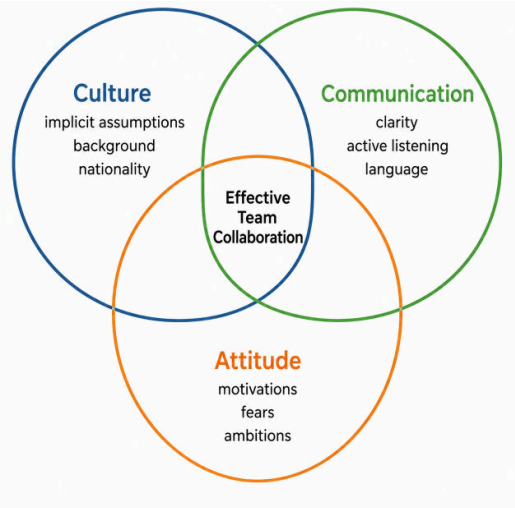


Figure 8.3: Culture, communication, and attitude are the three human factors that determine team effectiveness.

Leadership does not always come from the highest authority, but from whoever coordinates people effectively toward the goal. This requires clear objectives broken into subtasks, continuous monitoring, conflict management, and ensuring every team member understands how their work connects to the overall result.

8.1.4 Engineering Practice, Preparation and Modes

Real engineering problems are rarely clean or predictable. Decision-making under incomplete data, conflicting signals, and time pressure is the norm, and it requires combining quantitative data, simulation outputs, and qualitative inputs such as driver feedback. Modern tools like machine learning and simulation are powerful but cannot substitute for physical intuition and engineering judgment.

Success in a race environment depends on preparation rather than reaction: anticipating failure modes, planning contingencies in advance, and managing resources effectively. A key conceptual distinction is the difference between **test mode**, where experimentation and mistakes are expected, and **performance mode**, where reliable execution is critical. Confusing the two leads either to missed learning or to costly failures at critical moments.

8.1.5 Conclusion

The seminar's core argument is that the non-technical dimensions of engineering work, namely mindset, teamwork, communication, adaptability, and preparation, are what separate elite engineers from merely competent ones. These are rarely taught in university but are decisive in a competitive, fast-moving environment like motorsport.

8.2 Introduction to Racing Tires

Speaker: Davide Verga, a mechanical engineer who studeid at Politecnico di Milano and now working in the R&D department for cars and motorsport at Goodyear.
Topic: This seminar was focus on racing tires covering the design, their differences with road tires and the tire track data analysis.

8.2.1 Differences between road and racing tires

Racing tires operate in a completely different physical realm compared to road tires. While a standard sedan tire manages roughly 1g of lateral acceleration and peak forces of 7,000N, a racing tire must withstand over 2g and forces exceeding 13,000N. The lifecycle is also vastly different: a road tire is designed for 50,000 km, whereas a racing slick may only last between 100 and 600 km.

	Sedan	Racing
Power [CV]	200	6/700
Mass [kg]	1800	1300
Max. Lat. Acc. [g]	1	> 2
Max. Speed [km/h]	200	300
Lap time	2'35''	1'51''
Fy peak (single tire) [N]	6/7000	13000
Power peak (single tire) kW]	2-400	7-800

Figure 8.4: differences between a Sedan and a Racing car

The two types of tires are designed for different purposes:

- Racing tires have no grooves on the tread to maximize the contact area, the compound has to maximize the grip, the crown area is optimized to provide stiffness and high-speed resistance while the sidewalls have high stiffness to have steering precision and fast response.
- Road tires, instead, have tread with grooves to divide stiffness (longitudinal/lateral)and to evacuate water. The compound has a trade-off between wet, RR and wear and the crown area is optimized to last thousand of kms.

8.2.2 Tire shape design

A squarer footprint implies more direct engagement of external rib, Fy peak reached at lower slip angle and consequently better handling, dry traction, high speed stability, better braking performance and uniform wear. On the other hand with a rounder shape external ribs are gradually engaged and Fy peak is found at a larger slip angle, this implies better comfort, better wet behaviour and less noise.

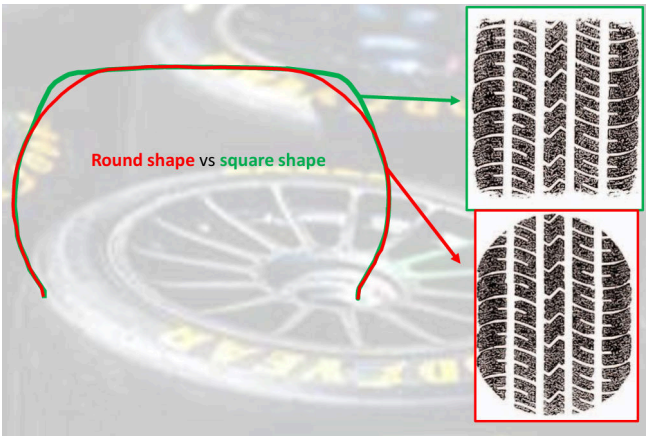


Figure 8.5: Tire footprints for round and square shapes

Tire reinforcement tensions are related to the tire's shape: ply tension is proportional to r1 (meridian radius of curvature) while belt tension decreases when r1 and sin(θ)increase.

8.2.3 Rubber compound

The main elements that form the tire compound are:

- Elastomer ,it is the backbone of the compound
- Filler, gives higher stiffness and better resistance against tear (properties of the vulcanized product)
- Oil, gives flexibility and processing aid
- Resins cocktail, to push the grip over a high temperature range
- Antidegradants, to keep the performance
- Sulfur, to make bonds between polymer (higher stiffness) and to cure system
- Accelerators, to cure the system

The rubber compound is viscoelastic: has an intermediate behaviour between an elastic solid and a fluid. The modulus of the material is the sum between the real elastic modulus *E'* and the imaginary viscous modulus *E"*.
$E^* = E' + E"$

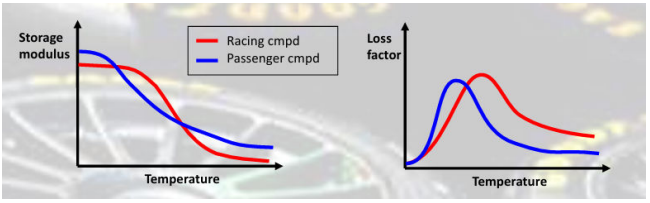


Figure 8.6: Storage modulus and loss energy comparison between racing and passenger compunds

The glass transition temperature is higher in racing compounds; they also use specific resins to shift the "Tan Delta" (or loss energy) peak to higher temperatures (80°C - 120°C) where maximum grip is required. The carbon black in the racing compounds dissipates more energy at high temperatures.

8.2.4 Tire track data analysis

Maximizing tire performance requires careful monitoring of telemetry data, different parameters are monitored:

- **Grip**, grip factors are overall grip, braking grip, acceleration grip, cornering grip, aerodynamic grip.
- **Inflation pressure and air temperature evolution**, tire pressure monitoring systems give inflation pressure but also air, rim and inner temperature.
- **Surface temperature**
- **Camber evaluation**, corner by corner camber evaluation with temperature sensors

8.3 Sustainability in Motorsport Engineering

Speaker: Emanuele Agnoli, Marelli Motorsport, Quality Manager and former Materials & Process Engineer.
Topic: Foundations of sustainability and their practical application to motorsport products and operations.

8.3.1 Fundamentals of Sustainability

Sustainability was defined by the UN in 1987 as meeting the needs of the current generation without compromising future generations. It rests on three pillars that must be balanced simultaneously: environmental protection, social well-being, and economic viability (Figure 8.7). The UN also set 17 Sustainable Development Goals (Figure 8.8) as a concrete roadmap. A common misconception is to reduce sustainability to only its environmental dimension.

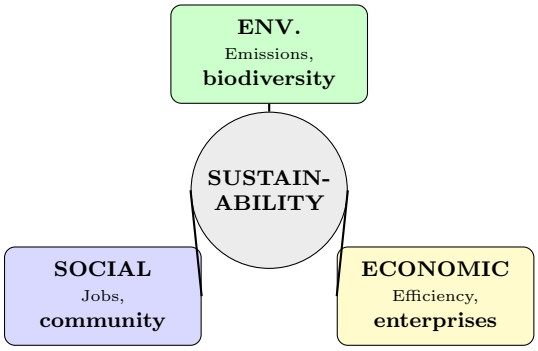


Figure 8.7: The three pillars of sustainability.



Figure 8.8: The 17 UN Sustainable Development Goals.

The **ecological footprint** measures the biologically active area required to sustain humanity's resource demand, while **biocapacity** is what the planet can actually supply. The current footprint exceeds biocapacity globally, with countries like the USA requiring the equivalent of 5 planets. The **Earth Overshoot Day** (August 1 in 2024) marks when the annual budget is exhausted and is advancing earlier each year.

8.3.2 GHG Emissions, Targets and Reporting

The greenhouse gas effect is the most critical global impact mechanism. All GHG emissions are normalised to CO_2 via the **Global Warming Potential (GWP)**, for example methane has GWP = 11, and expressed in CO_2-equivalent (CO_2e). The Paris Agreement targets limiting warming to below 2°C, ideally 1.5°C, through the Science-Based Targets initiative (SBTi). Companies distinguish between **carbon neutrality** (offsetting emissions) and the more demanding **net-zero** (eliminating them across the full supply chain).

The GHG Protocol classifies emissions into three scopes (Figure 8.9): Scope 1 (direct), Scope 2 (purchased energy), and Scope 3 (full supply chain). Scope 3 is typically the largest and most overlooked, meaning material selection and supplier choices have a direct impact on a product's total carbon footprint. **ESG** (Environmental, Social, Governance) is the framework that turns these commitments into auditable, time-bound targets backed by annual sustainability reports. Without this, sustainability risks becoming **greenwashing**: marketing without substantive action.

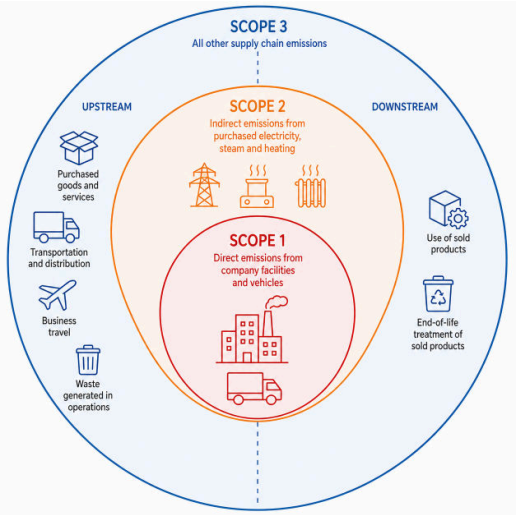


Figure 8.9: GHG Protocol emission scopes. Scope 3 covers the full supply chain and is the largest category for most companies.

8.3.3 Sustainability in Motorsport

Motorsport is often perceived negatively on environmental grounds. However, as Figure 8.10 shows, fuel combustion during racing represents less than 1% of Formula One's total 2022 carbon footprint; the dominant contributions are logistics and event operations. This illustrates the critical importance of evidence-based decision making: eliminating combustion from cars would have a negligible effect without acting on transport and energy. At Marelli, the entire motorsport department generated only 5 kton CO_2e in 2023, versus 4,605 kton for the whole company (0.1%).

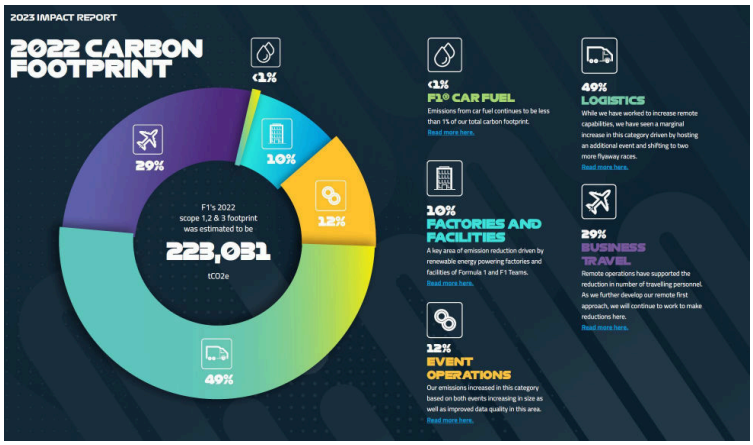


Figure 8.10: F1 2022 carbon footprint: logistics dominate; racing fuel accounts for less than 1% (source: FIA / F1 2023 Impact Report).

Motorsport's most significant sustainability contribution is as a **technology accelerator**: by maximising efficiency under fixed energy budgets, it drives innovations that transfer to road vehicles, including ABS, hybrid powertrains, regenerative braking, composites, and synthetic fuels. The FIA Environmental Accreditation Programme (3-level, based on ISO 14001) coordinates sustainability across all stakeholders; 3* accreditation is mandatory for World Championship teams by 2025.

8.3.4 Design for Sustainability and Case Studies

When developing new products, sustainability is integrated at the design phase using the framework in Figure 8.11, which maps improvement levers across five life cycle phases: material sourcing, manufacturing, transportation, use phase, and end of life.



Figure 8.11: Design for Sustainability framework applied across the product life cycle.

Three motorsport products illustrate this framework in practice (Figure 8.12). The **CFRP chassis** is lightweight but energy-intensive to produce and nearly impossible to recycle due to the thermoset resin matrix; thermoplastic alternatives and natural fibre (lignin) inclusions are the main mitigation paths. The **VCU** involves complex electronics with immature end-of-life recycling; the primary strategy is cascade reuse, redeploying units from F1 to lower championships, combined with modular design to replace individual boards. The **MGU** depends on rare earth magnets and copper windings with high extraction costs; alternative magnet materials (ferrite, alnico) and modular component replacement reduce this impact.



Figure 8.12: Case studies: CFRP chassis (left), Vehicle Control Unit (centre), and Motor Generator Unit (right).

8.3.5 Conclusion

Sustainability in motorsport operates at two levels: reducing direct operational impacts, where logistics and energy are the priority rather than fuel combustion, and leveraging motorsport as an innovation platform to accelerate efficiency technologies at automotive scale. For engineers, the practical tool is the Design for Sustainability framework, applied from the earliest design stage. The boundary between genuine sustainability and greenwashing is whether commitments are backed by data, targets, and transparent reporting.

8.4 Vehicle Performance in Motorsport: Scuderia Ferrari

Speakers: Luca de Bernardi – Vehicle Dynamics (Race Performance) deputy Team Leader at Ferrari Gestione Sportiva.
Topic: Vehicle performance in an F1 car starting from the objectives of a race weekend, analyzing car subsistems to arrive to the optimal setup using different simulation tools.

8.4.1 F1 car subsystems

An F1 car is made of different and numerous subsystems: **Aerodynamics, PU and gearbox, Electronics, Suspensions, Brakes, Tires, Driver**...
All subsystems are integrated and work together, a performance engineer must have knowledge of each subsystem and must take into account the relationship between the subsystems to comprehend the behaviour of the whole car.
Not all subsystems reach their optimum under the same conditions, so we have to accept some trade-offs, and it is not easy to find the best compromise to under the best performance of the car

8.4.2 Objectives

The final goal for an F1 car is to **WIN**. That means having the fastest car plus driver combination and being reliable.
There are two phases: Car Design, in which the objective is to build a car that is the quickest around all the calendar circuits; Race Weekend, the goal is to exploit the ultimate car performance. The methodology is similar for both phases but for the design phase there is:

- more time available
- fewer constraints
- more parameters that can be changed
- possible changes in rules, tires, drivers
- very little time on track (only one test before the start of the season)

During race weekend the only thing that can be changed is the setup.

8.4.3 Race Weekend

Optimal Setup

A deep optimization is required to achieve the optimal setup, since we look for very small time savings (cents of seconds) around a lap that lasts approximately 85 seconds. There are many tuning parameters.

Parameters

The following parameters are set to achieve **Performance, Drivability and Reliability.**

- Rear Wing level
- Front Wing angle
- Ride Heights
- Toe/camber angles
- Weight distribution
- Front and rear heave stiffness
- Front and rear roll stiffness
- Dampers settings
- Cooling (PU, gearbox, brakes)
- Control strategies (differential, brakes)
- ERS strategy
- Tyre temperature
- Tyre pressure (within limits)

Simulation tools

Every weekend only three free practice sessions are available, so teams have very little time to prepare the car. They need models and simulations to predict, quickly analyze and correct the behavior of the car.

QSS lap simulation

QSS (Quasi-Steady-State) lap time simulation is a vehicle dynamics method used to calculate a race car's theoretical minimum lap time. It evaluates limits on pre-defined centerlines by assuming steady-state behavior (ignoring suspension transient states) to rapidly iterate through various vehicle setups and track layouts.
The car's physical parameters (mass, downforce, aerodynamic drag, power limits, and tire slip) are summarized into a limit envelope, frequently utilizing a diagram, then the track is divided into segments based on curvature. The simulation solves for the speed profile in two steps:

- *Forward integration:* Starting from a low-speed corner, it accelerates forward up to the vehicle's maximum longitudinal limit.
- *Backward integration:* From a braking point or corner apex, it calculates backward at maximum braking limits.

The intersection of the forward and backward speed curves determines the exact braking/acceleration points for the theoretical limit of the car.

Downforce scan

A downforce scan evaluates how varying levels of aerodynamic downforce and drag impact a vehicle's lap time.

Driver in the Loop simulator

A Driver-in-the-Loop simulator is a development system where a human driver interacts in real time with a virtual vehicle. The system combines three main elements into a single real-time architecture:

- **Physical Model:** The software that calculates vehicle dynamics, suspension, tires, and aerodynamics.
- **Sensory Feedback:** The driver sits in a real cockpit and receives physical inputs via force-feedback steering, motion platform systems, and seat actuators.
- **Graphics Environment:** Visual and audio simulation of the track or road, often rendered using high-fidelity graphics engines.

Telemetry analysis

Telemetry captures dynamic physical responses: sensors generate detailed traces per lap, then engineers with drivers compare braking zones, apex minimum speeds, and throttle application to save cents of a second. The telemetry graphs help to validate mathematical vehicle models and tune setups.

Handling and stability analysis

There are no sensors able to measure understeering or oversteering, so they can be computed with other parameters: the stability derivatives: cornering stiffnesses, yaw stiffness, yaw damping.

8.5 Racing Bicycle Tyres: Engineering at Pirelli

Speakers: Michele and Lorenzo, Pirelli – Cycling Division, R&D Engineering.
Topic: Tyre construction, structural mechanics, and performance optimisation for competitive road cycling.

8.5.1 Anatomy and Manufacturing

A bicycle racing tyre is a pressure membrane: it carries no structural load on its own, and inflation pressure alone supports the rider's weight. The product is built from four elements: the **casing** (rubberised textile from bead to bead), the two **beads** (Kevlar or steel cords anchoring the tyre to the rim), a **puncture protection belt**, and the **tread** compound. Manufacturing splits into drum building, where layers are deposited on a rotating cylinder to form the green tyre, and **vulcanisation**, where heat and pressure in a mould chemically crosslink the rubber into its final geometry. The designer controls what goes on the drum; the intermediate shape during curing is dictated by rubber chemistry, not the engineer.

Cycling tyres are structurally unlike car or motorcycle tyres, as Table 8.1 shows. Running at 5–7 bar with a 1.5 mm wall and weighing just 300 g, they operate in a fundamentally different mechanical regime.

Table 8.1: Key tyre parameters across vehicle types (indicative values, Pirelli Cycling R&D).

	Road Bike	**Motorcycle**	**Car**
Typical pressure	5–7 bar	2–3 bar	2–3 bar
Wall thickness	∼1.5 mm	∼8 mm	∼9 mm
Ply architecture	Cross-ply	Cross-ply / Radial	Radial
Weight	300 g	7 kg	12 kg

8.5.2 Cross-Ply Architecture and Structural Mechanics

Car tyres use **radial** construction with cords perpendicular to the rolling direction and stiff belts at the tread. Cycling tyres use **cross-ply**: alternating layers of fabric crossed at roughly 45° (Figure 8.13). The resulting net structure deforms locally with the contact patch, giving better handling, vibration filtering, rolling resistance and impact absorption, with less weight and fewer material constraints than a radial construction.

The casing is a tension-only membrane. The mechanism mirrors a bicycle wheel: the rim is not pushed up by the spokes below, but held up by tension in the spokes above. Under load, tension drops locally at the contact patch and the load is redistributed through the casing arc. Designing this membrane correctly requires integrating hoop and meridional stresses across three different geometries: drum, mould exit, and inflated.

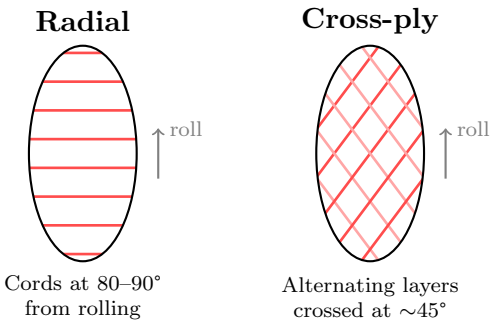


Figure 8.13: Radial vs cross-ply ply architecture: contact patch view showing cord orientation relative to the rolling direction. Pirelli Cycling R&D.

8.5.3 Rolling Resistance, Aerodynamics and Trade-offs

Rolling resistance coefficient (CRR) has two components: hysteretic loss in the rubber, which falls as pressure rises, and **impedance loss**, which rises as the tyre becomes too stiff to absorb road micro-asperities, transmitting energy as vibration into the frame and rider. Their sum has a minimum at an intermediate pressure. Maximising pressure to minimise deformation is counterproductive beyond that point.

On a flat 40 km time trial (70 kg, 390 W), aerodynamic drag absorbs over half the rider's power. On an uphill stage, gravity dominates and aerodynamics matters very little. The optimal tyre is therefore race-profile dependent. Tyre aerodynamics is evaluated efficiently in the wind tunnel using the front wheel alone, which reproduces the full-bike result. CFD is used to compare candidate tread shapes before committing to tooling, where relative drag differences are reliable even if absolute values are not.

Every target conflicts with another: lighter means less protection, lower CRR changes handling, aerodynamic tread shapes constrain compound choice. The designer's job is to define, for each product variant, which trade-offs are acceptable given the specific race demands, and to justify every choice with data.

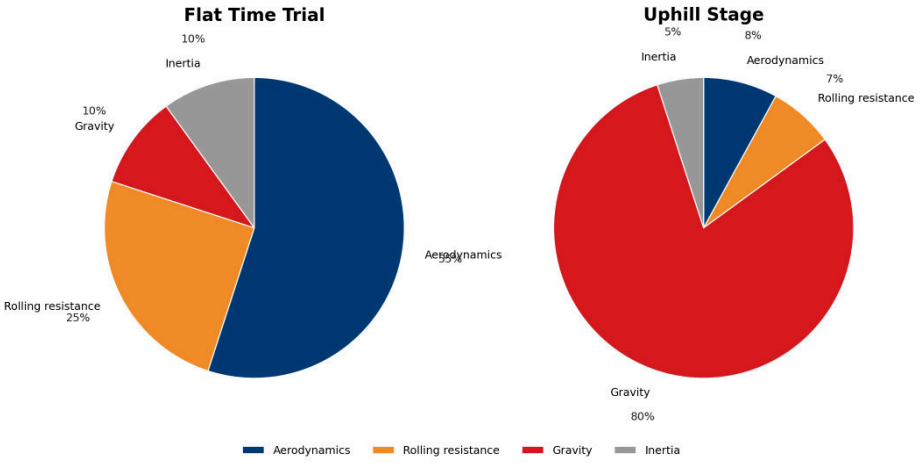


Figure 8.14: Energy balance breakdown for a flat time trial (left) and an uphill stage (right): aerodynamics dominates on the flat while gravity dominates on climbs. Pirelli Cycling R&D.

8.6 Experimental and Simulation approaches to racing motorcycles dynamics, with Ducati Corse

Speaker: Nicolò Mancinelli, Ducati – Vehicle Development Manager
Topic: creation, use, and setup of motorcycles models, for the whole vehicle and for the different subsystems, use of the models both in virtual and real environments.

8.6.1 Motorbike's susbsystems

Motorcycles are complex systems made of different subsystems

- Tires
- Brakes
- Suspensions

(these three subsystems are the only ones not produced by Ducati)

- Engine
- Aerodynamics
- Electronics
- Rider, is really important because it is very hard to manage to reach the physical limit of *MotoGP* motorbikes, only few drivers in the world have the necessary skills to minimize the lap time. The rider mass is approximately 1/3 of the mass of the system rider+motorbike, which implies that the COG of the driver also heavily affects the COG of the system. The rider's position also affects the aerodynamics of the system.

8.6.2 Virtual Models

How are virtual models built and how are they used to complete a correlation between the real and the virtual worlds?

- Engine: engine torque curves function of RMS (very simple model)
- Electronics: a simple model of the ECU, how the ECU interacts with the bike (i.e. traction control)
- Body: multibody system (made with VI-GRADE)
- Suspensions: not built in-house but on site supplied by suppliers. Dampers are characterized experimentally (an experimental model is used instead of a physical one)
- Aerodynamics: it is difficult to build an aerodynamic model because the rider continuously moves and the shape of the whole system changes, in addition the motorbike can reach a roll angle up to 60°, the pressure around the bike is not the same, these are challenging issues both in the real world (wind tunnel testing) and in the virtual world (CFD models). The goal is to have a simplified model of the aerodynamic fields of the motorbike + rider system in different conditions (braking, straight line, turns). The aerodynamic fields change with parameters like roll angle, pitch angle, yaw angle. Wind tunnels and automotive facilities are made for cars, the motorbike world must adapt them to its

necessities, the main differences are the number of the wheels that is only 2 and the roll angle that in cars can reach only low values, while in motorbikes is much higher. In addition, the rider has to be replaced by a mannequin.

- Rider: the rider is very difficult to model and it is crucial to have a customized bike for the riders in order allow them to exploit the maximum potential from the vehicle.
- Tires: the challenge is to start from the tire model created for cars and fix them, adapt them to *MotoGP* tires, it is done in an experimental way, but it is very difficult because indoor tires testing (made using test rigs) are not suitable for motorcycle tires.

Stability controller: it is really difficult to make a complex stability controller that can complete full race laps on different circuits as a real rider does.
The flexibility of the chassis as well as the vibrations on the motorbike, is studied running simulations around the track with a flexible multibody model.

8.6.3 Lap time simulations

Lap time simulations predict the laptime on a given circuit for a given motorbike. They have a low computation effort, they include some simplifications, some assumptions on the system, a twin model of the multibody model is used (some features are removed), the vehicle model features 6 rigid bodies, the transmission features 3 shafts.
Lap time simulations are usually used when new circuits are approached, for the engine configuration, the fuel consumption estimation, aerodynamics map creation, for gear ratio definition... Collocation methods are not traditional LTS, they allow also to individuate the best trackline, not to go on a defined trackline like traditional LTS.

8.6.4 Experimental Vehicle Dynamics

Kinematic and dynamic variables can be measured directly by sensors during tests, but during races, teams cannot mount all the sensors needed on the motorbike due to weight, cost,... The objective is to estimate these variables. Kinematic and dynamic variables are crucial to understand how the system and the rider behave and to correlate the system with virtual models

The tasks are the estimation of kinematic and dynamic variables (i.e. velocity, accelerations, tire slip, ground forces...) and the parameter identification on track test data (e.i. tire and aerodynamic parameters) These data are very important to understand bike and rider behavior but also for model validation and simulation correlation, they allow to create some KPIs for vehicle and rider performance analysis, and they are helpful to manage the tire wear during the weekend.

Sensors

Sensors usually mounted on motorbikes are IMU, suspension sensors, angular speed, braking pressure, and engine sensors; other sensors (heavy or more expensive) are mounted on bikes during test (i.e. speed and distance optical sensors, GPS, strain gauges for forces and moments, accelerometers, videocameras, etc. The goal is to tune the estimation algorithm in the sessions where these extra sensors are available, to have the same amount of information when they are not mounted.

Estimation

Kalman filters are mathematical algorithm that estimate the true state of a system over time, even when the data is obscured by noise or inaccuracies. It works by constantly combining a system's predicted behavior with new, real-world measurements to produce an estimate that is highly accurate, and it is used to estimate roll and pitch angles. Acceleration is primarily estimated using speed sensors to track changes in velocity over time or IMUs that directly measure G-force using accelerometers (g-g diagram). Tire slip is estimated by comparing the speed of the front and rear wheels, combining those data with a 6-axis Inertial Measurement Unit (IMU) and applying algorithms like the Extended Kalman Filter to filter out noise and compensate for tire lean. Kalman filters are used also in the estimation of other quantities such as tire forces, gyroscopic moments, etc... since the data are always affected by a noisy component.

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